

## 1.1. Python Libraries

Python libraries are collections of pre-written codes that we can use to perform common tasks, making our programming life easier. They are like toolkits that provide functions and methods to help us avoid writing code from scratch.

In the realm of data science and analytics, two powerful libraries stand out for their efficiency and versatility: NumPy and Pandas. These libraries form the backbone of data manipulation and analysis in Python, enabling users to handle large datasets with ease and precision.

Let's recap a few of these libraries (covered in Class XI) that are incredibly valuable in the realm of Artificial Intelligence, Data science and analytics.

### 1.1. 1 NumPy Library

NumPy, short for Numerical Python is a powerful library in Python used for numerical computing. It is a general-purpose array-processing package.

In NumPy, the number of dimensions of the array is called the rank of the array

```
# Creating a rank 1 Array
import numpy as np
arr = np.array([1, 2, 3])
print("Array with Rank 1: \n",arr)
```

```
Array with Rank 1:
[1 2 3]
```

```
# Creating a rank 2 Array
import numpy as np
arr = np.array([[1, 2, 3],[4,5,6]])
print("Array with Rank 2: \n",arr)
```

```
Array with Rank 2:
[[1 2 3]
 [4 5 6]]
```

```
# Creating an array from tuple
arr = np.array((1, 3, 2))
print("\nArray created using tuple:\n", arr)
```

```
Array created using tuple:
[1 3 2]
```

### 1.1.2 Pandas Library

#### Where and why do we use the Pandas library in Artificial Intelligence?

Suppose we have a dataset containing information about various marketing campaigns conducted by the company, such as campaign type, budget, duration, reach, engagement metrics, and sales performance. **Pandas** is used to load the dataset, display summary statistics, and perform group-wise analysis to understand the performance of different marketing campaigns. We can visualize the sales performance and average engagement metrics for each campaign type using **Matplotlib**, a popular plotting library in Python.

**Pandas** provides powerful data manipulation and aggregation functionalities, making it easy for us to perform complex analyses and generate insightful visualizations. This capability is invaluable in AI and data-driven decision-making processes, allowing businesses to gain actionable insights from their data.

### 1.1.2.1 Pandas Data Structures

Pandas generally provides two data structures for manipulating data, They are:

- Series
- Data Frame

i) Creation of a Series from Scalar Values- A Series can be created using scalar values as shown below:

```
import pandas as pd #import Pandas with alias pd
series1 = pd.Series([10,20,30]) #create a Series
print(series1) #Display the series
```

|   |    |
|---|----|
| 0 | 10 |
| 1 | 20 |
| 2 | 30 |

dtype: int64

ii) Creation of a DataFrame from NumPy arrays

```
array1=np.array([90,100,110,120])
array2=np.array([50,60,70])
array3=np.array([10,20,30,40])
marksDF = pd.DataFrame([array1, array2, array3], columns=[ 'A', 'B', 'C', 'D'])
print(marksDF)
```

|   | A  | B   | C   | D     |
|---|----|-----|-----|-------|
| 0 | 90 | 100 | 110 | 120.0 |
| 1 | 50 | 60  | 70  | NaN   |
| 2 | 10 | 20  | 30  | 40.0  |

iii) Creation of a DataFrame from dictionary of array/lists:

```
import pandas as pd
# initialise data of lists.
data = {'Name':['Varun', 'Ganesh', 'Joseph', 'Abdul','Reena'],
        'Age':[37,30,38, 39,40]}
# Create DataFrame
df = pd.DataFrame(data)
# Print the output.
print(df)
```

|   | Name   | Age |
|---|--------|-----|
| 0 | Varun  | 37  |
| 1 | Ganesh | 30  |
| 2 | Joseph | 38  |
| 3 | Abdul  | 39  |
| 4 | Reena  | 40  |

The dictionary keys become column labels by default in a DataFrame, and the lists become the columns of data.

#### iv) Creation of DataFrame from List of Dictionaries

```
# Create list of dictionaries
listDict = [{'a':10, 'b':20}, {'a':5, 'b':10, 'c':20}]
a= pd.DataFrame(listDict)
print(a)
```

|   | a  | b  | c    |
|---|----|----|------|
| 0 | 10 | 20 | NaN  |
| 1 | 5  | 10 | 20.0 |

There will be as many rows as the number of dictionaries present in the list.

#### 1.1.2.2 Dealing with Rows and Columns

##### i) Adding a New Column to a DataFrame:

```
ResultSheet={'Rajat': pd.Series([90, 91, 97],index=['Maths','Science','Hindi']),
             'Amrita': pd.Series([92, 81, 96],index=['Maths','Science','Hindi']),
             'Meenakshi': pd.Series([89, 91, 88],index=['Maths','Science','Hindi']),
             'Rose': pd.Series([81, 71, 67],index=['Maths','Science','Hindi']),
             'Karthika': pd.Series([94, 95, 99],index=['Maths','Science','Hindi'])}
```

```
Result = pd.DataFrame(ResultSheet)
```

```
print(Result)
```

|         | Rajat | Amrita | Meenakshi | Rose | Karthika |
|---------|-------|--------|-----------|------|----------|
| Maths   | 90    | 92     | 89        | 81   | 94       |
| Science | 91    | 81     | 91        | 71   | 95       |
| Hindi   | 97    | 96     | 88        | 67   | 99       |

To add a new column for another student 'Fathima', we can write the following statement:

```
Result['Fathima']=[89,78,76]
```

```
print(Result)
```

|         | Rajat | Amrita | Meenakshi | Rose | Karthika | Fathima |
|---------|-------|--------|-----------|------|----------|---------|
| Maths   | 90    | 92     | 89        | 81   | 94       | 89      |
| Science | 91    | 81     | 91        | 71   | 95       | 78      |
| Hindi   | 97    | 96     | 88        | 67   | 99       | 76      |

##### ii) Adding a New Row to a DataFrame:

```
Result.loc['English'] = [90, 92, 89, 80, 90, 88]
print(Result)
```

|         | Rajat | Amrita | Meenakshi | Rose | Karthika | Fathima |
|---------|-------|--------|-----------|------|----------|---------|
| Maths   | 90    | 92     | 89        | 81   | 94       | 89      |
| Science | 91    | 81     | 91        | 71   | 95       | 78      |
| Hindi   | 97    | 96     | 88        | 67   | 99       | 76      |
| English | 90    | 92     | 89        | 80   | 90       | 88      |

DataFrame.loc[] method can also be used to change the data values of a row to a particular value. For example, to change the marks of science.

```
Result.loc['Science'] = [92, 84, 90, 72, 96, 88]
print(Result)
```

|         | Rajat | Amrita | Meenakshi | Rose | Karthika | Fathima |
|---------|-------|--------|-----------|------|----------|---------|
| Maths   | 90    | 92     | 89        | 81   | 94       | 89      |
| Science | 92    | 84     | 90        | 72   | 96       | 88      |
| Hindi   | 97    | 96     | 88        | 67   | 99       | 76      |
| English | 90    | 92     | 89        | 80   | 90       | 88      |

### 1.1.2.3 Deleting Rows or Columns from a DataFrame

To delete a row, the parameter axis is assigned the value 0 and for deleting a column, the parameter axis is assigned the value 1.

```
Result = Result.drop('Hindi', axis=0) #delete the row "Hindi"
print(Result)
```

|         | Rajat | Amrita | Meenakshi | Rose | Karthika | Fathima |
|---------|-------|--------|-----------|------|----------|---------|
| Maths   | 90    | 92     | 89        | 81   | 94       | 89      |
| Science | 92    | 84     | 90        | 72   | 96       | 88      |
| English | 90    | 92     | 89        | 80   | 90       | 88      |

```
#delete multiple columns
```

```
Result = Result.drop(['Rajat', 'Meenakshi', 'Karthika'], axis=1)
print(Result)
```

|         | Amrita | Rose | Fathima |
|---------|--------|------|---------|
| Maths   | 92     | 81   | 89      |
| Science | 84     | 72   | 88      |
| English | 92     | 80   | 88      |

### 1.1.2.4 Attributes of DataFrames

We are going to use following data as example to understand the attributes of a DataFrame.

```
import pandas as pd
# creating a 2D dictionary
dict = {"Student": pd.Series(["Arnav", "Neha", "Priya", "Rahul"],
                             index=["Data 1", "Data 2", "Data 3", "Data 4"]),
        "Marks": pd.Series([85, 92, 78, 83],
                             index=["Data 1", "Data 2", "Data 3", "Data 4"]),
        "Sports": pd.Series(["Cricket", "Volleyball", "Hockey", "Badminton"],
                             index=["Data 1", "Data 2", "Data 3", "Data 4"])}
# creating a DataFrame
df = pd.DataFrame(dict)
# printing this DataFrame on the output screen
print(df)
```

|        | Student | Marks | Sports     |
|--------|---------|-------|------------|
| Data 1 | Arnav   | 85    | Cricket    |
| Data 2 | Neha    | 92    | Volleyball |
| Data 3 | Priya   | 78    | Hockey     |
| Data 4 | Rahul   | 83    | Badminton  |

#### i) DataFrame.index

```
>>>df.index
Index(['Data 1', 'Data 2', 'Data 3', 'Data 4'], dtype='object')
```

#### ii) DataFrame.columns

```
>>>df.columns
Index(['Student', 'Marks', 'Sports'], dtype='object')
```

#### iii) DataFrame.shape

```
>>>df.shape
(4,3)
```

#### iv) DataFrame.head(n)

```
>>>df.head(2)
```

|               | Student | Marks | Sports     |
|---------------|---------|-------|------------|
| <b>Data 1</b> | Arnav   | 85    | Cricket    |
| <b>Data 2</b> | Neha    | 92    | Volleyball |

#### v) DataFrame.tail(n)

```
>>>df.tail(2)
```

|               | Student | Marks | Sports    |
|---------------|---------|-------|-----------|
| <b>Data 3</b> | Priya   | 78    | Hockey    |
| <b>Data 4</b> | Rahul   | 83    | Badminton |

## 1.2. Import and Export Data between CSV Files and DataFrames

CSV files, which stand for Comma-Separated Values, are simple text files used to store tabular data. Each line in a CSV file represents a row in the table, and each value in the row is separated by a comma. This format is widely used because it is easy to read and write, both for humans and computers.

In Python, CSV files are incredibly important for data analysis and manipulation. We often use the Pandas library to load, manipulate, and analyze data stored in CSV files. Pandas provides powerful tools to read CSV files into Data Frames, which are data structures that allow us to perform complex operations on the data with ease. This makes CSV files a go-to format for data scientists and analysts working with Python.

### 1.2.1 Importing a CSV file to a DataFrame

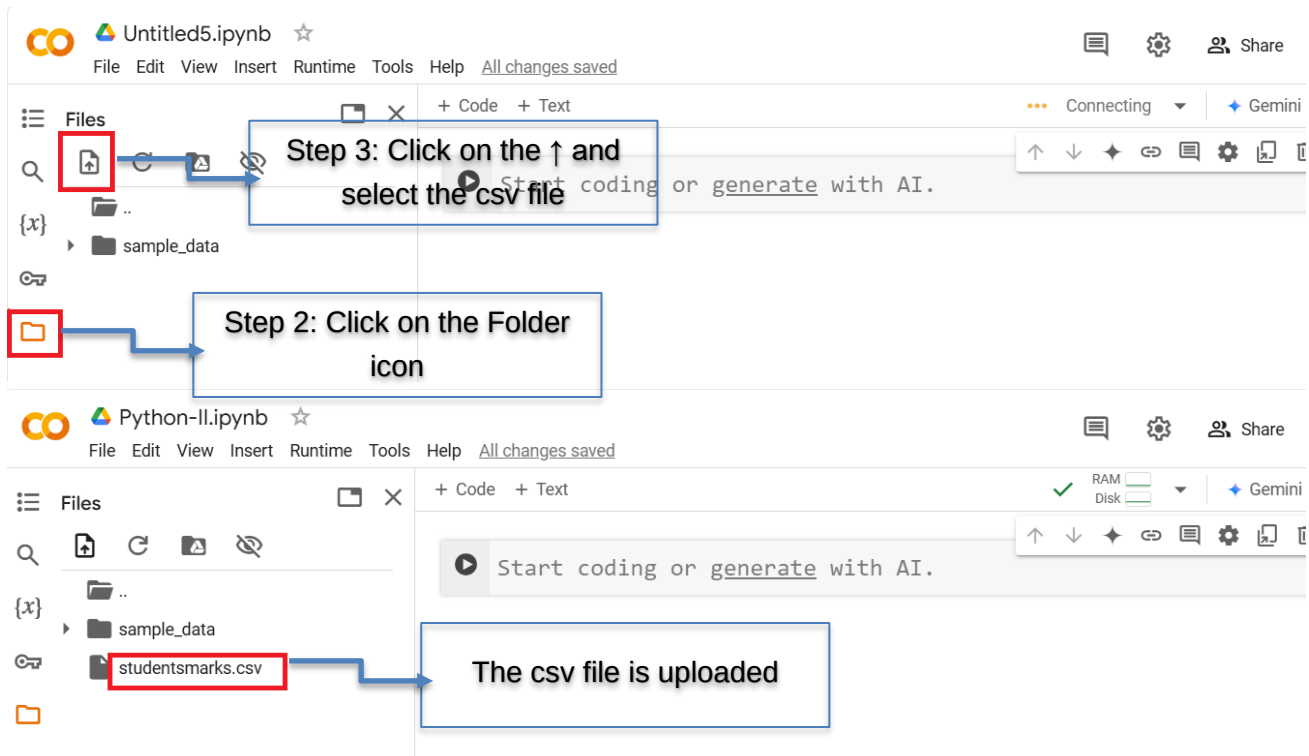
Using the `read_csv()` function, we can import tabular data from CSV files into pandas DataFrame by specifying a parameter value for the file name (e.g. `pd.read_csv("filename.csv")`).

Let us create a DataFrame from the “studentmarks.csv” file.

Follow the following steps to upload the csv file in the google colab file

**Step1:** Open the Google colab from the following link <https://colab.research.google.com/>

And create a new file from the File menu



Once the csv file is uploaded then we can execute the following code to convert the csv to DataFrame.

```
import pandas as pd
```

```
df=pd.read_csv("studentsmarks.csv")
```

```
print(df)
```

|   | Roll No | Name    | AI Marks | Maths Marks |
|---|---------|---------|----------|-------------|
| 0 | 1       | Akshita | 89       | 91          |
| 1 | 2       | Apoorva | 91       | 87          |
| 2 | 3       | Bhavik  | 88       | 76          |
| 3 | 4       | Deepti  | 78       | 71          |
| 4 | 5       | Farhan  | 84       | 84          |

On Python IDE we can directly give the complete path name with the csv file in parenthesis.

```
import pandas as pd
import io
df = pd.read_csv('C:/PANDAS/studentsmarks.csv', sep=",", header=0)
print(df)
```

|   | Roll No | Name    | AI Marks | Maths Marks |
|---|---------|---------|----------|-------------|
| 0 | 1       | Akshita | 89       | 91          |
| 1 | 2       | Apoorva | 91       | 87          |
| 2 | 3       | Bhavik  | 88       | 76          |
| 3 | 4       | Deepti  | 78       | 71          |
| 4 | 5       | Farhan  | 84       | 84          |

### 1.2.2 Exporting a DataFrame to a CSV file

We can use the `to_csv()` function to save a DataFrame to a text or csv file. For example, to save the DataFrame `df` created in above coding

```
print(df)
```

|   | Roll No | Name    | AI Marks | Maths Marks |
|---|---------|---------|----------|-------------|
| 0 | 1       | Akshita | 89       | 91          |
| 1 | 2       | Apoorva | 91       | 87          |
| 2 | 3       | Bhavik  | 88       | 76          |
| 3 | 4       | Deepti  | 78       | 71          |
| 4 | 5       | Farhan  | 84       | 84          |

Program to export this data-

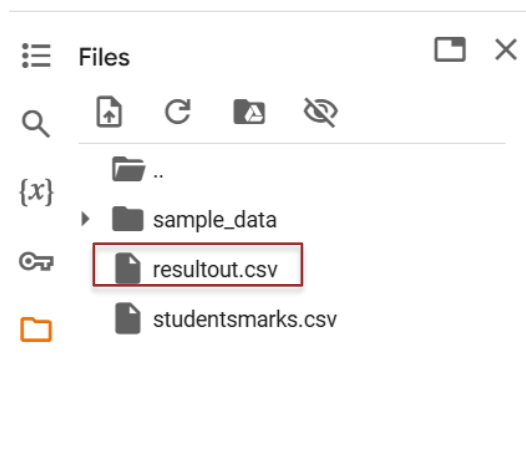
```
df.to_csv(path_or_buf='C:/PANDAS/resultout.csv', sep=',')
```

This creates a file by the name `resultout.csv` in the hard disk. When we open this file in any text editor or a spreadsheet, we will find the above data along with the row labels and the column headers, separated by comma.

On Google Colab we can write the following code

```
df.to_csv("resultout.csv", index=False)
```

The `resultout.csv` is created



### 1.3. Handling Missing Values

The two most common strategies for handling missing values explained in this section are:

- i) Drop the row having missing values **OR**
- ii) Estimate the missing value

#### Checking Missing Values

Pandas provide a function `isnull()` to check whether any value is missing or not in the DataFrame. This function checks all attributes and returns True in case that attribute has missing values, otherwise returns False

#### Drop Missing Values

Dropping will remove the entire row (object) having the missing value(s). This strategy reduces the size of the dataset used in data analysis, hence should be used in case of missing values on few objects.

The `dropna()` function can be used to drop an entire row from the DataFrame

#### Estimate the missing value

Missing values can be filled by using estimations or approximations e.g a value just before (or after) the missing value, average/minimum/maximum of the values of that attribute, etc. In some cases, missing values are replaced by zeros (or ones).

The `fillna(num)` function can be used to replace missing value(s) by the value specified in num.

For example, `fillna(0)` replaces missing value by 0. Similarly `fillna(1)` replaces missing value by 1.

### 1.4. CASE STUDY

Let's examine a scenario where we have the marks of certain students, but some data is missing in the columns due to specific circumstances. For example, Meera and Suhana couldn't attend the Science and Hindi respectively exam due to fever, Joseph participated in a national-level science exhibition on the day of the AI exam.

Let us feed the data in Python

```
ResultSheet={ 'Maths': pd.Series([90,91,97,89,65,93],
                                index=['Heena','Shefali','Meera','Joseph','Suhana','Bismmeet']),
               'Science':pd.Series([92,81,np.NaN,87,50,88],
                                index=['Heena','Shefali','Meera','Joseph','Suhana','Bismmeet']),
               'English': pd.Series([89, 91, 88,78,77,82],
                                index=['Heena','Shefali','Meera','Joseph','Suhana','Bismmeet']),
               'Hindi': pd.Series([81, 71, 67,82,np.NaN,89],
                                index=['Heena','Shefali','Meera','Joseph','Suhana','Bismmeet']),
               'AI': pd.Series([94, 95, 99,np.NaN,96,99],
                                index=['Heena','Shefali','Meera','Joseph','Suhana','Bismmeet'])}

marks = pd.DataFrame(ResultSheet)

print(marks)
```



|          | Maths | Science | English | Hindi | AI   |
|----------|-------|---------|---------|-------|------|
| Heena    | 90    | 92.0    | 89      | 81.0  | 94.0 |
| Shefali  | 91    | 81.0    | 91      | 71.0  | 95.0 |
| Meera    | 97    | NaN     | 88      | 67.0  | 99.0 |
| Joseph   | 89    | 87.0    | 78      | 82.0  | NaN  |
| Suhana   | 65    | 50.0    | 77      | NaN   | 96.0 |
| Bismmeet | 93    | 88.0    | 82      | 89.0  | 99.0 |

#check for missing values

```
>>>print(marks.isnull())
```

|          | Maths | Science | English | Hindi | AI    |
|----------|-------|---------|---------|-------|-------|
| Heena    | False | False   | False   | False | False |
| Shefali  | False | False   | False   | False | False |
| Meera    | False | True    | False   | False | False |
| Joseph   | False | False   | False   | False | True  |
| Suhana   | False | False   | False   | True  | False |
| Bismmeet | False | False   | False   | False | False |

We can see there are three “True” values. So three pieces of data are missing.

```
>>>print(marks['Science'].isnull().any())
```

True

```
print(marks['Maths'].isnull().any())
```

False

#To find the total number of NaN in the whole dataset

```
>>>marks.isnull().sum().sum()
```

3

#apply dropna() for the above case

```
drop=marks.dropna()
```

```
print(drop)
```

|          | Maths | Science | English | Hindi | AI   |
|----------|-------|---------|---------|-------|------|
| Heena    | 90    | 92.0    | 89      | 81.0  | 94.0 |
| Shefali  | 91    | 81.0    | 91      | 71.0  | 95.0 |
| Bismmeet | 93    | 88.0    | 82      | 89.0  | 99.0 |

#Estimate the missing value

```
FillZero = marks.fillna(0)
```

```
print(FillZero)
```

|          | Maths | Science | English | Hindi | AI   |
|----------|-------|---------|---------|-------|------|
| Heena    | 90    | 92.0    | 89      | 81.0  | 94.0 |
| Shefali  | 91    | 81.0    | 91      | 71.0  | 95.0 |
| Meera    | 97    | 0.0     | 88      | 67.0  | 99.0 |
| Joseph   | 89    | 87.0    | 78      | 82.0  | 0.0  |
| Suhana   | 65    | 50.0    | 77      | 0.0   | 96.0 |
| Bismmeet | 93    | 88.0    | 82      | 89.0  | 99.0 |

## 2.1. INTRODUCTION TO DATA SCIENCE METHODOLOGY

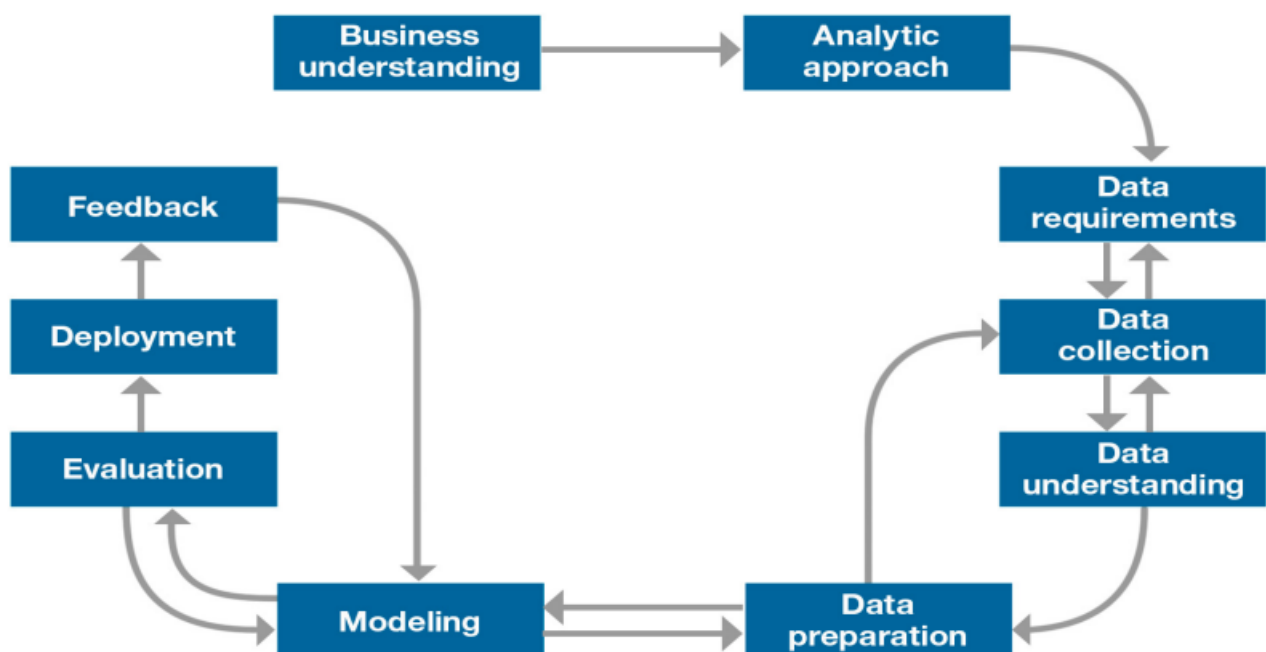
A Methodology gives the Data Scientist a **framework** for designing an AI Project. The framework will help the team to decide on the methods, processes and strategies that will be employed to obtain the correct output required from the AI Project. It is the best way to organize the entire project and finish it in a systematic way without losing time and cost.

**Data Science Methodology is a process with a prescribed sequence of iterative steps that data scientists follow to approach a problem and find a solution.**

Data Science Methodology enables the capacity to handle and comprehend the data.

In this unit, we discuss the steps of Data Science Methodology which was put forward by John Rollins, a Data Scientist at IBM Analytics. It consists of 10 steps. The foundation methodology of Data Science provides a deep insight on how every AI project can be solved from beginning to end. There are five modules, each going through two stages of the methodology, explaining the rationale as to why each stage is required.

1. From Problem to Approach
2. From Requirements to Collection
3. From Understanding to Preparation
4. From Modelling to Evaluation
5. From Deployment to Feedback



*Figure 1. Foundational methodology for data science.*

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## 1. From Problem to Approach

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### 2.1.1 Business understanding



#### What is the problem that you are trying to solve?

In this stage, first, we understand the problem of the customer by asking questions and try to comprehend what is exactly required for them. With this understanding we can figure out the objectives that support the customer's goal. This is also known as Problem Scoping and defining. The team can use 5W1H Problem Canvas to deeply understand the issue. This stage also involves using DT (Design Thinking) Framework.

To solve a problem, it's crucial to understand the customer's needs. This can be achieved by asking relevant questions and engaging in discussions with all stakeholders. Through this process, we will be able to identify the specific requirements and create a comprehensive list of business needs.



#### Activity 1:

Mr. Pavan Sankar visited a food festival and wants to sample various cuisines. But due to health concerns, he has to avoid certain dishes. However, since the dishes were not categorized by cuisine, he found it challenging and wished for assistance in identifying the cuisines offered.

Q1. Are you ready to help Pavan Sankar?

-----  
Q2. What do you think is the actual problem of Pavan?

-----  
Q3. Can we predict the cuisine of a given dish using the name of the dish only?

-----  
Q4. Let's check. Following names of dishes were taken from the menu of a restaurant in India.

- Bubble and Squeak
- Phan Pyat
- Jadoh

Are you able to tell the cuisines of these dishes?

-----  
Q5. Is it possible to determine the type of cuisine and dish name by analyzing an image of the dish?

-----  
Q6. Can you determine the cuisine of a dish based on its ingredients?

-----

Q7. What is the name of the dish prepared from the ingredients given in Fig 2.1?



Fig 2.1

### 2.1.2 Analytic Approach



**How can you use the data to answer the question?**

When the business problem has been established clearly, the data scientist will be able to define the analytical approach to solve the problem. This stage involves seeking clarification from the person who is asking the question, so as to be able to pick the most appropriate path or approach. Let us understand this in detail.

This stage involves asking more questions to the stakeholders so that the AI Project team can decide on the correct approach to solve the problem.

The different questions that can be asked now are:

1. Do I need to find how much or how many? (Regression)
2. Which category does the data belong to? (Classification)
3. Can the data be grouped? (Clustering)
4. Is there any unusual pattern in the data? (Anomaly detection)
5. Which option should be given to the customer? (Recommendation)

To solve a particular problem, there are four main types of data analytics as shown in Fig 2.2. They are:

1. Descriptive Analytics
2. Diagnostic Analytics
3. Predictive Analytics
4. Prescriptive Analytics

|                                          |
|------------------------------------------|
| <b>Descriptive</b>                       |
| • Current status                         |
| <b>Diagnostic (Statistical Analysis)</b> |
| • What happened?                         |
| • Why is this happening?                 |
| <b>Predictive (Forecasting)</b>          |
| • What if these trends continue?         |
| • What will happen next?                 |
| <b>Prescriptive</b>                      |
| • How do we solve it?                    |

Fig 2.2

## Let's understand each of them:

**Descriptive Analytics:** This summarizes past data to understand what has happened. It is the first step undertaken in data analytics to describe the trends and patterns using tools like graphs, charts etc. and statistical measures like mean, median, mode to understand the central tendency. This method also examines the spread of data using range, variance and standard deviation.

For example: To calculate the average marks of students in an exam or analyzing sales data from the previous year.

**Diagnostic Analytics:** It helps to understand the reason behind why some things have happened. This is normally done by analyzing past data using techniques like root cause analysis, hypothesis testing, correlation analysis etc. The main purpose is to identify the causes or factors that led to a certain outcome.

For example: If the sales of a company dropped, diagnostic analysis will help to find the cause for it, by analyzing questions like “Is it due to poor customer service” or “low product quality” etc.

**Predictive Analytics:** This uses the past data to make predictions about future events or trends, using techniques like regression, classification, clustering etc. Its main purpose is to foresee future outcomes and make informed decisions.

For example: A company can use predictive analytics to forecast its sales, demand, inventory, customer purchase pattern etc., based on previous sales data.

**Prescriptive Analytics:** This recommends the action to be taken to achieve the desired outcome, using techniques such as optimization, simulation, decision analysis etc. Its purpose is to guide decisions by suggesting the best course of action based on data analysis.

For example: To design the right strategy to increase the sales during festival season by analyzing past data and thus optimize pricing, marketing, production etc.

We can summarize each of these analytics as given in Table 2.1

|         | Descriptive Analytics                                 | Diagnostic Analytics                                              | Predictive Analytics                                                      | Prescriptive Analytics                                                                                                                    |
|---------|-------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Focus   | Questions on summarizing historical data              | Questions on understanding why certain events occurred            | Questions on predicting future outcomes based on historical data patterns | Questions on determining the best course of action                                                                                        |
| Purpose | Identify patterns, trends, and anomalies in past data | Uncover root causes and factors contributing to specific outcomes | Forecast future events or behaviors                                       | Recommend specific actions or interventions based on predictive insights. May indirectly influence classification through recommendations |

Table 2.1

[To know more about these analytics, you can go through this Coursera activity]

<https://www.coursera.org/learn/data-science-methodology/>



## Activity 2:

Mr. Pavan Sankar has set his goal to find the dish and its cuisine using its ingredients. He plans to proceed as shown in the flowchart in Fig 2.3.

Observe the flowchart and answer the questions.

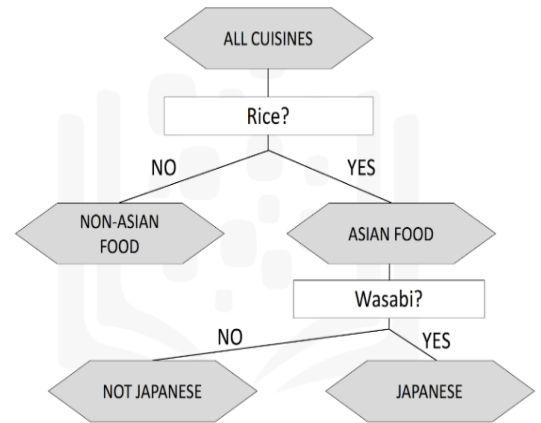


Fig 2.3

Q1. Which type of analytics questioning is being utilized here?

- A) Descriptive Analytics
- B) Diagnostic Analytics
- C) Predictive Analytics
- D) Prescriptive Analytics

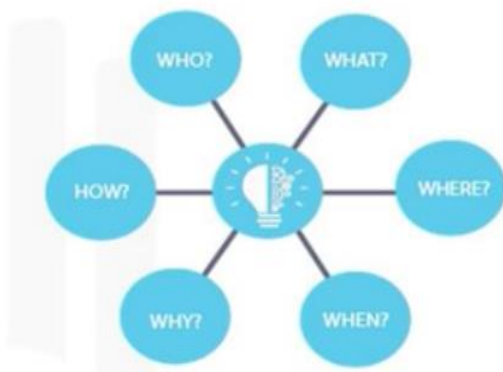
Q2. What type of approach is chosen here?

Q3. Which algorithm is depicted in the figure given here?

By thoroughly understanding the problem at hand and formulating a clear approach to address it, the stage is set for effective decision-making and problem-solving. This initial stage sets the direction for the entire project, ensuring that efforts are focused on solving the right problem in the most efficient and effective manner possible. As such, careful consideration and thoughtful planning during this stage are vital for achieving meaningful results and delivering value to stakeholders.

## 2. From Requirements to Collection

### 2.1.3 Data requirements



What are the data requirements?

The requirements of data are determined by the analytic approach chosen in the previous stage. The 5W1H questioning method can be employed in this stage also to determine the data requirements. It is necessary to identify the data content, formats, and sources for initial data collection, in this stage.

Determining the specific information needed for our analysis or project includes:

- identifying the types of data required, such as numbers, words, or images.
- considering the structure in which the data should be organized, whether it is in a table, text file, or database.
- identifying the sources from which we can collect the data, and
- any necessary cleaning or organization steps required before beginning the analysis.

This stage involves defining our data requirements, including the type, format, source, and necessary preprocessing steps to ensure the data is usable and accurate for our needs. Data for a project can be categorized into three types: **structured data** (organized in tables, e.g., customer databases), **unstructured data** (without a predefined structure, e.g., social media posts, images), and **semi-structured data** (having some organization, e.g., emails, XML files). Understanding these data types is essential for effective data collection and management in project development

### Activity 3:

Mr. Pavan Sankar is now ready with a classification approach. Now he needs to identify the data requirements.

Q1. Write down the name of two cuisines, five dishes from each cuisine and the ingredients needed for the five dishes separately.

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Q2. To collect the data on ingredients, in what format should the data be collected?

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### 2.1.4 Data collection



#### What occurs during data collection?

Data collection is a systematic process of gathering observations or measurements. In this phase, the data requirements are revised and decisions are made as to whether the collection requires more or less data. Today's high-performance database analytics enable data scientists to utilize large datasets. There are mainly two sources of data collection:

##### → **Primary data Source**

A primary data source refers to the original source of data, where the data is collected firsthand through direct observation, experimentation, surveys, interviews, or other methods. This data is raw, unprocessed, and unbiased, providing the most accurate and reliable information for research, analysis, or decision-making purposes. Examples include marketing campaigns, feedback forms, IoT sensor data etc.

##### → **Secondary data Source**

A secondary data source refers to the data which is already stored and ready for use. Data given in books, journals, websites, internal transactional databases, etc. can be reused for data analysis. Some methods of collecting secondary data are social media data tracking, web scraping, and satellite data tracking. Some sources of online data are data.gov, World bank open data, UNICEF, open data network, Kaggle, World Health Organization, Google etc. Smart forms are an easy way to procure data online.

DBA's and programmers often work together to extract data from both primary and secondary sources. Once the data is collected, the data scientist will have a good understanding of what they will be working with. The Data Collection stage may be revisited after the Data Understanding stage, where gaps in the data are identified, and strategies are developed to either collect additional data or make substitutions to ensure data completeness.



#### **Activity 4:**

Q1. If you need the names of American cuisine, how will you collect the data?

---

Q2. You want to try out some healthy recipes in the Indian culture. Mention the different ways you could collect the data.

---



Q3. How can you collect a large amount of data and where can it be stored?

---



---

### 3. From Understanding to Preparation

#### 2.1.5 Data Understanding



**Is the data collected representative of the problem to be solved?**

Data Understanding encompasses all activities related to constructing the dataset. In this stage, we check whether the data collected represents the problem to be solved or not. The relevance, comprehensiveness, and suitability of the data for addressing the specific problem or question at hand are evaluated. Techniques such as descriptive statistics (univariate analysis, pairwise correlation etc.) and visualization (Histogram) can be applied to the dataset, to assess the content, quality, and initial insights about the data.



#### Activity 5:

Q1. Semolina which is called *rava* or *suji* in Indian households is a by-product of durum wheat. Name a few dishes made from semolina. How will you differentiate the data of different dishes?

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Q2. Given below is a sample data collected during the data collection stage. Let us try to understand it.

Table -2.2

| Dish            | Country   | Salt | Chilli | Onion | Oil | Rice | Fish | Chicken | Potato | Milk | Sugar | Vegetable | Wasabi | Soy sauce |
|-----------------|-----------|------|--------|-------|-----|------|------|---------|--------|------|-------|-----------|--------|-----------|
| Chicken Biryani | Indian    | 1    | 1      | 1     | 1   | 1    | 0    | 2       | 0      | 0    | 0     | 0         | 0      | 0         |
| Kheer           | Indian    | 0    | 0      | 0     | N   | 1    | 0    | 0       | 0      | 1    | 1     | 0         | 0      | 0         |
| Pulao           | Indiana   | 1    | 1      | 1     | 1   | One  | 0    | 0       |        | 0    | 0     | 1         | 0      |           |
| Sushi           | Japane se | 1    | 0      | 1     | 1   | 1    | 1    | 0       | 0      | 0    | N     | 0         | 1      | 0         |
| Fried Rice      | Chinese   | Y    | 1      | 1     | 1   | 1    | 0    | 1       | 0      | 1    | 0     | 1         | 0      | 1         |

- a. Basic ingredients of sushi are rice, soy sauce, wasabi and vegetables. Is the dish listed in the data? Are all ingredients available?

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- b. Find out the ingredients for the dish “Pulao”. Check for invalid data or missing data.

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- c. Inspect all columns for invalid, incorrect or missing data and list them below.

---

- d. Which ingredients is common for all dishes? Which ingredient is not used for any dish?

---

### 2.1.6 Data preparation



**What additional work is required to manipulate and work with the data?**

This stage covers all the activities to build the set of data that will be used in the modelling step. Data is transformed into a state where it is easier to work with.

Data preparation includes

- cleaning of data (dealing with invalid or missing values, removal of duplicate values and assigning a suitable format)
- combine data from multiple sources (archives, tables and platforms)
- transform data into meaningful input variables

**Feature Engineering** is a part of Data Preparation. The preparation of data is the most time-consuming step among the Data Science stages.

Feature engineering is the process of selecting, modifying, or creating new features (variables) from raw data to improve the performance of machine learning models.

For example:

Suppose you're building an ML model to predict price of houses, and you have the following data:

- Raw Data: Area of the house (in sq.ft), number of bedrooms, and the year the house was built.

To improve the model's performance, you might create new features such as age of the house and price per square foot which can be derived from the raw data.

- New Features
  1. Age of the house = Current year - Year built.
  2. Price per square foot = Price of the house / Area.

These new features can help the model make more accurate predictions.



## Activity 6:

Q1. Are there any textual mistakes in the data given in the Table-1? Mention if any.

Q2. In Table-1, incorrect data was identified in the columns rice and chicken. Write the possible ways to rectify them.

Q3. Is the first column name appropriate? Can you suggest a better name?

Q4. First three values of the first column seem to be similar. Do we need to make any corrections to this data?

Q5. Do the dishes with common ingredients come under the same cuisine? Why?

Q6. Instead of mentioning whether the ingredient is present or not by using 0's and 1's, can you suggest any alternative ways to display the information?

## 4. From Modelling to Evaluation

### 2.1.7 AI modelling



**In what way can the data be visualized to get to the required answer?**

The modelling stage uses the initial version of the dataset prepared and focuses on developing models according to the analytical approach previously defined. The modelling process is usually iterative, leading to the adjustments in the preparation of data. For a determined technique, Data scientists can test multiple algorithms to identify the most suitable model for the Capstone Project.

Data Modelling focuses on developing models that are either descriptive or predictive.

1. **Descriptive Modeling:** It is a concept in data science and statistics that focuses on summarizing and understanding the characteristics of a dataset without making predictions or decisions. The goal of descriptive modeling is to describe the data rather

than predict or make decisions based on it. This includes summarizing the main characteristics, patterns, and trends that are present in the data. Descriptive modeling is useful when you want to understand what is happening within your data and how it behaves, but not necessarily why it happens.

Common Descriptive Techniques:

- **Summary Statistics:** This includes measures like:
  - Mean (average), Median, Mode
  - Standard deviation, Variance
  - Range (difference between the highest and lowest values)
  - Percentiles (e.g., quartiles)
- **Visualizations:** Graphs and charts to represent the data, such as:
  - Bar charts
  - Histograms
  - Pie charts
  - Box plots
  - Scatter plots

2. **Predictive modeling:** It involves using data and statistical algorithms to identify patterns and trends in order to predict future outcomes or values. It relies on historical data and uses it to create a model that can predict future behavior or trends or forecast what might happen next. It involves techniques like regression, classification, and time-series forecasting, and can be applied in a variety of fields, from predicting exam scores to forecasting weather or stock prices. While it is a powerful tool, students must also understand its limitations and the importance of good data.

The data scientist will use a training set for predictive modeling. A training set is a set of historical data in which the outcomes are already known. The training set acts like a gauge to determine if the model needs to be calibrated. In this stage, the data scientist will play around with different algorithms to ensure that the variables selected are actually required.



### Activity 7:

Q1. Name two programming languages which can be used to implement the Decision Tree Algorithm.

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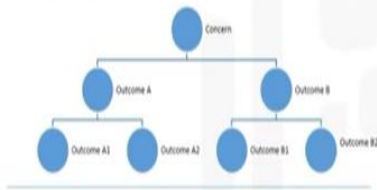
Q2. In the problem of identifying dish name and cuisine, if we chose the algorithm Decision Tree to solve the problem and choose Python as a tool, name some libraries which will help in the implementation.

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## 2.1.8 Evaluation

### Diagnostic measures

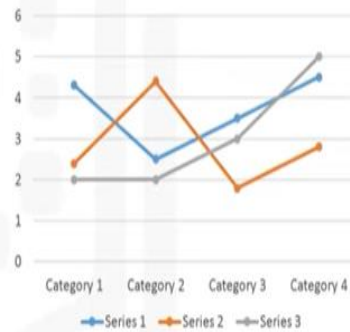
#### Predictive Model



#### Descriptive Model



#### Statistical Significance



**Does the model used really answer the initial question or does it need to be adjusted?**

<https://medium.com/ml-research-lab/part-4-data-science-methodology-from-modelling-to-evaluation-3fb3c0cdf805>

**Evaluation** in an AI project cycle is the process of assessing how well a model performs after training. It involves using test data to measure metrics like accuracy, precision, recall, or F1 score. This helps determine if the model is reliable and effective before deploying it in real-world situations.

Model evaluation can have two main phases.

#### First phase – Diagnostic measures

It is used to ensure the model is working as intended. If the model is a predictive model, a decision tree can be used to evaluate the output of the model, check whether it is aligned to the initial design or requires any adjustments. If the model is a descriptive model, one in which relationships are being assessed, then a testing set with known outcomes can be applied, and the model can be refined as needed.

#### Second phase – Statistical significance test

This type of evaluation can be applied to the model to verify that it accurately processes and interprets the data. This is designed to avoid unnecessary second guessing when the answer is revealed.



#### Activity 8:

Q1. In the cuisine identification problem, on which set will the Decision tree be built: Training or Test?

Q2. Name any diagnostic metric which can be used to determine an optimal classification model.

---

## 5. From Deployment to Feedback

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### 2.1.9 Deployment



**How does the solution reach the hands of the user?**

Deployment refers to the stage where the trained AI model is made available to the users in real-world applications. Data scientists must make the stakeholders familiar with the tool produced in different scenarios. Once the model is evaluated and the data scientist is confident it will work, it is deployed and put to the ultimate test. Depending on the purpose of the model, it may be rolled out to a limited group of users or in a test environment, to build up confidence in applying the outcome for use across the board.

Deploying a model into a live business process, frequently necessitates the involvement of additional internal teams, skills and technology.



#### **Activity 9:**

Q1. Mention some ways to embed the solution into mobiles or websites.

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### 2. 1.10 Feedback



**Is the problem solved?**

**Has the question been satisfactorily answered?**

The last stage in the methodology is feedback. This includes results collected from the deployment of the model, feedback on the model's performance from the users and clients, and observations from how the model works in the deployed environment. This process continues till the model provides satisfactory and acceptable results.

Feedback from the users will help to refine the model and assess it for performance and impact. The process from modelling to feedback is highly iterative. Data Scientists may automate

any or all of the feedback so that the model refresh process speeds up and can get quick improved results. Feedback from users can be received in many ways.



Throughout the Data Science Methodology, each step sets the stage for the next, making the methodology cyclical and ensuring refinement at each stage.

## 2.2. MODEL VALIDATION

Evaluating the performance of a trained machine learning model is essential. Model Validation offers a systematic approach to measure its accuracy and reliability, providing insights into how well it generalizes to new, unseen data.



<https://www.geeksforgeeks.org/what-is-model-validation-and-why-is-it-important/>

Model validation is the step conducted post Model Training, wherein the effectiveness of the trained model is assessed using a testing dataset. Validating the machine learning model during the training and development stages is crucial for ensuring accurate predictions. The benefits of Model Validation include

- Enhancing the model quality.
- Reduced risk of errors
- Prevents the model from overfitting and underfitting.

## Model Validation Techniques

The commonly used Validation techniques are Train-test split, K-Fold Cross Validation, Leave One out Cross Validation, Time Series Cross Validation etc. Let's discuss the Train test split and K-Fold Cross Validation

### 2.2.1 Train Test Split

The train-test split is a technique for evaluating the performance of a machine learning algorithm. It can be used for classification or regression problems and can be used for any supervised learning algorithm.

The procedure involves taking a dataset and dividing it into two subsets, as shown in Fig 2.4. The first subset is used to fit/train the model and is referred to as the **training dataset**. The second subset is used to test the model. It is with the testing data that predictions are made and compared to the expected values. This second dataset is referred to as the **test dataset**.

**Train Dataset:** Used to fit the machine learning model.

**Test Dataset:** Used to evaluate the fit machine learning model.

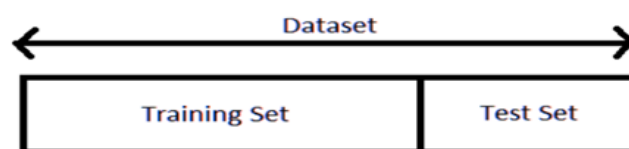


Fig 2.4

The objective is to estimate the performance of the machine learning model on new data, i.e., data not used to train the model. This is how we expect to use the model in practice. The procedure is to fit it on available data with known inputs and outputs, then make predictions on new examples in the future where we do not have the expected output or target values. The train-test procedure is appropriate when there is a sufficiently large dataset available.

#### How to Configure the Train-Test Split

The procedure has one main configuration parameter, which is the size of the train and test sets. This is most commonly expressed as a percentage between 0 and 1 for either the train or test datasets. For example, a training set with the size of 0.67 (67 percent) means that the remaining percentage 0.33 (33 percent) is assigned to the test set. There is no optimal split percentage. You must choose a split percentage that meets your project's objectives with considerations that include:

- Computational cost in training the model.
- Computational cost in evaluating the model.
- Representation of the Train set.
- Representation of the Test set.

Nevertheless, common split percentages include:

- Train: 80%, Test: 20%
- Train: 70%, Test: 30%
- Train: 67%, Test: 33%



## 2.2.2 K-Fold Cross Validation

Cross Validation is a technique used to evaluate a model's performance. It splits the data into multiple parts or folds. It trains the model on some folds and tests it on other folds and repeats this process for a number fixed by the data scientist. In cross-validation, we run our modeling process on different subsets of the data to get multiple measures of model quality.

### k-fold cross validation

In k-fold cross validation we will be working with k subsets of datasets. For example, if we divide the data into 5 folds or 5 pieces, as shown in Fig 2.5, each being 20% of the full dataset, then  $k=5$ .

We run an experiment called experiment 1 which uses the first fold as a holdout set (validation), and remaining four folds as training data. This gives us a measure of model quality based on a 20% holdout set. We then run a second experiment, where we hold out data from the second fold. This gives us a second estimate of model quality. We repeat this process, using every fold once as the holdout. Putting this together, 100% of the data is used as a holdout at some point.

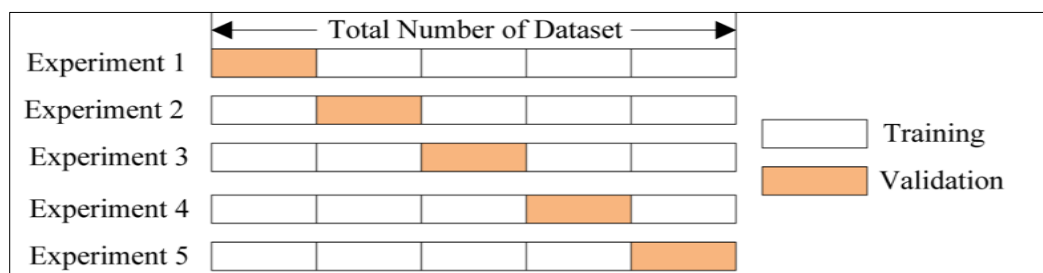


Fig 2.5

Cross-validation gives a more accurate measure of model quality, which is especially important if you are making a lot of modelling decisions. However, it can take more time to run, because it estimates models once for each fold.

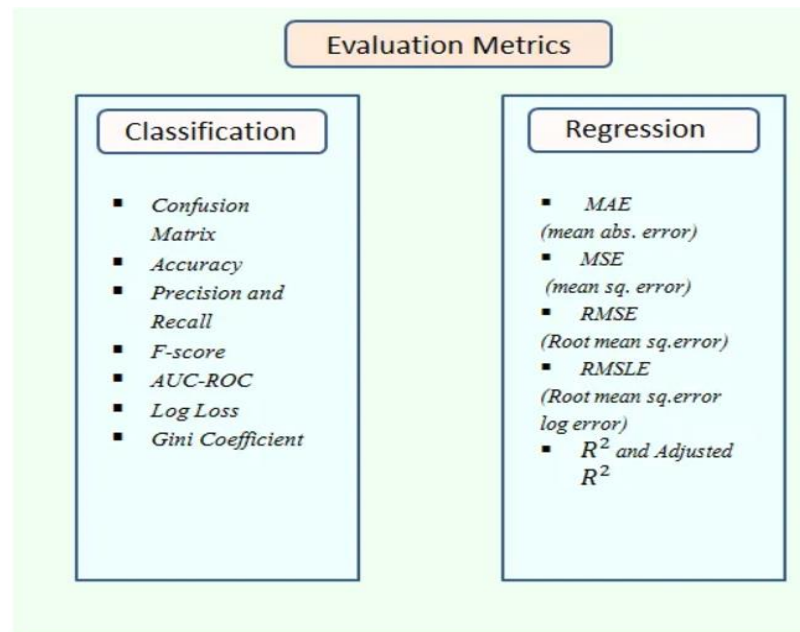
### Difference between Train-Test Split and Cross Validation

| Train-Test Split                                             | Cross Validation                                                                                                             |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Normally applied on large datasets                           | Normally applied on small datasets                                                                                           |
| Divides the data into training data set and testing dataset. | Divides a dataset into subsets (folds), trains the model on some folds, and evaluates its performance on the remaining data. |
| Clear demarcation on training data and testing data.         | Every data point at some stage could be in either testing or training data set.                                              |

## 2.3. MODEL PERFORMANCE - EVALUATION METRICS

Evaluation metrics help assess the performance of a trained model on a test dataset, providing insights into its strengths and weaknesses. These metrics enable comparison of different models, including variations of the same model, to select the best-performing one for a specific task.

In classification problems, we categorize the target variable into a finite number of classes, while in regression problems, the target variable has continuous values. Hence, we have different evaluation metrics for each type of supervised learning, as depicted in Fig 2.6.



Evaluation metrics for classification and regression models

Fig 2.6

<https://medium.com/@ladkarsamisha123/most-popular-machine-learning-performance-metrics-part-1-ab7189dce555>

### 2.3.1 Evaluation Metrics for Classification

#### 1. Confusion Matrix

A Confusion Matrix is a table (Fig 2.7) used to evaluate the performance of a classification model. It summarizes the predictions against the actual outcomes. It creates an  $N \times N$  matrix, where  $N$  is the number of classes or categories that are to be predicted. Suppose there is a problem, which is a binary classification, then  $N=2$  (Yes/No). It will create a  $2 \times 2$  matrix.

**True Positives:** It is the case where the model predicted Yes and the real output was also yes.

**True Negatives:** It is the case where the model predicted No and the real output was also No.

**False Positives:** It is the case where the model predicted Yes but it was actually No.

False Negatives: It is the case where the model predicted No but it was actually Yes.

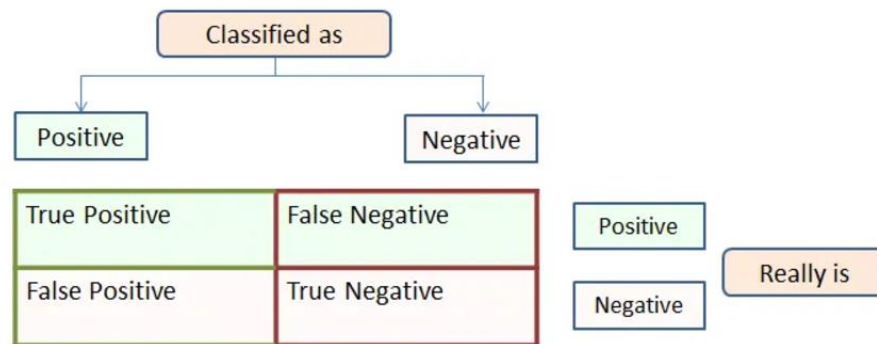


Fig 2.7

## 2. Precision and Recall

Precision measures “What proportion of predicted Positives is truly Positive?”

Precision =  $(TP)/(TP+FP)$ .

Precision should be as high as possible.

Recall measures “What proportion of actual Positives is correctly classified?”

Recall =  $(TP)/(TP+FN)$

## 3. F1-score

A good F1 score means that you have low false positives and low false negatives, so you’re correctly identifying real threats, and you are not disturbed by false alarms. An F1 score is considered perfect when it is 1, while the model is a total failure when it is 0.

$F1 = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$

## 4. Accuracy

Accuracy = Number of correct predictions / Total number of predictions

Accuracy =  $(TP+TN)/(TP+FP+FN+TN)$

### 2.3.2 Evaluation Metrics for Regression

#### 1. MAE

Mean Absolute Error is a sum of the absolute differences between predictions and actual values.

A value of 0 indicates no error or perfect predictions

#### 2. MSE

Mean Square Error (MSE) is the most commonly used metric to evaluate the performance of a regression model. MSE is the mean(average) of squared distances between our target variable and predicted values.

$$MSE = \frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}$$

### 3. RMSE

Root Mean Square Error (RMSE) is the standard deviation of the residuals (prediction errors). RMSE is often preferred over MSE because it is easier to interpret since it is in the same units as the target variable.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$

## 2.4. PRACTICAL ACTIVITIES

### 2.4.1. Calculate MSE and RMSE values for the data given below using MS Excel.

|                 |    |    |    |    |    |    |    |    |    |    |
|-----------------|----|----|----|----|----|----|----|----|----|----|
| Predicted value | 14 | 19 | 17 | 13 | 12 | 7  | 24 | 23 | 17 | 18 |
| Actual Value    | 17 | 18 | 18 | 15 | 18 | 11 | 20 | 18 | 13 | 19 |

| Predicted | Actual | Residual (actual - predicted) | Squared Residuals |
|-----------|--------|-------------------------------|-------------------|
| 14        | 17     | 3                             | 9                 |
| 19        | 18     | -1                            | 1                 |
| 17        | 18     | 1                             | 1                 |
| 13        | 15     | 2                             | 4                 |
| 12        | 18     | 6                             | 36                |
| 7         | 11     | 4                             | 16                |
| 24        | 20     | -4                            | 16                |
| 23        | 18     | -5                            | 25                |
| 17        | 13     | -4                            | 16                |
| 18        | 19     | 1                             | 1                 |

Total 125  
MSE 12.5  
RMSE 3.54

### 2.4.2. Given a confusion matrix, calculate Precision, Recall, F1 score and Accuracy

Confusion Matrix:

$$\begin{bmatrix} TN = 50 & FP = 10 \\ FN = 5 & TP = 35 \end{bmatrix}$$

From the confusion matrix:

True Positives (TP) = 35, True Negatives (TN) = 50, False Positives (FP) = 10, False Negatives (FN) = 5

With the rapid expansion of social media platforms such as Facebook, Instagram, and Twitter, smartphones have emerged as pivotal tools, thanks to their integrated cameras facilitating effortless sharing of photos and videos. While the Internet predominantly consists of text-based content, indexing and searching images present a distinct challenge. Indexing and searching images involve organizing image data for quick retrieval based on specific features like colour, texture, shape, or metadata. During indexing, key attributes are extracted and stored in a searchable format. Searching uses this index to match query parameters with stored image features, enabling efficient retrieval. Unlike text, which can be easily processed, algorithms require additional capabilities to interpret image content.

Traditionally, the information conveyed by images and videos has relied heavily on manually provided meta descriptions. To overcome this limitation, there is a growing need for computer systems to visually perceive and comprehend images to extract meaningful information from them. This involves enabling computers to "see" images and decipher their content, thereby bridging the gap in understanding and indexing visual data. This poses a simple challenge for humans, evident in the common practice of teaching children to associate an image, such as an apple, with the letter 'A'.



Humans can easily make this connection. However, enabling computers to comprehend images presents a different dilemma. Similarly to how children learn by repeatedly viewing images to memorize objects or people, we need computers to develop similar capabilities to effectively analyse our images and videos.

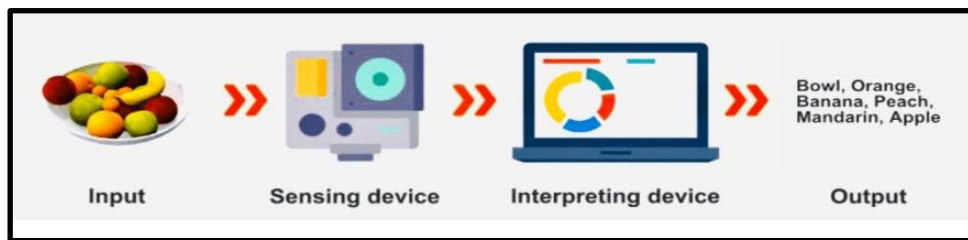
### 3.1. HOW MACHINES SEE?

Computer Vision, commonly referred to as CV, enables systems to see, observe, and understand. Computer Vision is similar to human vision as it trains machines with cameras, data, and algorithms similar to retinas, optic nerves, and a visual cortex as in human vision. CV derives meaningful information from digital images, videos and other visual input and makes recommendations or takes actions accordingly.

Computer Vision systems are trained to inspect products, watch infrastructure, or a production asset to analyse thousands of products or processes in real-time, noticing defects or issues. Due to its speed, objectivity, continuity, accuracy, and scalability, it can quickly surpass human capabilities. The latest deep learning models achieve above human-level accuracy and performance in real-world image recognition tasks such as facial recognition, object detection, and image classification.

Computer Vision is a field of artificial intelligence (AI) that uses Sensing devices and deep learning models to help systems understand and interpret the visual world.

Computer Vision is sometimes called Machine Vision.



**Fig.3.1: Process flow of computer vision**

image source: <https://www.ciopages.com/wp-content/uploads/2020/07/vision-work.jpg>

## 3.2. WORKING OF COMPUTER VISION

At its core, computer vision is the field of study that focuses on processing and analysing digital images and videos to comprehend their content. A fundamental aspect of computer vision lies in understanding the basics of digital images.

### 3.2.1. Basics of digital images

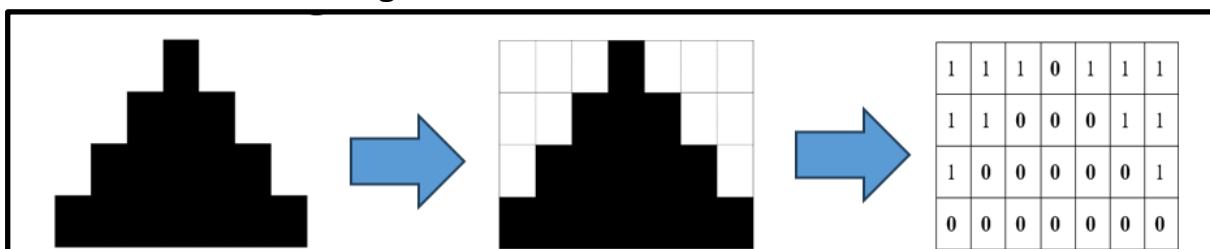
A digital image is a picture that is stored on a computer in the form of a sequence of numbers that computers can understand. Digital images can be created in several ways like using design software (like Paint or Photoshop), taking one on a digital camera, or scan one using a scanner.

### 3.2.2. Interpretation of Image in digital form



**Fig.3.2: How pixel affects the image**

When a computer processes an image, it perceives it as a collection of tiny squares known as *pixels*. Each pixel, short for "picture element," represents a specific color value. These pixels collectively form the digital image. During the process of digitization, an image is converted into a grid of pixels. The resolution of the image is determined by the number of pixels it contains; the higher the resolution, the more detailed the image appears and the closer it resembles the original scene.



**Fig. 3.3: Representing an image using 0's and 1's**

In representing images digitally, each pixel is assigned a numerical value. For monochrome images, such as black and white photographs, a pixel's value typically ranges from 0 to 255. A value of 0 corresponds to black, while 255 represents white.

### Step 6: Paste into a Word Document

- Open a Word document (Google Docs or Microsoft Word).
- Paste the copied pixel values into the document.

### Step 7: Adjust the Font Size

- Select all the pasted pixel values in the document.
- Change the font size to **1** for better visualization.
- **Observe the image formation as 0s and 1s recreate the original grayscale image.**

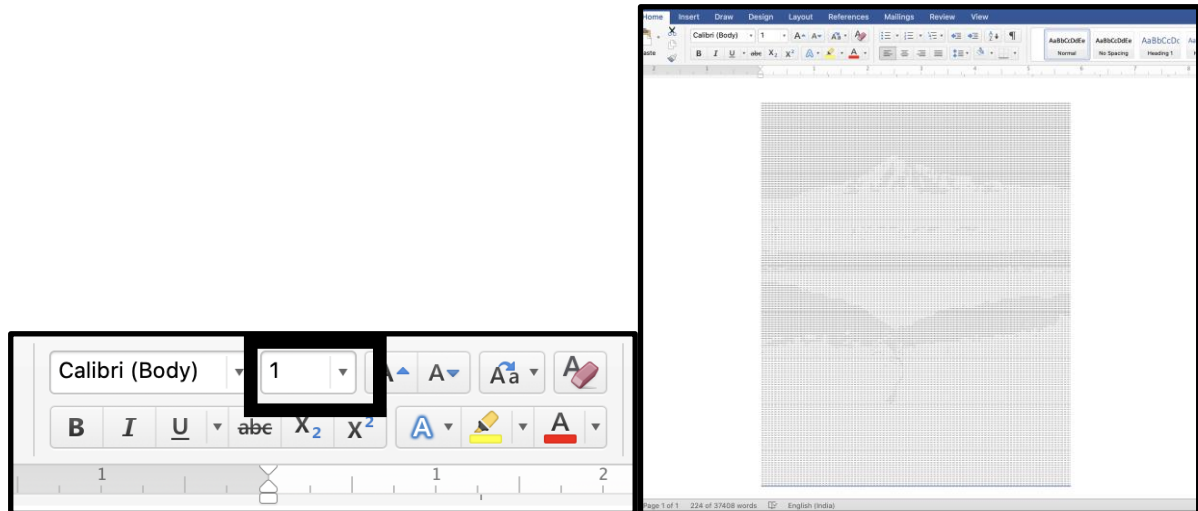


Fig. 3.9: image formation as 0s and 1s recreate the original grayscale image

In coloured images, each pixel is assigned a specific number based on the RGB colour model, which stands for Red, Green, and Blue.

1 byte= 8 bits so the total number of binary numbers formed will be  $2^8=256$ .

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = 2^7 \times 0 + 2^6 \times 0 + 2^5 \times 0 + 2^4 \times 0 + 2^3 \times 0 + 2^2 \times 0 + 2^1 \times 0 + 2^0 \times 0 = 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} = 2^7 \times 1 + 2^6 \times 1 + 2^5 \times 1 + 2^4 \times 1 + 2^3 \times 1 + 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 1 = 255$$

By combining different intensities of red, green, and blue, a wide range of colours can be represented in an image, each colour channel can have a value from 0 to 255, resulting in over 16 million possible colours.

## 3.3. COMPUTER VISION – PROCESS:

The Computer Vision process often involves five stages. They are explained below.

**3.3.1. Image Acquisition:** *Image acquisition is the initial stage in the process of computer vision, involving the capture of digital images or videos.* This step is crucial as it provides the raw data upon which subsequent analysis is based. Digital images can be acquired through various means, including capturing them with digital cameras, scanning physical photographs or documents, or even generating them using design software.

The quality and characteristics of the acquired images greatly influence the effectiveness of subsequent processing and analysis. It is important to understand that the capabilities and resolutions of different imaging devices play a significant role in determining the quality of acquired images. Higher-resolution devices can capture finer details and produce clearer images compared to those with lower resolutions. Moreover,



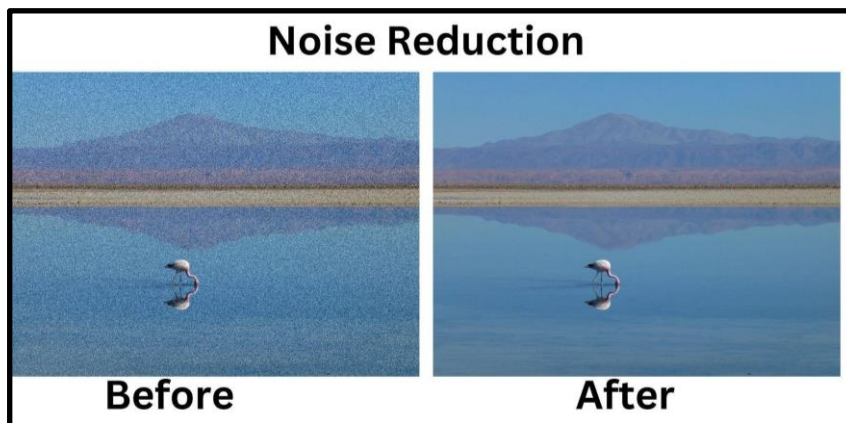
various factors such as lighting conditions and angles can influence the effectiveness of image acquisition techniques. For instance, capturing images in low-light conditions may result in poorer image quality, while adjusting the angle of capture can provide different perspectives of the scene.

In scientific and medical fields, specialized imaging techniques like MRI (Magnetic Resonance Imaging) or CT (Computed Tomography) scans are employed to acquire highly detailed images of biological tissues or structures. These advanced imaging modalities offer insights into the internal composition and functioning of biological entities, aiding in diagnosis, research, and treatment planning.

### 3.3.2. Preprocessing:

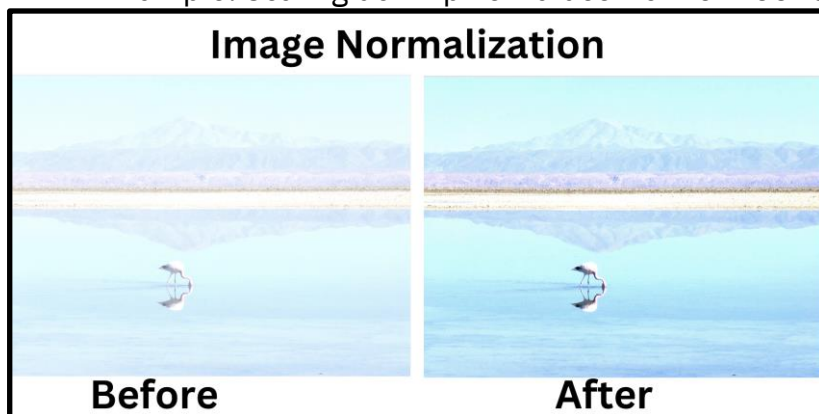
Preprocessing in computer vision aims to enhance the quality of the acquired image. Some of the common techniques are-

- a. **Noise Reduction:** Removes unwanted elements like blurriness, random spots, or distortions. This makes the image clearer and reduces distractions for algorithms.  
Example: Removing grainy effects in low-light photos.



*Fig. 3.10: before image is noise image, after image is noise reduced*

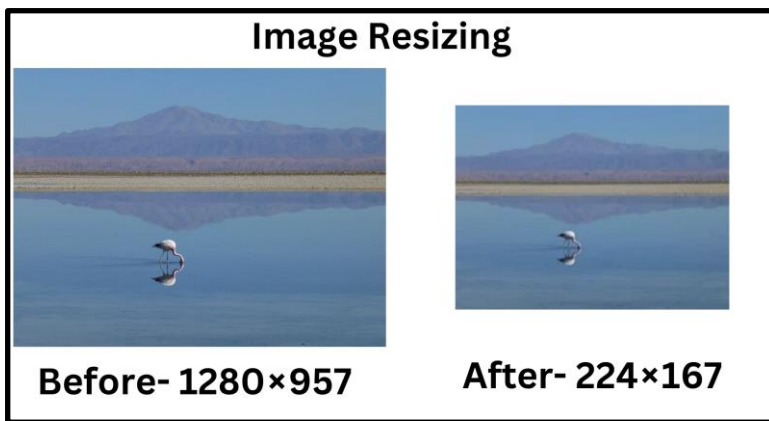
- b. **Image Normalization:** Standardizes pixel values across images for consistency. Adjusts the pixel values of an image so they fall within a consistent range (e.g., 0–1 or -1 to 1).  
Ensures all images in a dataset have a similar scale, helping the model learn better.  
Example: Scaling down pixel values from 0–255 to 0–1.



*Fig. 3.11: before is distorted with no normalization*

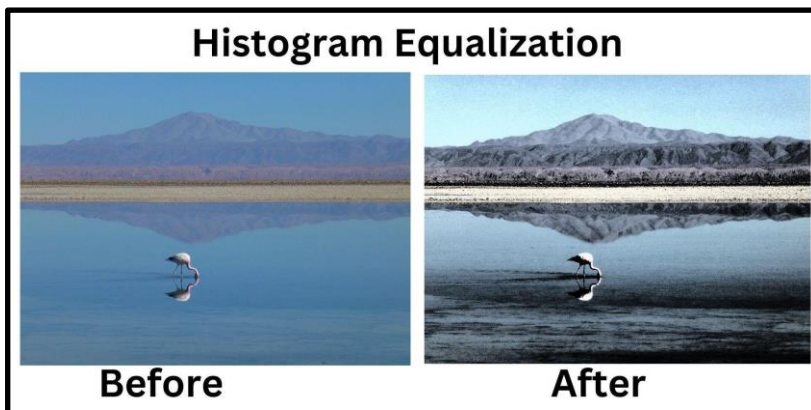
- c. **Resizing/Cropping:** Changes the size or aspect ratio of the image to make it uniform. Ensures all images have the same dimensions for analysis.  
Example: Resizing all images to 224×224 pixels before feeding them into a neural network.





*Fig. 3.12: Resizing of image*

- d. **Histogram Equalization:** Adjusts the brightness and contrast of an image. Spreads out the pixel intensity values evenly, enhancing details in dark or bright areas. Example: Making a low-contrast image look sharper and more detailed.



*Fig. 3.13*

The main goal for preprocessing is to prepare images for computer vision tasks by:

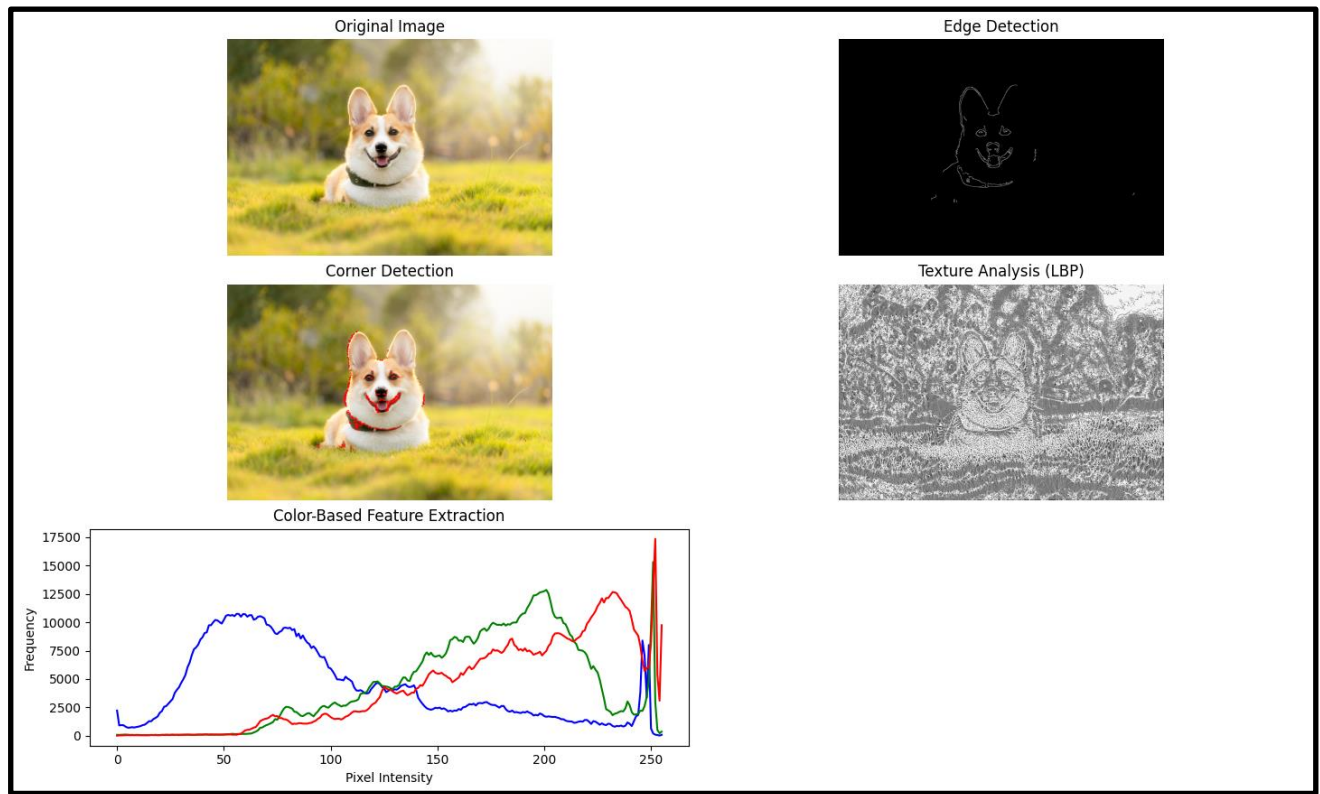
- Removing noise (disturbances).
- Highlighting important features.
- Ensuring consistency and uniformity across the dataset.

### 3.3.3. Feature Extraction:

Feature extraction involves identifying and extracting relevant visual patterns or attributes from the pre-processed image. Feature extraction algorithms vary depending on the specific application and the types of features relevant to the task. The choice of feature extraction method depends on factors such as the complexity of the image, the computational resources available, and the specific requirements of the application.

- **Edge detection** identifies the boundaries between different regions in an image where there is a significant change in intensity
- **Corner detection** identifies points where two or more edges meet. These points are areas of high curvature in an image, focused on identifying sharp changes in image gradients, which often correspond to corners or junctions in objects.
- **Texture analysis** extracts features like smoothness, roughness, or repetition in an image

- **Colour-based feature extraction** quantifies colour distributions within the image, enabling discrimination between different objects or regions based on their colour characteristics.



**Fig.3.14-Edge detection, corner detection, texture analysis, color-based feature extraction**

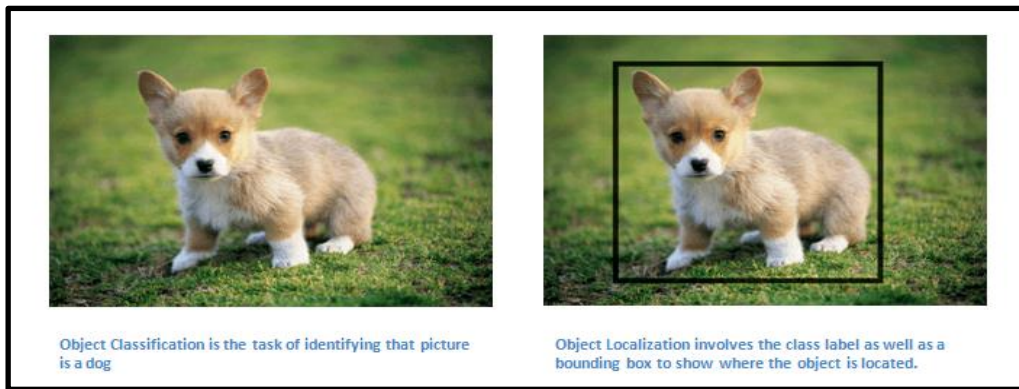
In deep learning-based approaches, feature extraction is often performed automatically by convolutional neural networks (CNNs) during the training process.

#### **3.3.4. Detection/Segmentation:**

Detection and segmentation are fundamental tasks in computer vision, focusing on identifying objects or regions of interest within an image. These tasks play a pivotal role in applications like autonomous driving, medical imaging, and object tracking. This crucial stage is categorized into two primary tasks:

1. Single Object Tasks
2. Multiple Object Tasks

**Single Object Tasks:** Single object tasks focus on analysing/or delineate individual objects within an image, with two main objectives:



**Fig.3.15: classification, classification+localization**

- i) **Classification:** This task involves determining the category or class to which a single object belongs, providing insights into its identity or nature. KNN(K-Nearest Neighbour)algorithm may be used for supervised classification while K-means clustering algorithm can be used for unsupervised classification.
- ii) **Classification + Localization:** In addition to classifying objects, this task also involves precisely localizing the object within the image by predicting bounding boxes that tightly enclose it.

**Multiple Object Tasks:** Multiple object tasks deal with scenarios where an image contains multiple instances of objects or different object classes. These tasks aim to identify and distinguish between various objects within the image, and they include:

- i) **Object Detection:** Object detection focuses on identifying and locating multiple objects of interest within the image. It involves analysing the entire image and drawing bounding boxes around detected objects, along with assigning class labels to these boxes. The main difference between classification and detection is that classification considers the image as a whole and determines its class whereas detection identifies the different objects in the image and classifies all of them.

In detection, bounding boxes are drawn around multiple objects and these are labelled according to their particular class. Object detection algorithms typically use extracted features and learning algorithms to recognize instances of an object category. Some of the algorithms used for object detection are: R-CNN (Region-Based Convolutional Neural Network), R-FCN (Region-based Fully Convolutional Network), YOLO (You Only Look Once) and SSD (Single Shot Detector).



**Fig.3.16- object detection**

- ii) **Image segmentation:** It creates a mask around similar characteristic pixels and identifies their class in the given input image. Image segmentation helps to gain a better understanding of the image at a granular level. Pixels are assigned a class and for each object, a pixel-wise mask is created in the image. This helps to easily identify each object separately from the other. Techniques like Edge detection which works by detecting discontinuities in brightness is used in Image segmentation. There are different types of Image Segmentation available.

Two of the popular segmentation are:

**a. Semantic Segmentation:** It classifies pixels belonging to a particular class. Objects belonging to the same class are not differentiated. In this image for example the pixels are identified under class animals but do not identify the type of animal.

**b. Instance Segmentation:** It classifies pixels belonging to a particular instance. All the objects in the image are differentiated even if they belong to the same class. In this image for example the pixels are separately masked even though they belong to the same class.

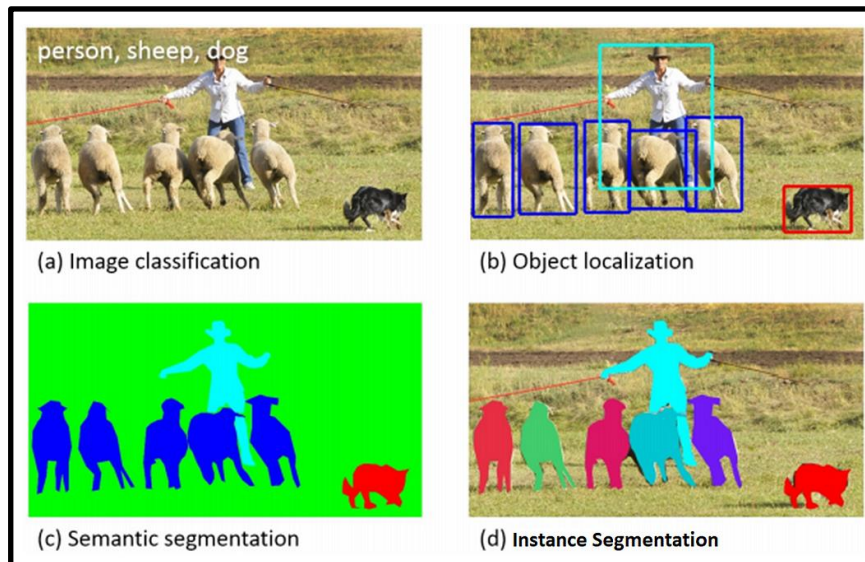


Fig.3.17

**3.3.5. High-Level Processing:** In the final stage of computer vision, high-level processing plays a crucial role in interpreting and extracting meaningful information from the detected objects or regions within digital images. This advanced processing enables computers to achieve a deeper understanding of visual content and make informed decisions based on the visual data. Tasks involved in high-level processing include recognizing objects, understanding scenes, and analysing the context of the visual content. Through sophisticated algorithms and machine learning techniques, computers can identify and categorize objects, infer relationships between elements in a scene, and derive insights from complex visual data. Ultimately, high-level processing empowers computer vision systems to extract valuable insights and drive intelligent decision-making in various applications, ranging from autonomous driving to medical diagnostics.

### 3.4. APPLICATIONS OF COMPUTER VISION

Computer vision is one of the areas in Machine Learning whose principle is already integrated into major products that we use every day. Some of the applications are listed below which you might have already learned in lower classes.

- **Facial recognition:** Popular social media platforms like Facebook uses facial recognition to detect and tag users.
- **Healthcare:** Helps in evaluating cancerous tumours, identifying diseases or abnormalities. Object detection & tracking in medical imaging.
- **Self-driving vehicles:** Makes sense of the surroundings by capturing video from different angles around the car. Detect other cars and objects, read traffic signals, pedestrian paths, etc.
- **Optical character recognition (OCR):** Extract printed or handwritten text from visual data such as images or documents like invoices, bills, articles, etc.
- **Machine inspection:** Detects a machine's defects, features, and functional flaws, determines inspection goals, chooses lighting and material-handling techniques, and other irregularities in manufactured products.
- **3D model building:** Constructing 3D computer models from existing objects which has a variety of applications in various places, such as Robotics, Autonomous driving, 3D tracking, 3D scene reconstruction, and AR/VR.
- **Surveillance:** Live footage from CCTV cameras in public places helps to identify suspicious behaviour, identify dangerous objects, and prevent crimes by maintaining law and order.
- **Fingerprint recognition and biometrics:** Detects fingerprints and biometrics to validate a user's identity.

### 3.5. CHALLENGES OF COMPUTER VISION

Computer vision, a vital part of artificial intelligence, faces several hurdles as it strives to make sense of the visual world around us. These challenges include:

1. **Reasoning and Analytical Issues:** Computer vision relies on more than just image identification; it requires accurate interpretation. Robust reasoning and analytical skills are essential for defining attributes within visual content. Without such capabilities, extracting meaningful insights from images becomes challenging, limiting the effectiveness of computer vision systems.
2. **Difficulty in Image Acquisition:** Image acquisition in computer vision is hindered by various factors like lighting variations, perspectives, and scales. Understanding complex scenes with multiple objects and handling occlusions adds to the complexity. Obtaining high-quality image data amidst these challenges is crucial for accurate analysis and interpretation.
3. **Privacy and Security Concerns:** Vision-powered surveillance systems raise serious privacy concerns, potentially infringing upon individuals' privacy rights. Technologies like facial recognition and detection prompt ethical dilemmas regarding privacy and security. Regulatory scrutiny and public debate surround the use of such technologies, necessitating careful consideration of privacy implications.
4. **Duplicate and False Content:** Computer vision introduces challenges related to the proliferation of duplicate and false content. Malicious actors can exploit vulnerabilities in image and video processing algorithms to create misleading or fraudulent content. Data breaches pose a significant threat, leading to the dissemination of duplicate images and videos, fostering misinformation and reputational damage.

### 3.6. THE FUTURE OF COMPUTER VISION

Over the years, computer vision has evolved from basic image processing tasks to complex systems capable of understanding and interpreting visual data with human-like precision. Breakthroughs in deep learning algorithms, coupled with the availability of vast amounts of labelled training data, have propelled the field forward, enabling machines to perceive and analyse images and videos in ways previously thought impossible.

As we look to the future, the possibilities by computer vision are awe-inspiring. From personalized healthcare diagnostics to immersive AR experiences, the impact of computer vision on society is set to be profound and far-reaching. By embracing innovation, fostering collaboration, and prioritizing ethical considerations, we can unlock the full potential of computer vision and harness its transformative power for the benefit of humanity.



18. Copy the URL and paste it into a new browser window to check the working of your model.

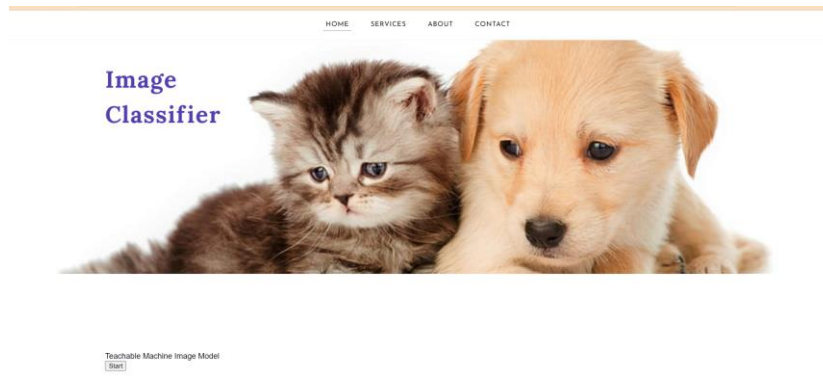


Fig.3.32

19. Click on start and you can show pictures of kitten and puppy to check the predictions of your model.

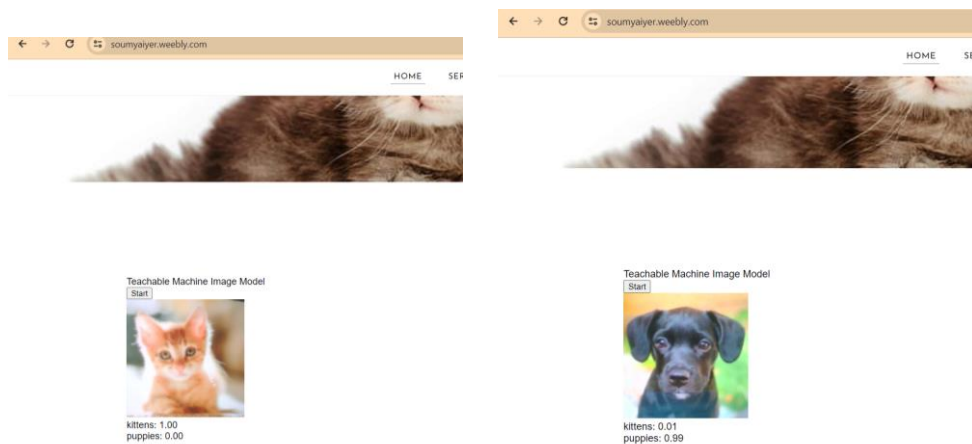


Fig.3.33

### 3.7. Working with OpenCV: (\*\*For Advanced Learners)

**3.7.1. Introduction to OpenCV:** OpenCV or Open-Source Computer Vision Library is a cross-platform library using which we can develop real-time computer vision applications. It mainly focuses on image processing, video capture, and analysis including features like face detection and object detection. It is also capable of identifying objects, faces, and even handwriting.

To use OpenCV in Python, you need to install the library. Use the following command in your terminal or command prompt:

**pip install opencv-python**

**3.7.2. Loading and Displaying an Image:** Let us understand the loading and displaying using a scenario followed by a question.

**Scenario-** You are working on a computer vision project where you need to load and display an image. You decide to use OpenCV for this purpose.

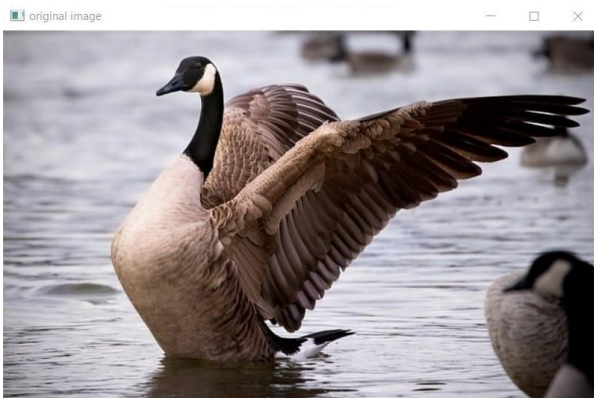
**Question:**

What are the necessary steps to load and display an image using OpenCV? Write a Python code snippet to demonstrate this.

**sol -**

Here's a simple Python script to load and display an image using OpenCV:

```
import cv2
image = cv2.imread('example.jpg') # Replace 'example.jpg' with the path to
your image
cv2.imshow('original image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



*Fig.3.34: loading and displaying image*

- `cv2.imread('example.jpg')` loads the image into a variable. Replace `example.jpg` with your image's file name or path.
- `cv2.imshow()` opens a new window to display the image.
- `cv2.waitKey(0)` waits indefinitely for a key press to proceed
- `cv2.destroyAllWindows()` closes any OpenCV windows.

**3.7.3. Resizing an image:**

**Scenario-** Imagine we are working on an application where we need to process images of varying sizes. To standardize the input, all images must be resized to 300x300 pixels before further processing. We are using OpenCV to achieve this. Additionally, we want to ensure the aspect ratio is maintained in some cases.

**Question:**

1. Write a Python script using OpenCV to resize an image named “example.jpg” to a fixed size of 300x300 pixels.

**Sol -**

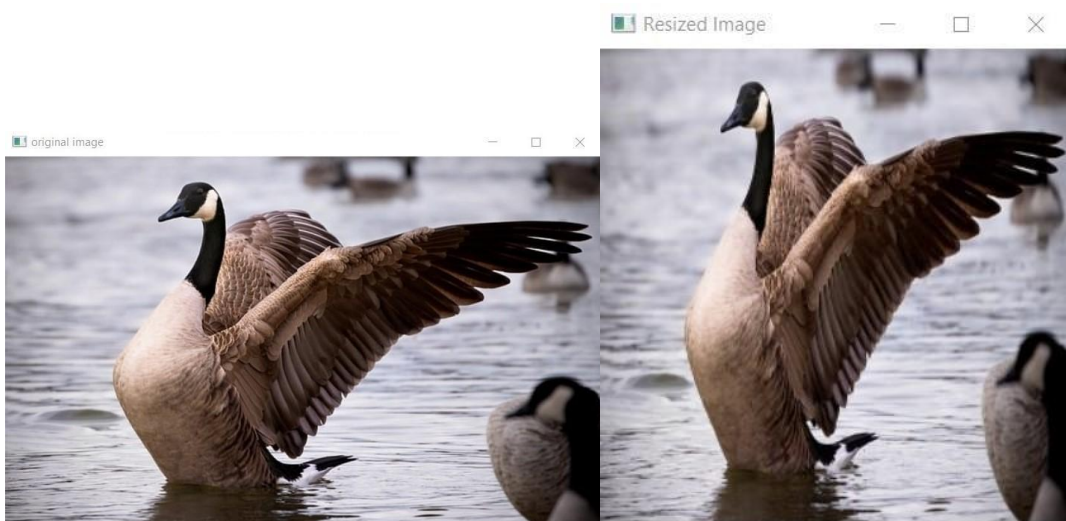
Here we will use the “`cv2.resize()`” function to resize the image. We can directly specify the width and height of the new image



```
new_width = 300
new_height = 300
# Resize the image to the new dimensions
resized_image = cv2.resize(image, (new_width, new_height))
```

So, the full code will look like this -

```
import cv2
image = cv2.imread('example.jpg') # Replace 'example.jpg' with the path
to your image
new_width = 300
new_height = 300
# Resize the image to the new dimensions
resized_image = cv2.resize(image, (new_width, new_height))
cv2.imshow('Resized Image', resized_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



*Fig.3.35: Resizing an Image*

### **3.7.4. Converting an Image to Grayscale**

#### **Scenario:**

Imagine we are working on an application where color images need to be converted to grayscale before further processing, as grayscale images reduce computational complexity. We are using OpenCV to achieve this.

**Question:** Write a Python script using OpenCV to convert an image named “example.jpg” into a grayscale.

#### **Solution:**

Here we will use the `cv2.cvtColor()` function to convert the image into grayscale. The function requires the source image and a color conversion code as inputs. The code for grayscale conversion is `cv2.COLOR_BGR2GRAY`.

```
grayscale_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

## 4.1. What is Data Mining?

Data mining is the process of discovering trends, useful information, and patterns from large datasets. It involves analyzing and interpreting data to extract meaningful insights that can aid in decision-making.

## 4.2. Introduction to Orange Data Mining Tool

In this unit, we embark on an exhilarating journey into the realms of Data Science, Computer Vision, and Natural Language Processing (NLP) using the capabilities of Orange data mining. Orange is a component-based visual programming software package designed for various tasks such as data visualization, machine learning, data mining, and data analysis. Within Orange, components are referred to as widgets, offering a diverse array of functionalities. These widgets range from basic data visualization, subset selection, and preprocessing to the empirical evaluation of learning algorithms and predictive modeling. The essence of visual programming is manifested through an intuitive interface, where workflows are constructed by interconnecting predefined or user-designed widgets. Throughout this unit, we will delve into the applications and challenges of AI techniques and explore how they can be applied to tackle real-world problems and extract valuable insights from data using Orange data mining.

## 4.3. Beneficiaries of Orange data mining

|                              |                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Analysts and Scientists | Orange provides a user-friendly interface for data analysis, making it accessible to professionals who may not have extensive programming skills.                         |
| Researchers                  | Orange provides tools for exploring and analyzing research data, facilitating hypothesis testing, and generating insights from experimental results                       |
| Educators and Students       | Educators can leverage Orange's intuitive interface and visual programming environment to introduce students to complex topics in a more approachable way.                |
| Business Professionals       | Orange's visualizations and predictive modeling capabilities can help businesses identify trends, predict customer behavior, optimize processes, and improve performance. |
| Open-Source Community        | Orange is an open-source software tool, meaning its source code is freely available to the public.                                                                        |

## 4.4. Getting started with Orange tool

To install the Orange tool on your system, follow these simple steps:

### 4.4.1. Visit the Orange Website:

Go to the official Orange website using the link: <https://orangedatamining.com/download/>  
Scan the QR code.



Fig.4.1: Scan the QR code

### 4.4.2. Choose the Correct Version:

- **For Windows Users:** Download the "Standalone installer" tailored for Windows users.
- **For Mac Users:** Select "Orange for Apple silicon" for Mac users.

### 4.4.3. Download the Installer:

Click on the respective download link to initiate the download process.

### 4.4.4. Install the Software:

Once the download is complete, locate the downloaded file and run it.

- For Windows: Double-click the installer file and follow the on-screen instructions to install Orange on your system.
- For Mac: Double-click the downloaded file to mount the disk image. Drag the Orange application to your Applications folder to install it.

### 4.4.5. Launch Orange:

After installation is complete, launch the Orange tool from your system's applications menu or by double-clicking its icon.

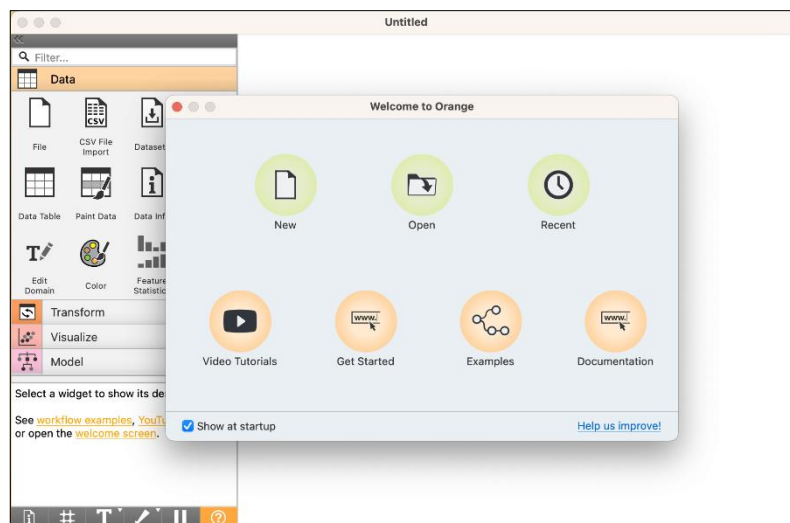


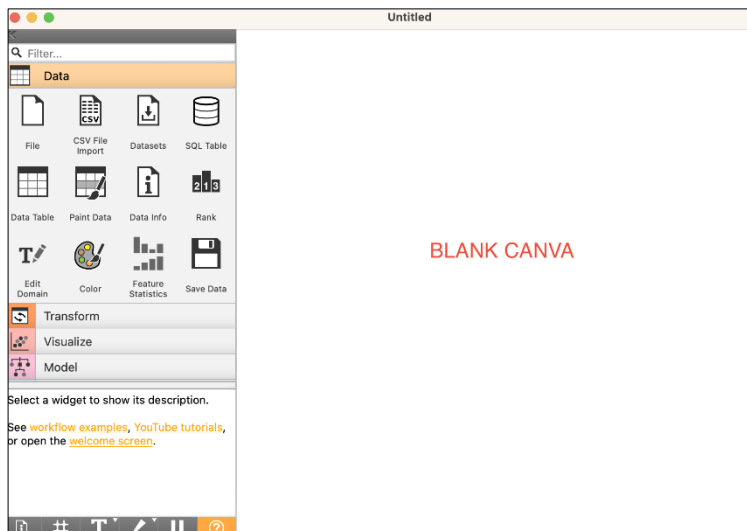
Fig.4.2: Orange Data Mining Tool

## 4.5. COMPONENTS OF ORANGE DATA MINING TOOL :-

The main components of Orange tools are-

1. Blank Canva
2. Widgets
3. Connectors

1. **Blank Canva:** The Blank Canva is where you build your analysis workflows by dragging and dropping widgets. It serves as the workspace where you connect widgets together to form a data analysis pipeline. You can add widgets to the canvas, rearrange them, and connect them to create a flow of data processing from input to output.

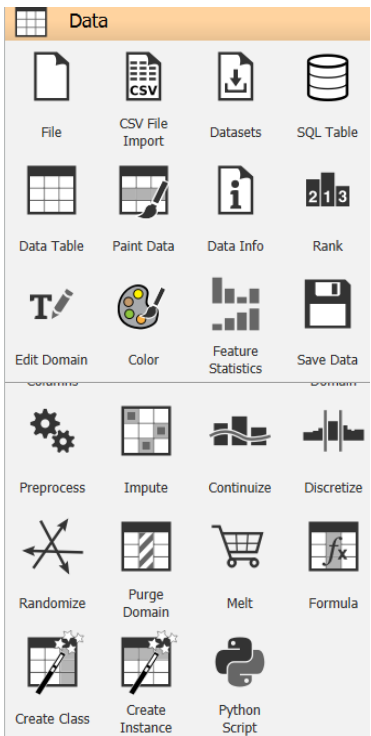
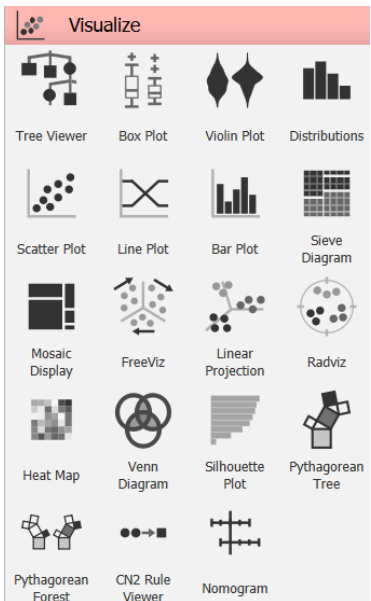


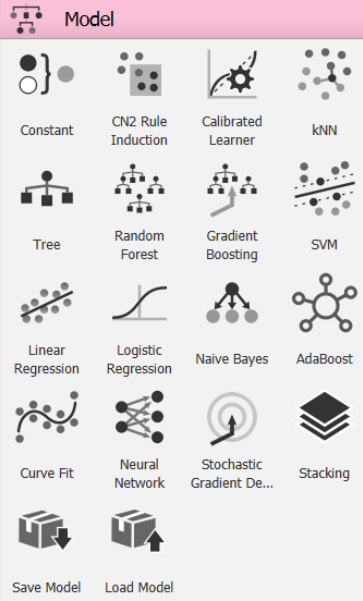
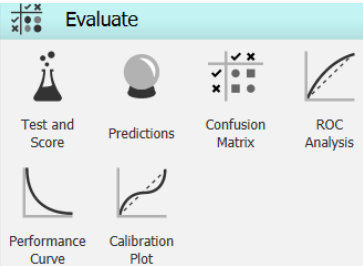
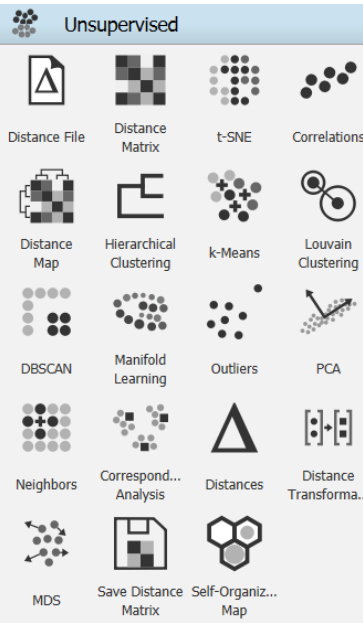
**Fig.4.3: Blank Canva**

3. **Connectors:-** Connectors are lines that link widgets together on the canvas. They represent the flow of data from one widget to another, indicating how the output of one widget is used as input for another.

## 4.6. DEFAULT WIDGET CATALOG

|                            |                       |                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Data Widgets</b></p> | <p><b>Fig.4.4</b></p> | <p>These widgets are used for data manipulation, i.e.,</p> <ul style="list-style-type: none"> <li>• File widget reads the input data file (data table with data instances) and sends the dataset to its output channel.</li> <li>• Data Table- Displays attribute-value data in a spreadsheet.</li> <li>• SQL Table - Reads data from an SQL database.</li> </ul> |
|----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                 |                                                                                                           |                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Transform Widgets</b></p> |  <p><b>Fig. 4.5</b></p>  | <p>The Transform Widget in Orange is a powerful tool used for data transformation tasks within your analysis workflow. It allows you to apply various transformations to your dataset</p> |
| <p><b>Visualize Widgets</b></p> |  <p><b>Fig. 4.6</b></p> | <p>Widgets for visualizing data, including scatter plots, bar charts, and heatmaps.</p>                                                                                                   |

|                                    |                                                                                                            |                                                                                                                                                                                                       |
|------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Model Widgets</b></p>        |  <p><b>Fig. 4.7</b></p>   | <p>Widgets that enable users to apply machine learning algorithms, including classification, regression, clustering, and anomaly detection, to build predictive models and analyze data patterns.</p> |
| <p><b>Evaluate Widgets</b></p>     |  <p><b>Fig. 4.8</b></p>  | <p>Widgets for evaluating model performance through techniques like cross-validation, confusion matrices etc.</p>                                                                                     |
| <p><b>Unsupervised Widgets</b></p> |  <p><b>Fig. 4.9</b></p> | <p>These widgets facilitate exploratory data analysis and pattern recognition without labeled data, enabling tasks such as clustering, dimensionality reduction, and association rule mining.</p>     |

## 4.7. KEY DOMAINS OF AI WITH ORANGE DATA MINING TOOL

1. Data science with Orange Data Mining
2. Computer Vision with Orange Data Mining
3. Natural Language Process with Orange Data Mining

### 4.7.1. Data Science with Orange

Let us begin our journey by diving into the vast ocean of data, where we will learn the fundamental principles of Data Science. Through hands-on exercises with Orange data mining, we will uncover the secrets hidden within datasets, data visualization and how data is classified using classification models.

You want to add a special touch to someone's day with a bouquet, but wouldn't it be even more delightful if you could surprise with their favorite kind of flower? While you can easily visit a flower shop and pick out some iris flowers, there is a catch - iris flowers come in various shapes and sizes. Specifically, when it comes to the violet-colored iris, there are three main types: *Iris Setosa*, *Iris Versicolor*, and *Iris Virginica*.

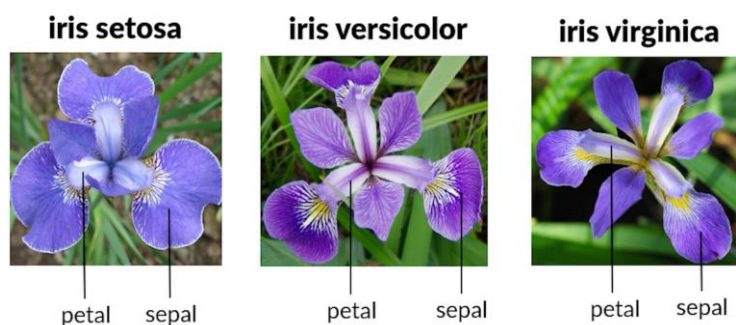


Fig. 4.10: Iris Flower Types

At first glance, distinguishing between these varieties isn't straightforward. However, the key differences lie in the dimensions of the petal and sepal of each flower. To make an informed choice and ensure the surprise is just right, we need data - specifically, measurements of the length and width of the sepal and petal for each iris variety. This data will help us differentiate between the different types of iris flowers accurately. To gather this data efficiently and visualize it for better understanding, we'll turn to Orange data mining tool. With its data visualization capabilities, we can explore the dimensions of iris flowers. Let's dive in and use Orange to enhance our flower surprise with a personal touch!



## Data Visualization: - Exploring Iris Flower Dimensions with Orange Data Mining

### Step 1- Launch Orange Data Mining Software

Once the software is launched, you will be greeted with a blank canvas where you can construct your analysis workflow.

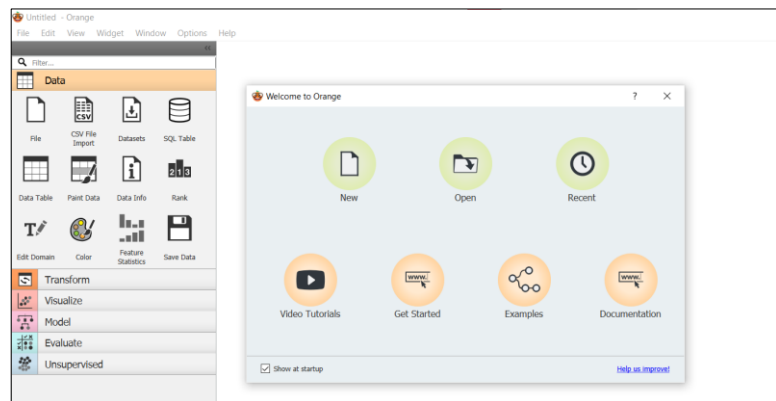


Fig. 4.11: Launch orange data mining

### Step 2 – Select the Data Widget

Load the Iris Dataset, Drag the "File" widget from the widget panel onto the blank canvas

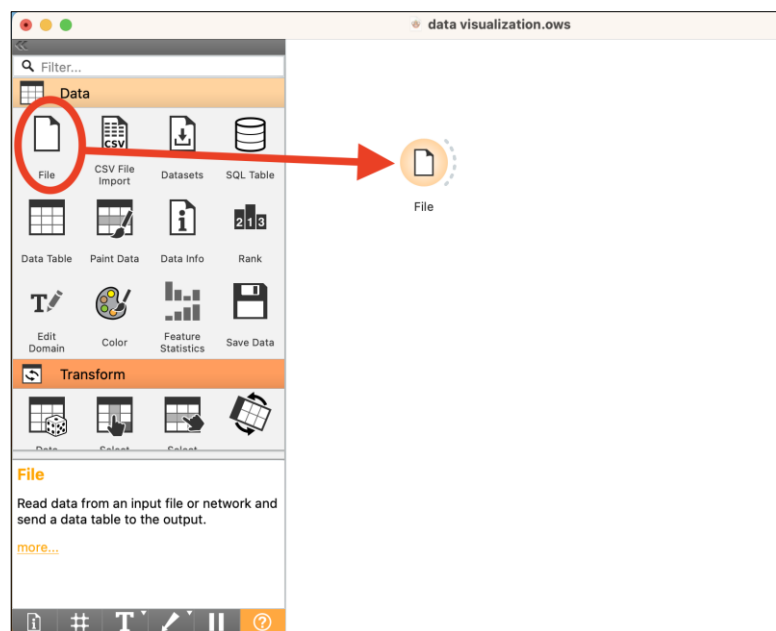


Fig. 4.12: select the data widget

### Step 3 - Load the iris Dataset

Double Click on the "File" widget and select the iris data set.

Change the Role of the column "iris" which has the values of flowers as "target"

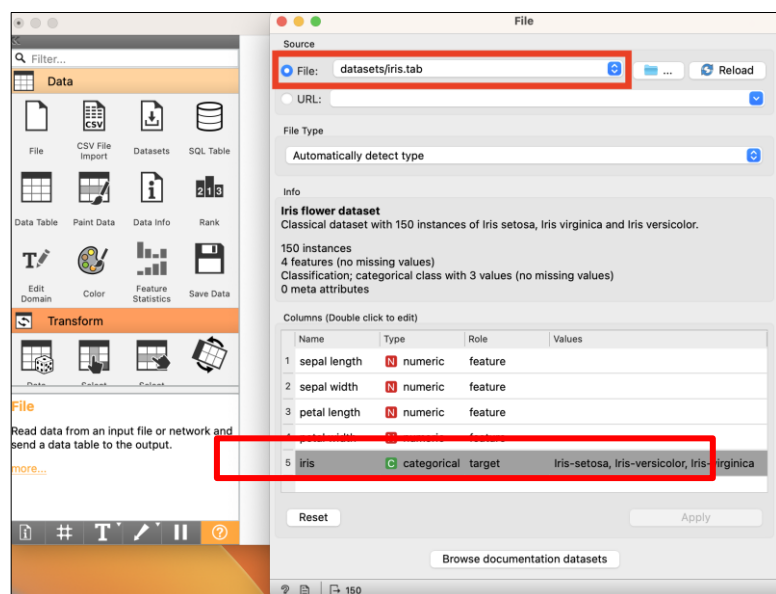


Fig. 4.13: Load Iris dataset

#### Step 4 - Display the Dataset in a Data Table

Drag and drop the "Data Table" widget onto the canvas. Connect the output of the "File" widget to the input of the "Data Table" widget by dragging the connector from one widget to the other.

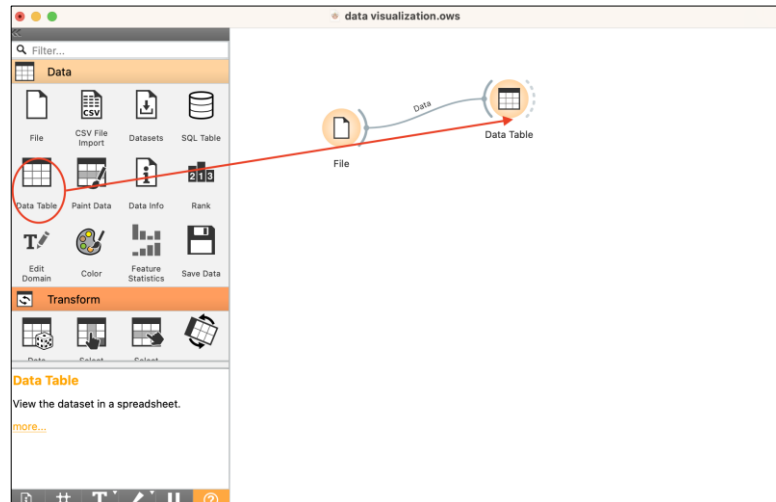


Fig. 4.14: Display the dataset

#### Step 5 – Explore the dataset

Double-click on the "Data Table" widget to open it. You will now see a tabular view of the iris dataset, displaying 150 samples of iris flower dimensions, including the length and width of the sepal and petal.

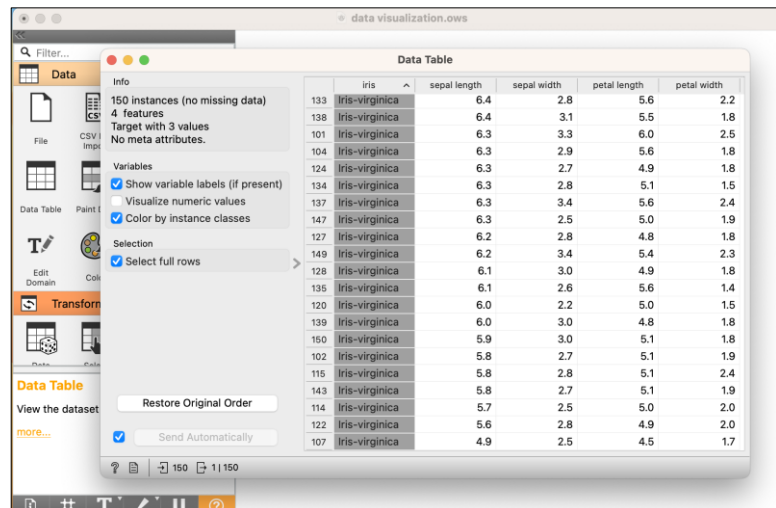


Fig. 4.15: Explore the dataset

#### Step 6- Visualize the Data with a Scatter Plot

Drag and drop the "Scatter Plot" widget onto the canvas. Connect the output of the "File" widget to the input of the "Scatter Plot" widget. By doing so, you are instructing Orange to use the iris dataset as input for generating a scatter plot.

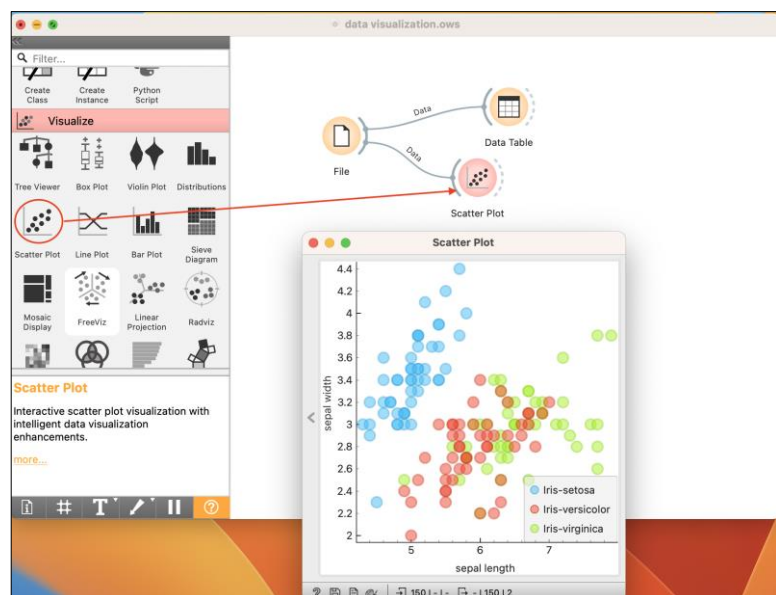


Fig. 4.16: Visualize the data

#### Step 7- Interpret the Scatter Plot

Once the scatter plot is generated, you'll see a graphical representation of the iris Dataset. The scatter plot will display the relationship between two selected variables, such as sepal length

and sepal width, with each point representing an individual iris sample. You can further customize the scatter plot and explore other visualizations using additional widgets available in Orange.

You can experiment with other widgets available in Orange for data visualization, such as histograms, box plots, or parallel coordinate plots. Each visualization offers unique insights into the iris dataset, helping you understand the distribution and patterns of flower dimensions.

### 4.7.1.1. CLASSIFICATION: -

Imagine you have just returned from the flower shop with a bunch of irises, and you have meticulously measured the dimensions of their petals and sepals. Now, armed with these data features, you are ready to determine the type of iris flower - whether it is iris setosa, iris versicolor, or iris virginica. We will collect this testing data and feed it into a spreadsheet for further analysis.

### Step 1: Prepare Testing Data in Spreadsheet

| A            | B           | C            | D           |
|--------------|-------------|--------------|-------------|
| sepal length | Sepal width | petal length | petal width |
| 5.8          | 4           | 1.2          | 0.2         |
| 5.5          | 2.4         | 3.7          | 1           |
| 6            | 3           | 4.8          | 1.8         |
| 5.9          | 3           | 5.1          | 1.8         |

Fig. 4.17: creating Testing dataset in excel

- Create columns in a new spreadsheet for sepal length, sepal width, petal length, and petal width, matching the field names in the training data (e.g., "sepal length" in lowercase).
- Then, enter the measured dimensions of the petals and sepals for each iris sample, ensuring consistency with the units used in the training data (e.g., centimeters).

### Step 2: Classification with Orange Data Mining

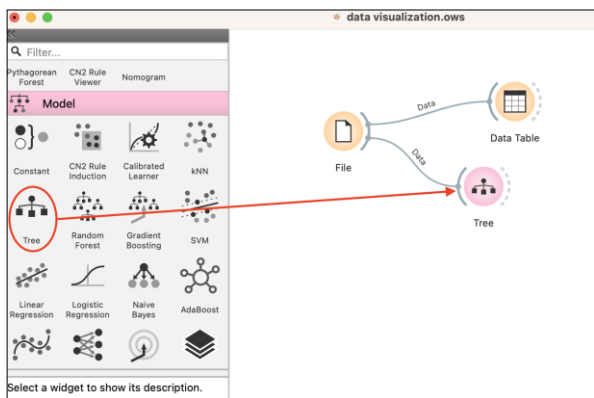


Fig. 4.18: Display the dataset

Note:- “Data table” widget just shows the content of the data in tabular form, it is optional to connect or not.

### Step 3: Perform Classification

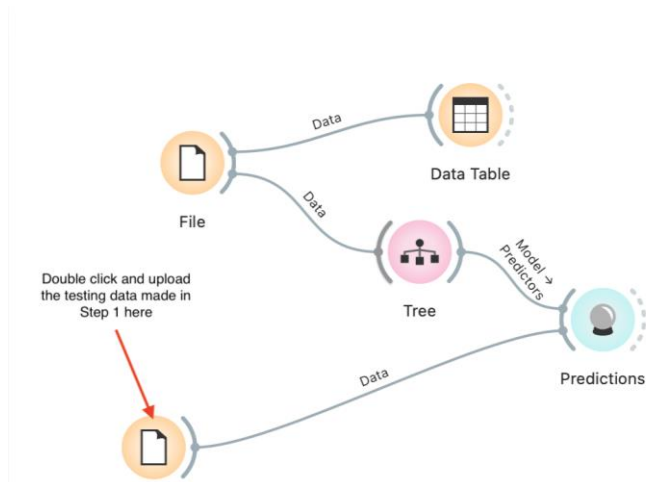


Fig. 4.19

- Drag and drop the "Predictions" widget onto the canvas. Connect the output of the "File" widget (containing the training data) to the input of the "Predictions" widget.
- Drag another "File" widget onto the canvas and upload the iris testing dataset created in the spreadsheet.
- Connect the output of the second "File" widget (containing the testing data) to the input of the "Predictions" widget. This instructs Orange to use the trained model to classify the samples in testing dataset.

### Step 4: Interpret Results

|   | Tree            | sepal length | Sepal width | petal length | petal width |
|---|-----------------|--------------|-------------|--------------|-------------|
| 1 | Iris-setosa     | 5.8          | 4.0         | 1.2          | 0.2         |
| 2 | Iris-versicolor | 5.5          | 2.4         | 3.7          | 1.0         |
| 3 | Iris-virginica  | 6.0          | 3.0         | 4.8          | 1.8         |
| 4 | Iris-virginica  | 5.9          | 3.0         | 5.1          | 1.8         |

Fig. 4.20: Interpret Results

- Click on the prediction widget
- The "Predictions" widget displays the predicted class labels for each iris sample in the testing dataset, allowing you to assess the accuracy of the classification model.

By following the above steps, you have successfully employed Orange data mining software to classify iris flower types using testing data. This demonstrates the practical application of machine learning techniques in real-world scenarios, offering insights into the classification process and its accuracy.

#### 4.7.1.2. Evaluating the Classification Model with Orange

Now that we have classified iris flower types using our model, it is crucial to assess its performance. Evaluation metrics such as accuracy, precision, recall, and F1 score provide insights into the effectiveness of the classification model. Additionally, we will utilize the confusion matrix with cross-validation to gain a deeper understanding of the model's performance. **Cross-validation**, sometimes called **rotation estimation**, includes resampling and sample splitting methods that use different portions of the data to test and train a model on different iterations.

### Step 1: Perform Evaluation: -

- Begin by adding the "File" widget to the canvas and connect it to the output of the classification tree model. This will provide the dataset used in the earlier training phase.
- Connect the same "File" widget to the input of the "Test and Score" widget. This widget evaluates the model's performance and calculates metrics such as accuracy, precision, recall, and F1 score.
- Double-click on the "Test and Score" widget to view the evaluation metrics. Orange utilizes cross-validation, splitting the data into 10 subsets for training and testing. This process is repeated multiple times to ensure robust evaluation.

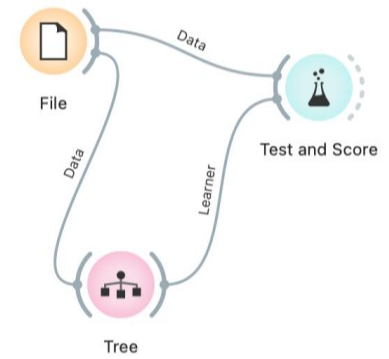


Fig. 4.21

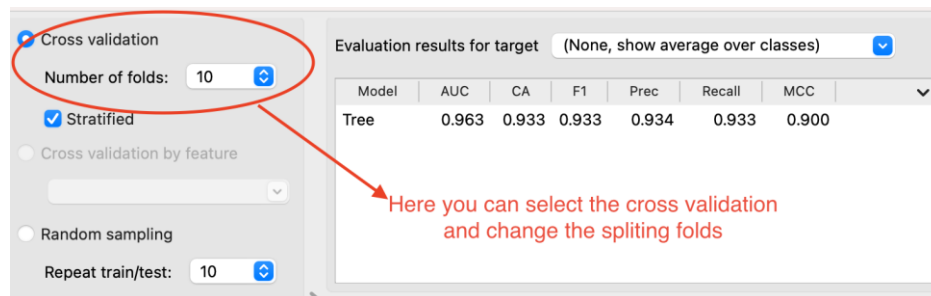


Fig. 4.22

### Step 2: Interpret Evaluation Metrics

- Analyze the evaluation metrics displayed by the "Test and Score" widget. The accuracy metric indicates the percentage of correctly classified instances, which in our case is 93%.
- However, to gain deeper insights, we will examine precision, recall, and F1 score. Precision measures the proportion of true positives among all instances classified as positive, while recall measures the proportion of true positives correctly identified. The F1 score is the harmonic mean of precision and recall.
- These metrics collectively provide a comprehensive understanding of the model's performance across different classes.

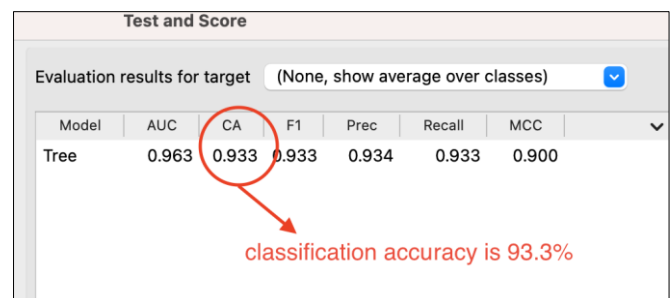
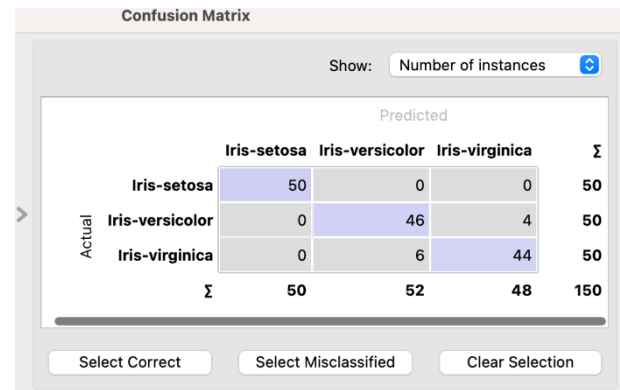
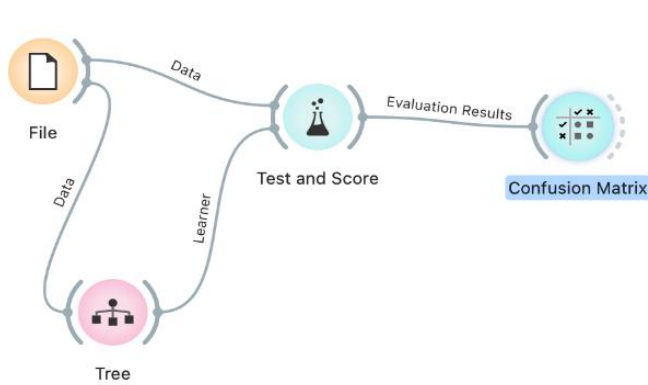


Fig. 4.23: Evaluation Metric

### Step 3: Utilize Confusion Matrix

- Add the "Confusion Matrix" widget to the canvas and connect it to the output of the "Test and Score" widget.
- Double-click on the "Confusion Matrix" widget to view the matrix. This matrix provides a detailed breakdown of the model's classifications, showing the number of true positives, true negatives, false positives, and false negatives for each class.



**Fig. 4.24: Confusion Matrix**

- Interpret the confusion matrix to identify any patterns or areas where the model may be struggling. For instance, while the model accurately identifies iris setosa, it may struggle with distinguishing between versicolor and virginica.

By evaluating our classification model using Orange, we've gained valuable insights into its performance. Through metrics such as accuracy, precision, recall, and F1 score, as well as the confusion matrix with cross-validation, we can assess the model's strengths and weaknesses. This evaluation process empowers us to make informed decisions and further refine our model for improved performance.

### Practical Activity

- Your task is to conduct a Data Science project aiming to differentiate between fruits and vegetables based on their nutritional characteristics.
- Begin by gathering a dataset containing nutritional information for a variety of fruits and vegetables, including features such as energy (kcal/kJ), water content (g), protein (g), total fat (g), carbohydrates (g), fiber (g), sugars (g), calcium (mg), iron (mg), magnesium (mg), phosphorus (mg), potassium (mg), and sodium (g).
- You can explore reputable sources like Kaggle for this data.
- Next, split the dataset into training and testing subsets. Utilize classification algorithms available in Orange to train models on the training data, then evaluate their performance using the testing data.
- Utilize evaluation metrics such as accuracy, precision, recall, and F1-score to assess the models' effectiveness and compare their performance to identify the most effective one.

## 4.7.2. Computer Vision with Orange

In the vast landscape of data, images represent a rich and diverse source of information. With advancements in computer vision and machine learning, we can now extract valuable insights from images using tools like Orange. Let us now explore how Orange transforms images into numerical representations and enables machine learning on image data.

### Step 1: Install Image Analytics Add-On

- To get started, we need to install the Image Analytics add-on in Orange. Go to the "Options" menu, click on "Add-ons," and install Image Analytics. Restart Orange to enable the add-on to appear in the widget panel.

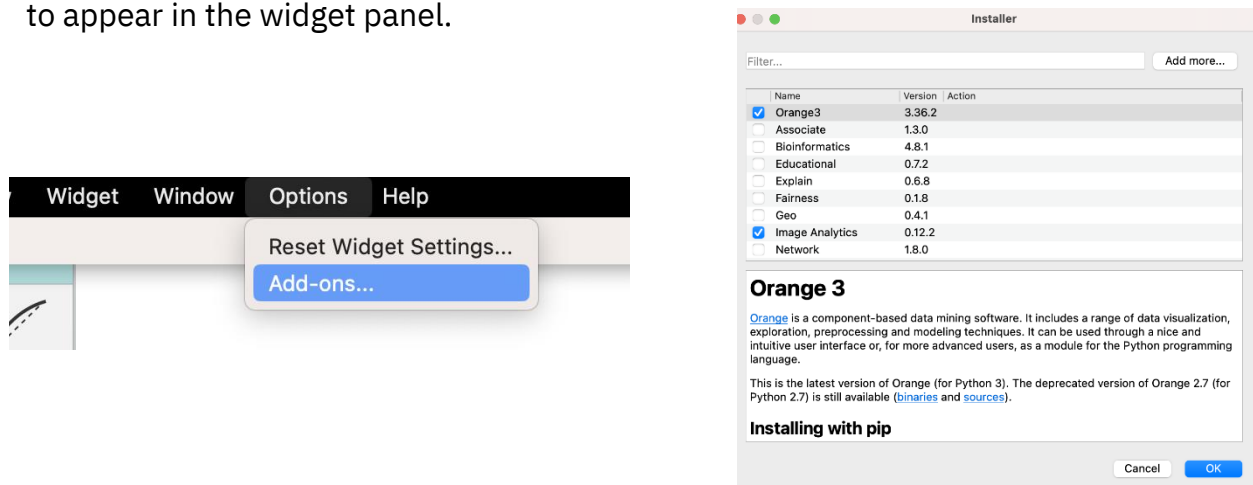


Fig. 4.25: Install Image Analytics Add on

### Step 2: Example Scenario

- Let us dive into an example scenario where we want to build a model to cluster unlabeled images of dogs and cats. We will start by obtaining a dataset containing images of dogs and cats and saving it on our computer. Click on the link below to obtain the dataset:

[https://drive.google.com/drive/folders/1tLbVXWkKzcp6O\\_-FAvn9usEWuof3jTp3?usp=sharing](https://drive.google.com/drive/folders/1tLbVXWkKzcp6O_-FAvn9usEWuof3jTp3?usp=sharing)

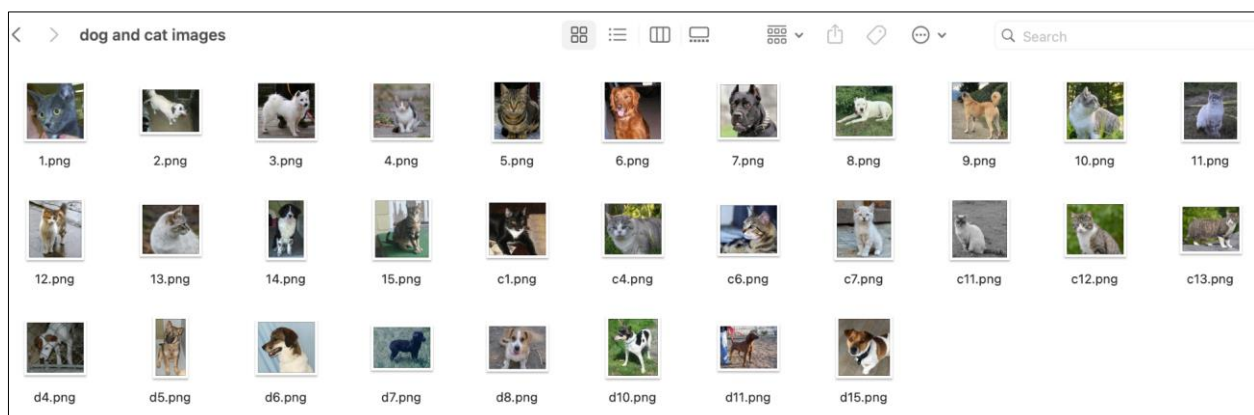


Fig. 4.26: Dog and Cat images

### Step 3: Import Images

- Drag and drop the "Import Images" widget onto the canvas and upload the dataset folder containing images of dogs and cats.



## Step 4: Image Visualization

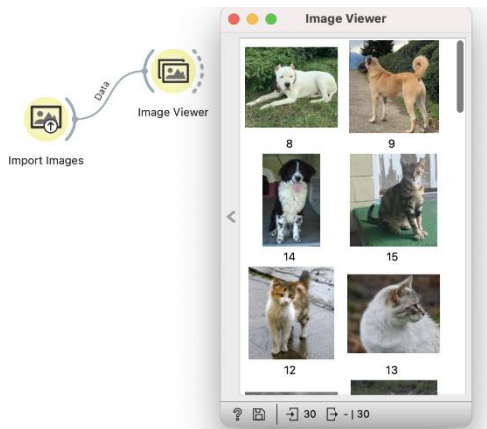


Fig. 4.27: Image Visualization

- Add the "Image Viewer" widget to the canvas and connect it to the output of the "Import Images" widget. Double-click on the "Image Viewer" to visualize the images in the dataset.

## Step 5: Image Embedding

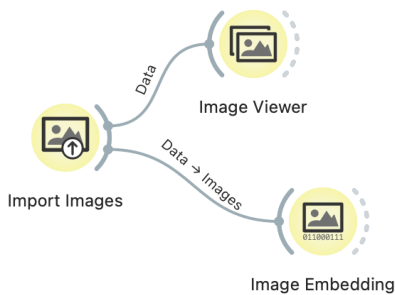


Fig. 4.28: Image Embedding

Now, we need to transform the raw images into numerical representations using image embeddings. Connect the "Image Embedding" widget to the "Import Images" widget. This widget sends information to the server and computes embeddings remotely using a deep neural network trained on millions of real-life images.

## Step 6: Computing Similarities

- We will compare the embeddings of images and compute their similarities. Connect the "Distance" widget to the "Image Embedding" output. Double-click on the "Distance" widget and select the cosine option, which typically works best with images.

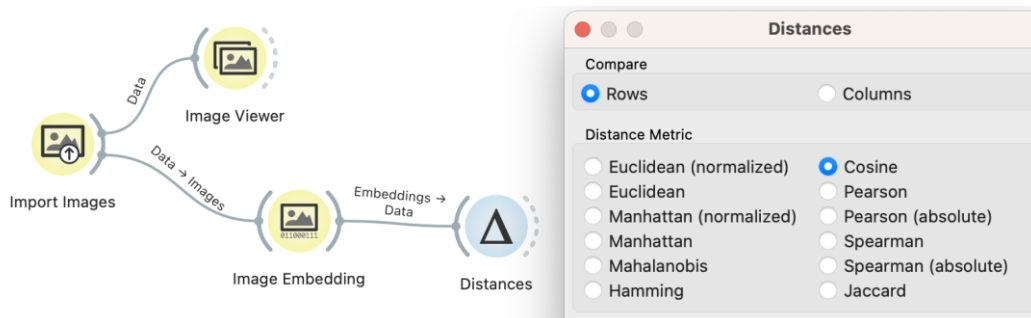
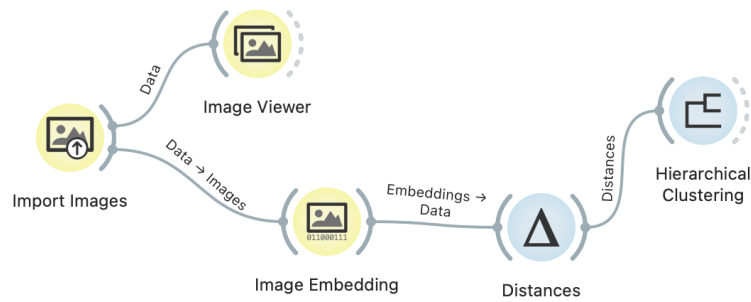


Fig. 4.29: Computing Similarities

## Step 7: Hierarchical Clustering

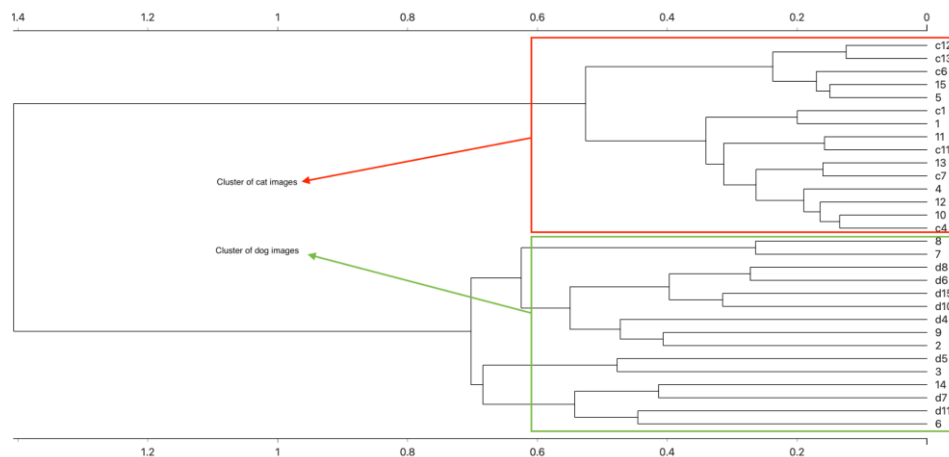
- Pass the distance matrix to the "Hierarchical Clustering" algorithm by dragging and dropping the widget onto the canvas.





**Fig. 4.30: Hierarchical Clustering**

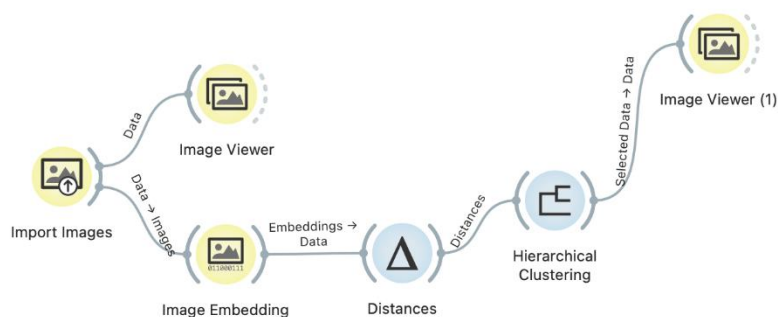
- Double-click on the "Hierarchical Clustering" widget to observe the dendrogram, where images are grouped based on their similarities.



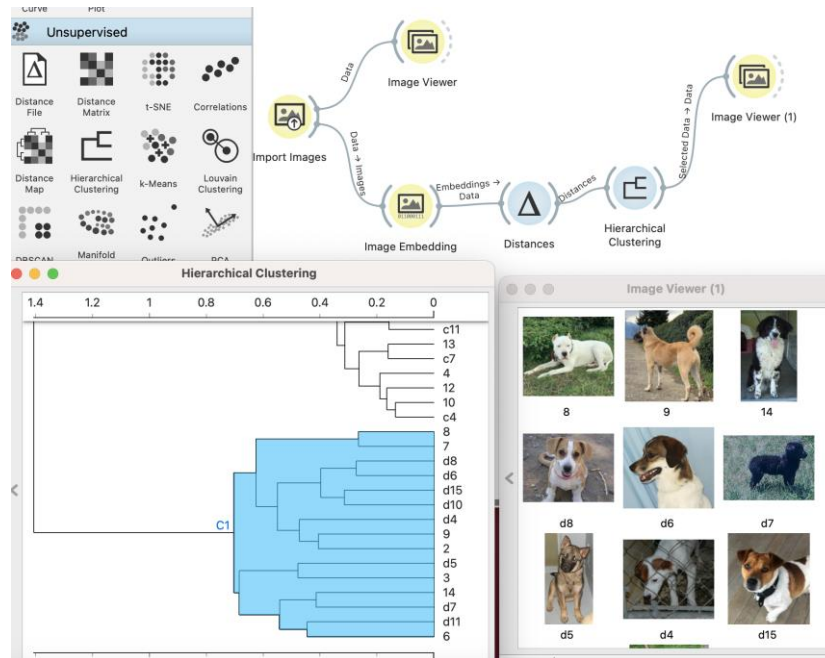
**Fig. 4.31: Dendrogram**

## Step 8: Visualization and Interpretation

- Use the "Image Viewer" widget to visualize each cluster by selecting the corresponding area in the dendrogram.



You will notice that images of dogs and cats are grouped together, demonstrating the effectiveness of image analytics using Orange data mining.



**Fig. 4.32: Visualization and Interpretation**

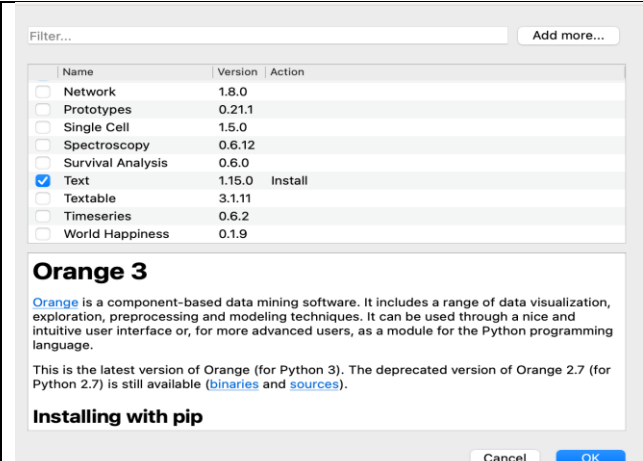
With Orange's Image Analytics capabilities, we can unlock the potential of image data and perform sophisticated analysis and machine learning tasks. By leveraging image embeddings and clustering algorithms, we can gain insights and make informed decisions from image datasets, paving the way for innovative applications in various domains.

### Practical Activity:

- Cluster images of birds and animals into distinct groups based on their visual characteristics.
- Collect datasets of images containing various species of birds and animals.
- Ensure that each dataset contains a sufficient number of images representing different species within the respective categories.
- Import the collected image datasets into Orange Data Mining.
- Apply clustering algorithms to group similar images together based on their numerical representations.
- Analyze the clustering results and interpret the grouping of images into different clusters.
- Identify any patterns or similarities observed within each cluster and between clusters.

### 4.7.3. Natural Language Processing with Orange

Now that we have learned about analyzing data in spreadsheets and images, let us move on to working with text using Orange Data Mining. Natural Language Processing (NLP) helps us understand and learn from written words, like finding patterns in documents. With Orange, we will do different tasks in NLP, which can help us in lots of ways, like analyzing and understanding text better.



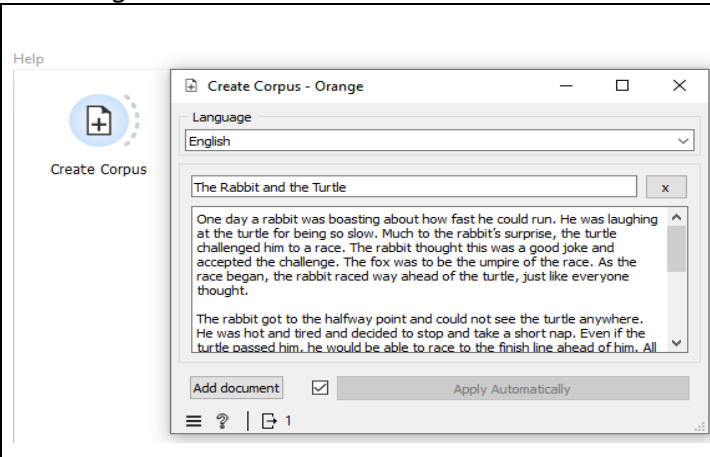
The screenshot shows the 'Add more...' dialog in Orange 3. It features a table with columns 'Name', 'Version', and 'Action'. The 'Text' widget is selected with a blue checkmark and has an 'Install' button next to it. Below the table, there is a section titled 'Orange 3' with a description of the software and a link to 'Installing with pip'. At the bottom, there are 'Cancel' and 'OK' buttons.

| Name                                       | Version | Action  |
|--------------------------------------------|---------|---------|
| <input type="checkbox"/> Network           | 1.8.0   |         |
| <input type="checkbox"/> Prototypes        | 0.21.1  |         |
| <input type="checkbox"/> Single Cell       | 1.5.0   |         |
| <input type="checkbox"/> Spectroscopy      | 0.6.12  |         |
| <input type="checkbox"/> Survival Analysis | 0.6.0   |         |
| <input checked="" type="checkbox"/> Text   | 1.15.0  | Install |
| <input type="checkbox"/> Textable          | 3.1.11  |         |
| <input type="checkbox"/> Timeseries        | 0.6.2   |         |
| <input type="checkbox"/> World Happiness   | 0.1.9   |         |

**Step 1: Install Text Add-On**

First, we need to install the Text add-on. Navigate to the "Options" menu, select "Add-ons," and choose "Text." Restart Orange Data Mining to activate the add-on.

Fig. 4.33: Install Text Add-on



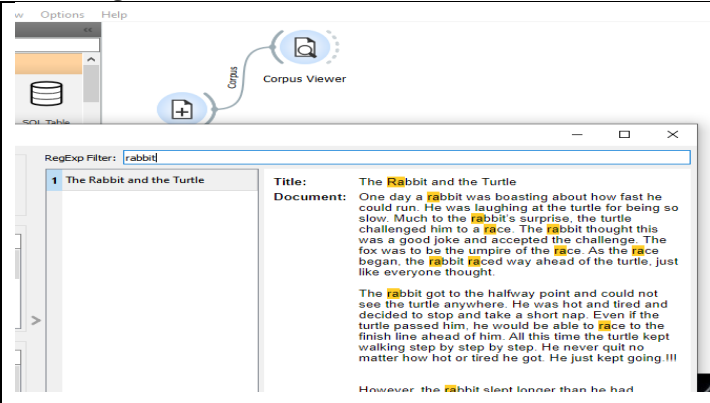
The screenshot shows the 'Create Corpus - Orange' dialog box. It has a 'Language' dropdown set to 'English'. Below it, there is a text area containing the story 'The Rabbit and the Turtle'. At the bottom, there are 'Add document' and 'Apply Automatically' buttons, along with a checkbox for 'Add document'.

**Step 2: Load or Create Textual Data**

Whether we have existing textual data or want to create your own corpus, Orange provides the tools to handle both scenarios. Drag and drop the "Corpus" widget to load data (or) use the "Create Corpus" widget to input your own text.

Double-click on the "Create Corpus" widget to add textual data. You can input any text you wish to analyze, such as articles, reviews, or documents.

Fig. 4.34: Load or create Textual Data



The screenshot shows the Orange 3 interface with the 'Corpus Viewer' widget. The widget displays the text from the 'Create Corpus' dialog, with the word 'rabbit' highlighted in yellow. The interface includes a 'RegEx Filter' bar at the top and a list of documents on the left.

**Step 3: Visualize Text with Corpus Viewer**

Add the "Corpus Viewer" widget to the canvas and connect it with the output of the "Create Corpus" widget. The Corpus Viewer allows us to browse through the text and search for specific words, which it highlights within the corpus.

Fig. 4.35: Visualize text with corpus viewer

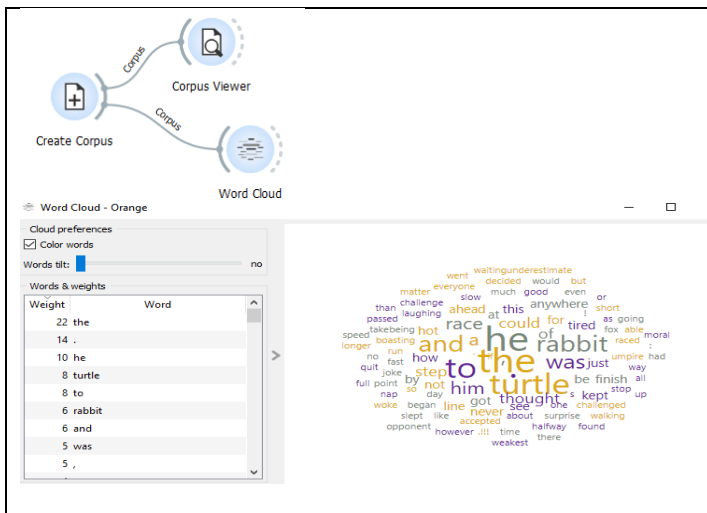


Fig. 4.36: Visualize word frequencies with cloud word

#### Step 4: Visualize Word Frequencies with Word Cloud

Connect the output of the "Corpus" widget to the "Word Cloud" widget. The Word Cloud visually represents word frequencies in a cloud format, with more frequent words appearing larger. This visualization provides an initial glimpse into the prominent themes or topics within the text.

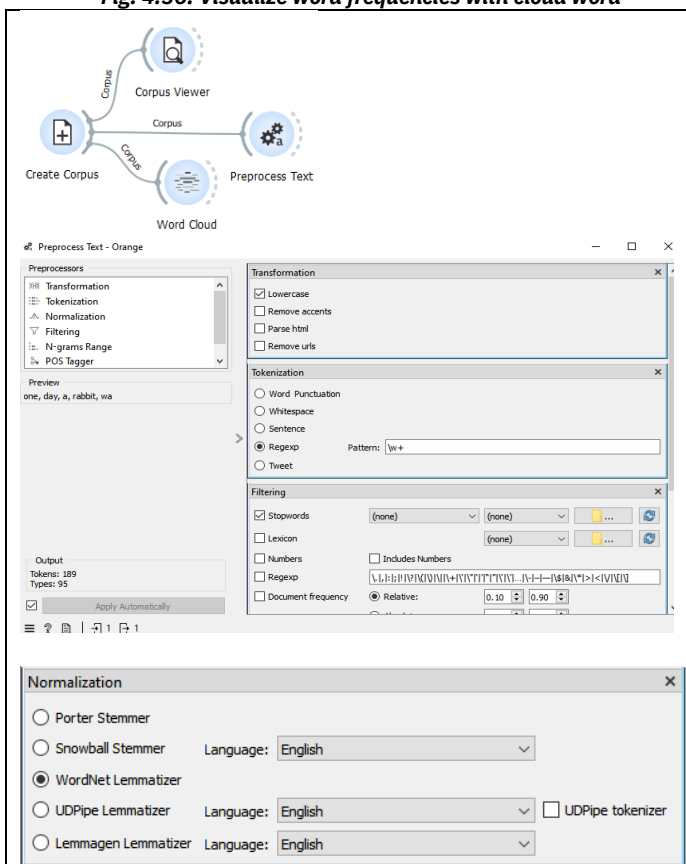


Fig. 4.37: Preprocess Text

#### Step 5: Preprocess Text

It is essential to preprocess the text to remove noise and irrelevant information (.,!@#\$). Use the "Preprocess Text" widget for this task. Connect this widget to the output of the "Corpus" widget.

"Preprocess text" widget performs text normalization by converting text to lowercase, tokenizing it into individual words, removing punctuation, and filtering out stop words.

Optionally, you can choose to perform stemming or lemmatization to further refine the text.

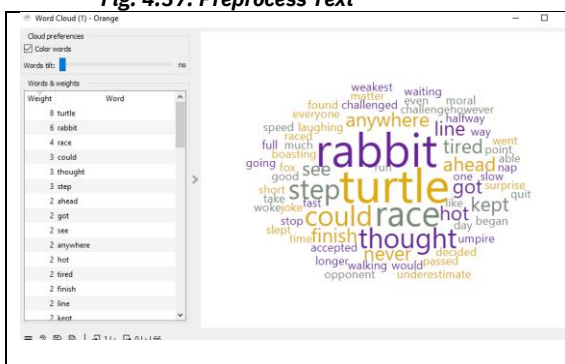


Fig. 4.38: Visualize cleaned text with word cloud

#### Step 6: Visualize Cleaned Text with Word Cloud

Connect the output of the "Preprocess Text" widget to the "Word Cloud" widget to visualize the cleaned text data.

- Now, the Word Cloud displays only meaningful words, allowing us to better understand the main themes or topics within the corpus. Here you can see Turtle and rabbit are larger in size because these words are more frequent in the corpus.

With Orange Data Mining, we can leverage the power of NLP to analyze textual data effectively. Through the steps outlined above, we have seen how to load, visualize, preprocess, and analyze textual data effectively. From exploring word frequencies with the Word Cloud to cleaning and refining text through preprocessing, Orange empowers users to derive meaningful insights from text effortlessly.

### Practical Activity

Your task is to create your own corpus by selecting a story or article of your choice. Ensure the text is sufficiently long to provide meaningful insights. Once you have your corpus, use Orange Data Mining to perform: -

1. Text normalization techniques like Converting all text to lowercase to ensure consistency, tokenization to split the text into individual words or tokens, removal of stop words to eliminate common words that do not contribute to the meaning of the text
2. Analyze the word cloud to identify which words appear most frequently and gain insights into the main themes or topics of the text.
3. Compare the results of the word cloud analysis before and after applying lemmatization and stemming to observe any differences in the most commonly used words.

## EXERCISES

### A. Multiple Choice Questions

1. Which widget helps in knowing the Accuracy, Precision, recall and F1 score values?
  - a. Confusion matrix
  - b. Test and score
  - c. Word cloud
  - d. Corpus
2. Which widget provides the detailed breakdown of the model's classifications, showing the number of true positives, true negatives, false positives, and false negatives for each class?
  - a. Test and score
  - b. Preprocess text
  - c. Confusion Matrix
  - d. Corpus
3. \_\_\_\_\_ widget performs text normalization by converting text to lowercase, tokenizing it into individual words, removing punctuation, and filtering out stop words.
4. Cross validation can also be called as \_\_\_\_\_
  - a. Rotation estimation
  - b. Cross hybrid
  - c. Revolution Estimation
  - d. Estimation
5. In word cloud, more the frequency of word in corpus larger the size of word would look.
  - a. True
  - b. False
6. Which widget is used to transform the raw images into numerical representations?
  - a. Image viewer
  - b. Image analytics
  - c. Image Embedding
  - d. Distance
7. Which widget is used to compare the embeddings of images and compute their similarities?
  - a. Distance
  - b. Image Embedding
  - c. Preprocess Text
  - d. Image viewer
8. \_\_\_\_\_ are lines that link widgets together on the canvas. They represent the flow of data from one widget to another, indicating how the output of one widget is used as input for another.





| Aspect        | Structured Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Semi-Structured Data                    | Unstructured Data                                 |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------|-----|--------|---|------|----|-------|---|-------|----|-------|---|--------|----|-------|---|------|----|-------|---|---------|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organization  | Organized in clearly defined columns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Less organized than structured data     | No organization exhibits variability over time    |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Accessibility | Easily accessible and searchable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Accessible but may be harder to analyze | Accessibility depends on the specific data format |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Examples      | <p>Customer information, transaction records, product directories</p> <p>Structured data</p> <table border="1"> <thead> <tr> <th>ID</th><th>Name</th><th>Age</th><th>Degree</th></tr> </thead> <tbody> <tr> <td>1</td><td>John</td><td>18</td><td>B.Sc.</td></tr> <tr> <td>2</td><td>David</td><td>31</td><td>Ph.D.</td></tr> <tr> <td>3</td><td>Robert</td><td>51</td><td>Ph.D.</td></tr> <tr> <td>4</td><td>Rick</td><td>26</td><td>M.Sc.</td></tr> <tr> <td>5</td><td>Michael</td><td>19</td><td>B.Sc.</td></tr> </tbody> </table> | ID                                      | Name                                              | Age | Degree | 1 | John | 18 | B.Sc. | 2 | David | 31 | Ph.D. | 3 | Robert | 51 | Ph.D. | 4 | Rick | 26 | M.Sc. | 5 | Michael | 19 | B.Sc. | <p>XML files, CSV files, JSON files, HTML files, semi-structured documents</p> <p>Semi-structured data</p> <pre> &lt;University&gt;   &lt;Student ID="1"&gt;     &lt;Name&gt;John&lt;/Name&gt;     &lt;Age&gt;18&lt;/Age&gt;     &lt;Degree&gt;B.Sc.&lt;/Degree&gt;   &lt;/Student&gt;   &lt;Student ID="2"&gt;     &lt;Name&gt;David&lt;/Name&gt;     &lt;Age&gt;31&lt;/Age&gt;     &lt;Degree&gt;Ph.D. &lt;/Degree&gt;   &lt;/Student&gt;   .... &lt;/University&gt; </pre> | <p>Audio files, images, video files, emails, PDFs, social media posts</p> <p>Unstructured data</p> <div style="border: 1px solid black; padding: 5px;"> <p>The university has 5600 students. John's ID is number 1, he is 18 years old and already holds a B.Sc. degree. David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.</p> </div> |
| ID            | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Age                                     | Degree                                            |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1             | John                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 18                                      | B.Sc.                                             |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2             | David                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 31                                      | Ph.D.                                             |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3             | Robert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 51                                      | Ph.D.                                             |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4             | Rick                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 26                                      | M.Sc.                                             |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5             | Michael                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 19                                      | B.Sc.                                             |     |        |   |      |    |       |   |       |    |       |   |        |    |       |   |      |    |       |   |         |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 5.3. Advantages and Disadvantages of Big Data:

Big Data is a key to modern innovation. It has changed how organizations analyze and use information. While it offers great benefits, it also comes with challenges that affect its use in different industries. In this section, we will be discussing a few pros and cons of big data.

#### Advantages:

- **Enhanced Decision Making:** Big Data analytics empowers organizations to make data-driven decisions based on insights derived from large and diverse datasets.
- **Improved Efficiency and Productivity:** By analyzing vast amounts of data, businesses can identify inefficiencies, streamline processes, and optimize resource allocation, leading to increased efficiency and productivity.
- **Better Customer Insights:** Big Data enables organizations to gain a deeper understanding of customer behavior, preferences, and needs, allowing for personalized marketing strategies and improved customer experiences.
- **Competitive Advantage:** Leveraging Big Data analytics provides organizations with a competitive edge by enabling them to uncover market trends, identify opportunities, and stay ahead of competitors.
- **Innovation and Growth:** Big Data fosters innovation by facilitating the development of new products, services, and business models based on insights derived from data analysis, driving business growth and expansion.



### Disadvantages:

- **Privacy and Security Concerns:** The collection, storage, and analysis of large volumes of data raise significant privacy and security risks, including unauthorized access, data breaches, and misuse of personal information.
- **Data Quality Issues:** Ensuring the accuracy, reliability, and completeness of data can be challenging, as Big Data often consists of unstructured and heterogeneous data sources, leading to potential errors and biases in analysis.
- **Technical Complexity:** Implementing and managing Big Data infrastructure and analytics tools require specialized skills and expertise, leading to technical challenges and resource constraints for organizations.
- **Regulatory Compliance:** Organizations face challenges in meeting data protection laws like GDPR (General Data Protection Regulation) and The Digital Personal Data Protection Act, 2023. These laws require strict handling of personal data, making compliance essential to avoid legal risks and penalties.
- **Cost and Resource Intensiveness:** The cost of acquiring, storing, processing, and analyzing Big Data, along with hiring skilled staff, can be high. This is especially challenging for smaller organizations with limited budgets and resources.

**Activity:** Find the sources of big data using the link [UNSTATS](#)



## 5.4. Characteristics of Big Data

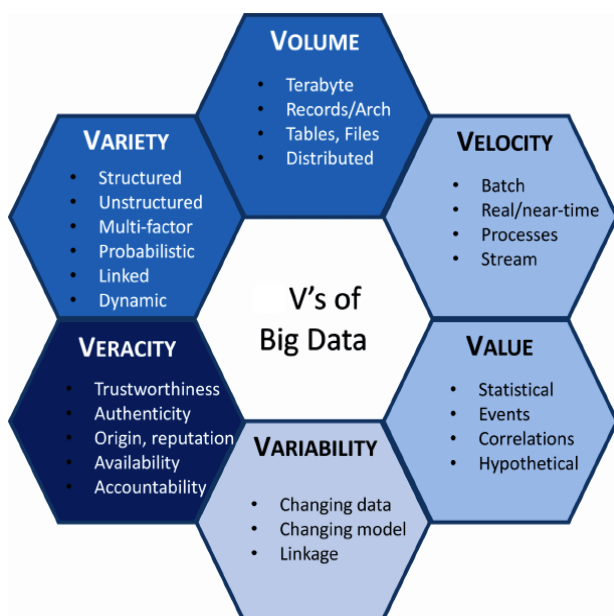


Fig. 5.3 Characteristics of Big Data

The “**characteristics of Big Data**” refer to the defining attributes that distinguish large and complex datasets from traditional data sources. These characteristics are commonly described using the “**3Vs**” framework: **Volume, Velocity, and Variety**. The **6Vs framework** provides a holistic view of Big Data, emphasizing not only its volume, velocity, and variety but also its veracity, variability, and value. Understanding and addressing these six dimensions are essential for effectively managing, analyzing, and deriving value from Big Data in various domains.

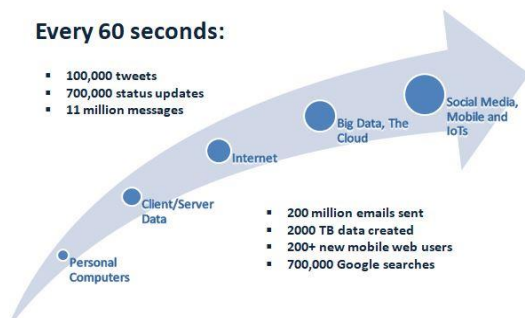


Fig. 5.4 Speed of data generation from various sources

**5.4.1. Velocity:** Velocity refers to the speed at which data is generated, delivered, and analyzed. In the present world, where millions of people are accessing and storing information online, the speed at which the data gets stored or generated is huge. For example: Google alone generates more than 40,000 search queries per second. See the statistics in the picture provided. Isn't it huge!

**5.4.2. Volume:** Every day a huge volume of data is generated as the number of people using online platforms has increased exponentially. Such a huge volume of data is considered Big Data. Typically, if the data volume exceeds gigabytes, it falls into the realm of big data. This volume can range from petabytes to terabytes or even exabytes, based on surveys conducted by various organizations. According to the latest estimates, 328.77 million terabytes of data are created each day.

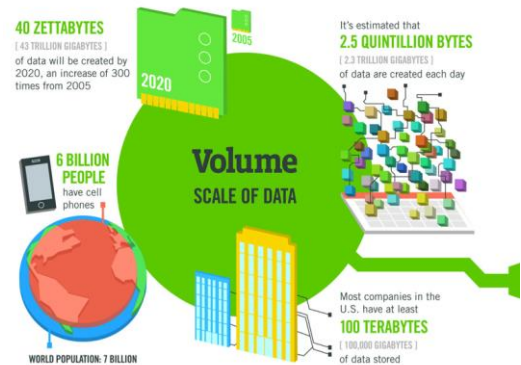


Fig.5.5 Volume of data

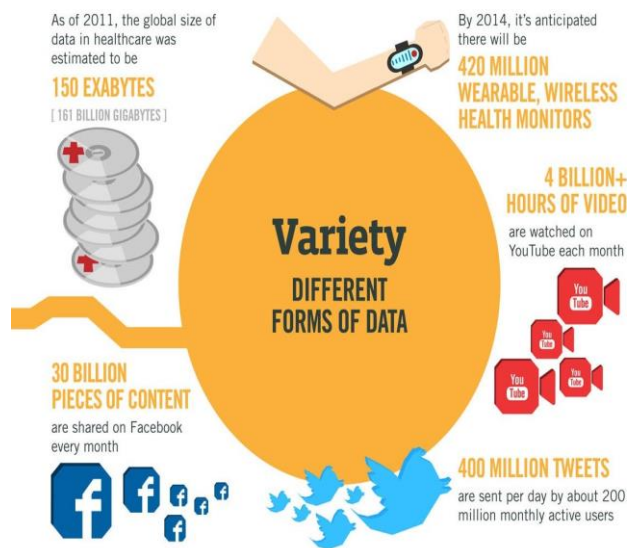


Fig.5.6 Varieties in Big data

**5.4.3. Variety:** Big data encompasses data in various formats, including structured, unstructured, semi-structured, or highly complex structured data. These can range from simple numerical data to complex and diverse forms such as text, images, audio, videos, and so on. Storing and processing unstructured data through RDBMS is challenging. However, unstructured data often provides valuable insights that structured data cannot offer. Additionally, the variety of data sources within big data provides information on the diversity of data.

**5.4.4. Veracity:** Veracity is a characteristic in Big Data related to consistency, accuracy, quality, and trustworthiness. Not all data that undergoes processing holds value. Therefore, it is essential to clean data effectively before storing or processing it, especially when dealing with massive volumes. Veracity addresses this aspect of big data, focusing on the accuracy and reliability of the data source and its suitability for analytical models.

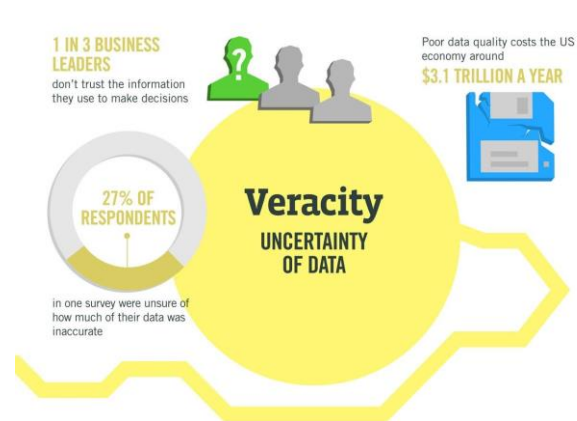


Fig. 5.7



Fig. 5.8 The value of Big Data

**5.4.5. Value:** The goal of big data analysis lies in extracting business value from the data. Hence, the business value derived from big data is perhaps its most critical characteristic. Without obtaining valuable insights, the other characteristics of big data hold little significance. So, in simple terms Value of Big Data refers to the benefits the big data can provide.

**5.4.6. Variability:** This refers to establishing if the contextualizing structure of the data stream is regular and dependable even in conditions of extreme unpredictability. It defines the need to get meaningful data considering all possible circumstances.



Fig. 5.9

#### Case Study: How a Company Uses 3V and 6V Frameworks for Big Data

Company: An OTT Platform 'OnDemandDrama'

##### 3V Framework:

**Volume:** OnDemandDrama processes huge amounts of data from millions of users, including watch history, ratings, searches, and preferences to offer personalized content recommendations.

**Velocity:** Data is processed in real-time, allowing OnDemandDrama to immediately adjust recommendations, track the patterns of the users, and offer trending content based on their activity.

**Variety:** The platform handles diverse data such as user profiles, watch lists, video content, and user reviews which are categorized as structured, semi-structured, and unstructured data.

##### 6V Framework:

Along with the above 3 V of big data, the 6V Framework involves 3 more features of big data named Veracity, Value, and Variability.

**Veracity:** OnDemandDrama filters out irrelevant or low-quality data (such as incomplete profiles) to ensure accurate content recommendations.

**Value:** OnDemandDrama uses the data to personalize user experiences, driving engagement and retention by recommending shows and movies that match individual tastes.

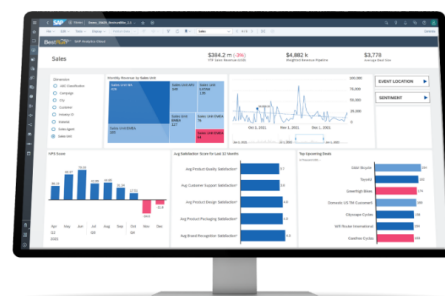
**Variability:** OnDemandDrama handles changes or inconsistencies in data streams caused by factors like user behavior, trends, or any other external events. For example, user preferences can vary based on region, time, or trends.

By using the 3V and 6V frameworks, OnDemandDrama can manage, process, and derive valuable insights from its Big Data, which enhances customer satisfaction and drives business decisions.

## 5.5. Big Data Analytics

### Data Analytics

Data analytics involves analyzing datasets to uncover insights, trends, and patterns. It can be applied to datasets of any size, from small to moderate volumes. Technologies commonly used in data analytics include statistical analysis software, data visualization tools, and relational database management systems (RDBMS).



Big data analytics uses advanced analytic techniques against huge, diverse datasets that include structured, semi-structured, and unstructured data, from different sources, and in various sizes from terabytes to zettabytes.

Big-Data Analytics encompasses the methodologies, tools, and practices involved in analyzing and managing data, covering tasks such as data collection, organization, and storage. The primary objective of data analytics is to utilize statistical analysis and technological methods to uncover patterns and address challenges. In the business realm, big data analytics has gained significance as a means to assess and refine business processes, as well as enhance decision-making and overall business performance. It provides valuable insights and forecasts that help businesses make informed decisions to improve their operations and outcomes. Different types of Big Data Analytics can help businesses and organizations find insights from large and complex datasets. Some of the common types are: Descriptive analytics, Diagnostic analytics, Predictive analytics, and Prescriptive analytics, which we have discussed in Unit 2 of Data Science Methodology.



Big Data Analytics emerges as a consequence of four significant global trends:

1. **Moore's Law:** The exponential growth of computing power as per Moore's Law has enabled the handling and analysis of massive datasets, driving the evolution of Big Data Analytics.
2. **Mobile Computing:** With the widespread adoption of smartphones and mobile devices, access to vast amounts of data is now at our fingertips, enabling real-time connectivity and data collection from anywhere.
3. **Social Networking:** Platforms such as Facebook, Foursquare, and Pinterest facilitate extensive networks of user-generated content, interactions, and data sharing, leading to the generation of massive datasets ripe for analysis.
4. **Cloud Computing:** This paradigm shift in technology infrastructure allows organizations to access hardware and software resources remotely via the Internet on a pay-as-you-go basis, eliminating the need for extensive on-premises hardware and software investments.

## 5.6. Working on Big Data Analytics

Big data analytics involves collecting, processing, cleaning, and analyzing enormous datasets to improve organizational operations. The working process of big data analytics includes the following steps –

### Step 1. Gather data

Each company has a unique approach to data collection. Organizations can now collect structured and unstructured data from various sources, including cloud storage, mobile apps, and IoT sensors.

### Step 2. Process Data

Once data is collected and stored, it must be processed properly to get accurate results on analytical queries, especially when it's large and unstructured. Various processing options are available:

- **Batch processing** which looks at large data blocks over time.
- **Stream processing** looks at small batches of data at once, shortening the delay time between collection and analysis for quicker decision-making.

### Step 3. Clean Data

Scrubbing all data, regardless of size, improves quality and yields better results. Correct formatting and elimination of duplicate or irrelevant data are essential. Erroneous and missing data can lead to inaccurate insights.

### Step 4. Analyze Data

Getting big data into a usable state takes time. Once it's ready, advanced analytics processes can turn big data into big insights.

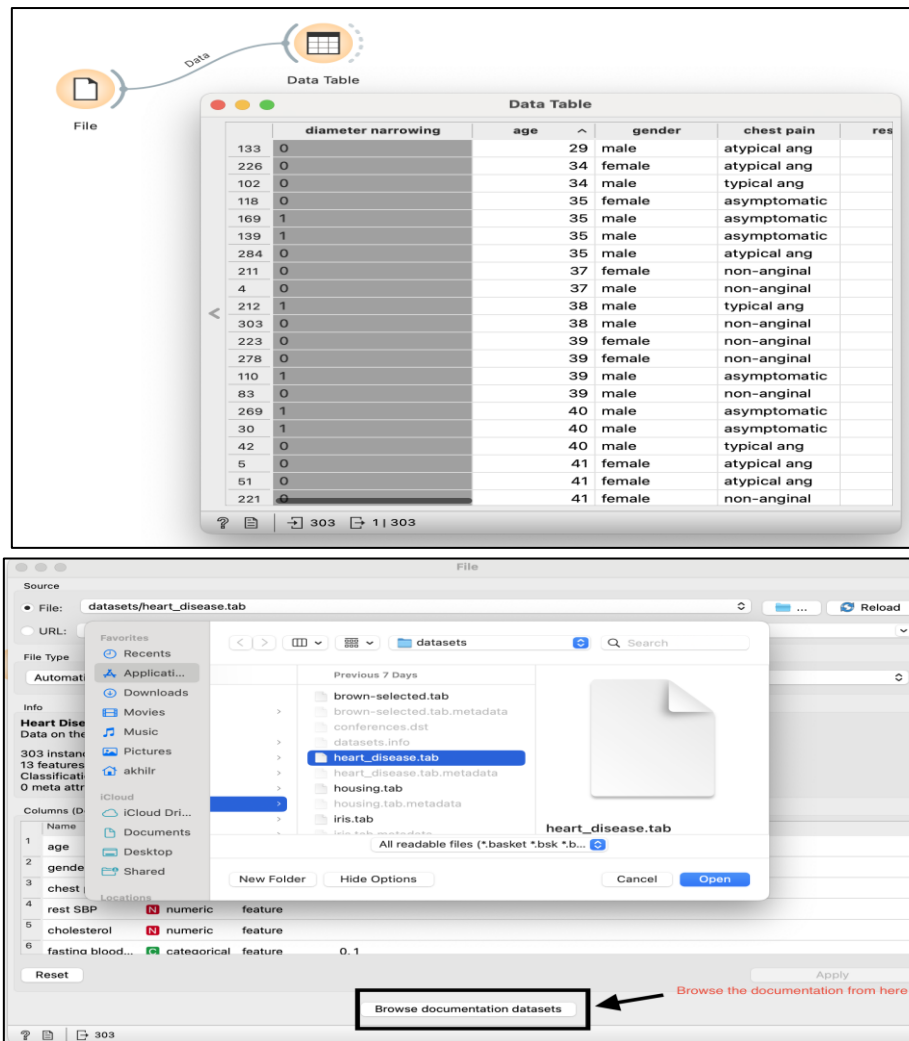
Example: **Data Analytics Tools – Tableau, APACHE Hadoop, Cassandra, MongoDB, SaS**

## Using Orange Data Mining for Big Data Analytics

We will explore how big data analysis can be performed using Orange Data Mining.

### Step 1: Gather Data

1. Use the **File** widget to load data into Orange.
2. Load the desired dataset. For demonstration, we will use the built-in **Heart Disease** dataset.



It is important to carefully study the dataset and understand the **features** and **target** variable.

- **Features:** age, gender, chest pain, resting blood pressure (rest\_sbp), cholesterol, resting ECG (rest\_ecg), maximum heart rate (max\_hr), etc.
- **Target:** diameter narrowing.

If the value for **diameter narrowing** is **1**, it signifies significant narrowing of the arteries, which is a risk factor for heart disease. If the value is **0**, it indicates healthier arteries with minimal or no narrowing.



## Step 2: Process Data

Data processing involves preparing the data for accurate analysis. There are two methods:

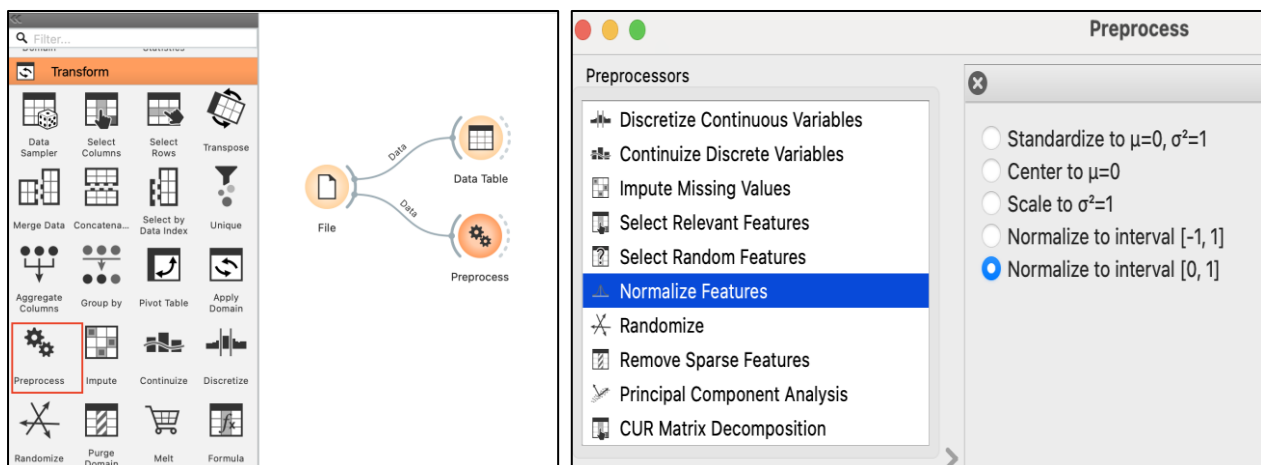
1. **Batch Processing:** Use the **Preprocess** widget to normalize large chunks of structured data at once.
2. **Stream Processing (near-real-time):** While Orange does not natively support live stream data, you can incrementally process smaller subsets of the data in parallel workflows.

Here, we will focus on the **Normalization** technique.

Normalization in data preprocessing refers to scaling numerical values to a specific range (e.g., 0–1 or -1–1), making them comparable and improving the performance of machine learning algorithms.

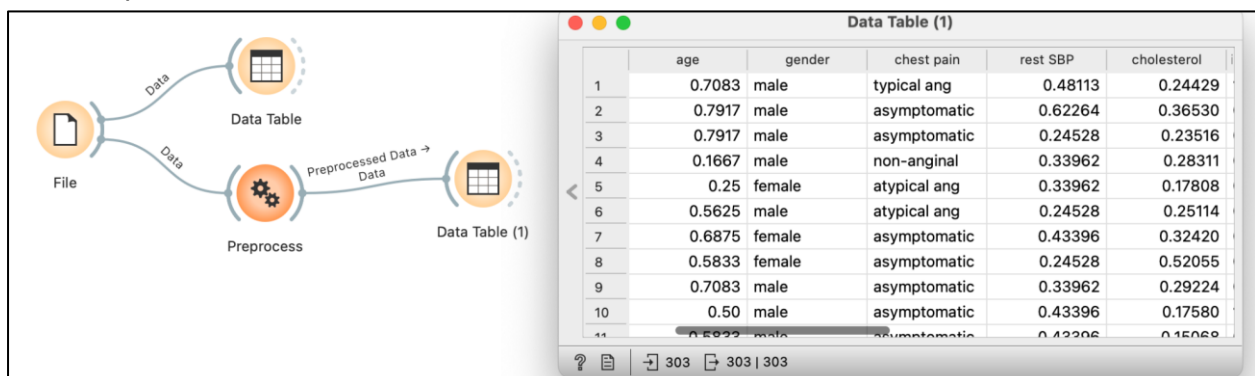
### Step 2.1: Normalize Data

1. Connect the **Preprocess** widget to the **File** or **Data Table** widget.
2. Double-click on the **Preprocess** widget and select "**Normalize Features**".
3. Choose an interval, such as **0–1** or **-1–1**.



### Step 2.2: Verify Normalized Data

1. Connect the **Data Table** widget to the **Preprocess** widget.
2. Open the Data Table to observe the differences in values.



You will see that all numerical values are now scaled between **0 and 1**.



### Step 3: Clean Data

Data cleaning is essential to ensure quality results. We will use the **Impute** widget to handle missing values by replacing them with the **mean, median, mode**, or a custom value. In this data we all can see that some values are missing in the figure below. This missing value data set is being saved as heart data.xlsx in the computer folder.

|    | A           | B   | C      | D            | E        | F           | G             | H             | I      | J          | K               | L              |
|----|-------------|-----|--------|--------------|----------|-------------|---------------|---------------|--------|------------|-----------------|----------------|
|    | diameter na | age | gender | chest pain   | rest SBP | cholesterol | fasting blood | rest ECG      | max HR | exerc incd | ar ST by exerci | slope peak e m |
| 1  | 0           | 29  | male   | atypical ang | 130      | 204         | 0             | left vent hyp | 202    | 0          | 0               | upsloping      |
| 2  | 0           | 34  | female | atypical ang | 118      |             | 0             | normal        | 192    | 0          | 0.7             | upsloping      |
| 3  | 0           | 34  | male   | typical ang  |          | 182         | 0             | left vent hyp | 174    | 0          | 0               | upsloping      |
| 4  | 0           | 35  | female | asymptomat   | 138      | 183         | 0             | normal        | 182    | 0          | 1.4             | upsloping      |
| 5  | 1           | 35  | male   | asymptomat   | 126      | 282         | 0             | left vent hyp | 156    | 1          | 0               | upsloping      |
| 6  | 1           | 35  | male   | asymptomat   | 120      | 198         |               | normal        | 130    | 1          | 1.6             | flat           |
| 7  | 0           | 35  | male   | atypical ang | 122      | 192         | 0             | normal        | 174    | 0          | 0               | upsloping      |
| 8  | 0           | 37  | female | non-anginal  | 120      | 215         | 0             | normal        | 170    | 0          | 0               | upsloping      |
| 9  | 0           | 37  | male   | non-anginal  | 130      | 250         | 0             | normal        | 187    | 0          | 3.5             | downsloping    |
| 10 | 1           | 38  | male   | typical ang  | 120      | 231         | 0             | normal        | 182    | 1          | 3.8             | flat           |
| 11 | 0           | 38  | male   | non-anginal  | 138      | 175         | 0             | normal        | 173    | 0          | 0               | upsloping      |
| 12 | 0           | 39  | female | non-anginal  | 94       | 199         | 0             | normal        | 179    | 0          | 0               | upsloping      |
| 13 | 0           | 39  | female | non-anginal  | 138      | 220         | 0             | normal        | 152    | 0          | 0               | flat           |

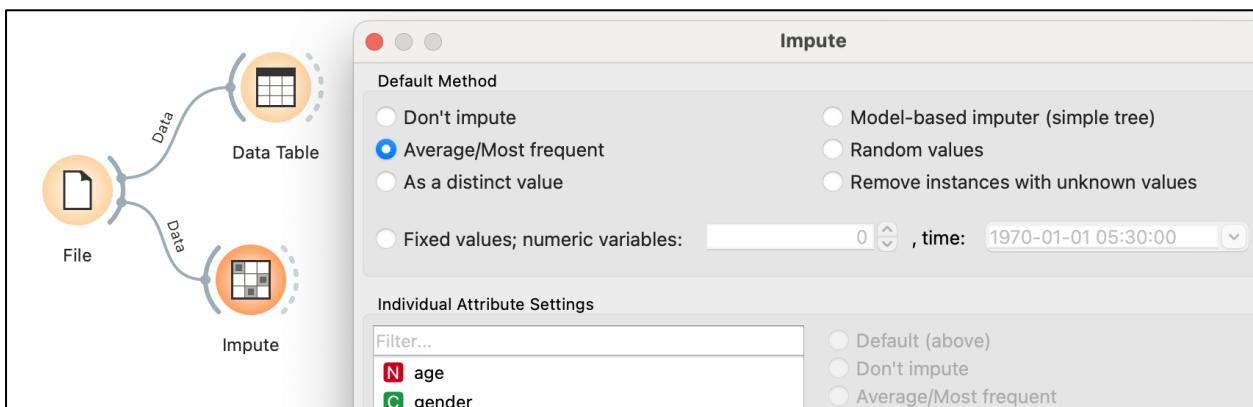
#### Step 3.1: Upload Data

1. Use the **File** widget to upload a dataset with missing values.
2. Assign the role of **"Target"** to the feature you want to predict.



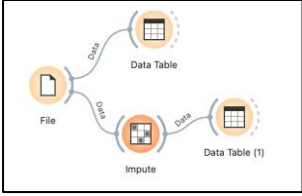
#### Step 3.2: Handle Missing Values

1. Connect the **Impute** widget to the **File** widget.
2. Double-click the **Impute** widget and select an imputation strategy: Average (mean), Most frequent (mode), Fixed value, Random value



### Step 3.3: Verify Cleaned Data

1. Connect the **Data Table** widget to the **Impute** widget.
2. Open the Data Table to confirm the missing values have been replaced.



|    | arterial narrowin | age | gender | chest pain   | rest SBP | cholesterol | fasting blood sugar> | rest ECG         | max HR |
|----|-------------------|-----|--------|--------------|----------|-------------|----------------------|------------------|--------|
| 1  | 0                 | 29  | male   | atypical ang | 130      | 204         | 0                    | left vent hyp... |        |
| 2  | 0                 | 34  | female | atypical ang | 118      | 220.76      | 0                    | normal           |        |
| 3  | 0                 | 34  | male   | typical ang  | 125.10   | 182         | 0                    | left vent hyp... |        |
| 4  | 0                 | 35  | female | asymptomatic | 138      | 183         | 0                    | normal           |        |
| 5  | 1                 | 35  | male   | asymptomatic | 126      | 282         | 0                    | left vent hyp... |        |
| 6  | 1                 | 35  | male   | asymptomatic | 120      | 198         | 0                    | normal           |        |
| 7  | 0                 | 35  | male   | atypical ang | 122      | 192         | 0                    | normal           |        |
| 8  | 0                 | 37  | female | non-anginal  | 120      | 215         | 0                    | normal           |        |
| 9  | 0                 | 37  | male   | non-anginal  | 130      | 250         | 0                    | normal           |        |
| 10 | 1                 | 38  | male   | typical ang  | 120      | 231         | 0                    | normal           |        |
| 11 | 0                 | 38  | male   | non-anginal  | 138      | 175         | 0                    | normal           |        |
| 12 | 0                 | 39  | female | non-anginal  | 94       | 199         | 0                    | normal           |        |

Missing values are now filled with the chosen method (e.g., average values).

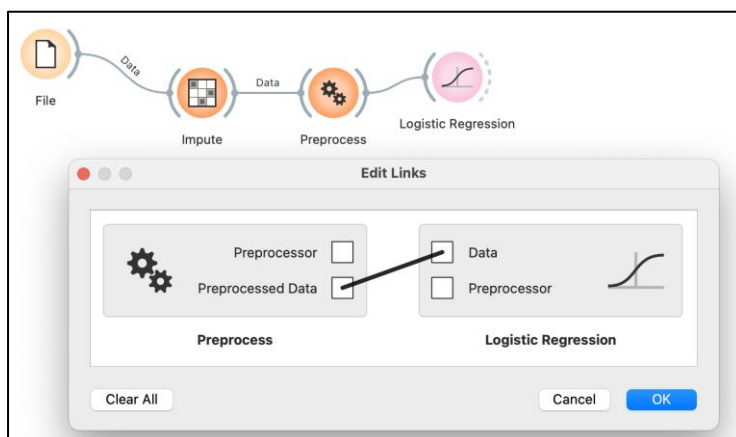
### Step 4: Analyze Data

After cleaning, Orange provides various advanced analytics tools to extract insights:

- **K-Means:** For segmenting data into clusters.
- **Logistic Regression / Decision Tree:** For predicting outcomes using labeled data.
- **Scatter Plot / Box Plot / Heat Map:** For visualizing data patterns and relationships.

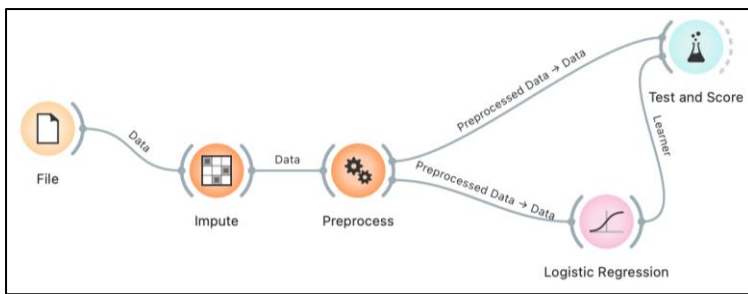
#### Step 4.1: Build a Logistic Regression Model

1. Drag and drop the **Logistic Regression** widget.
2. Connect it to the cleaned and normalized data.



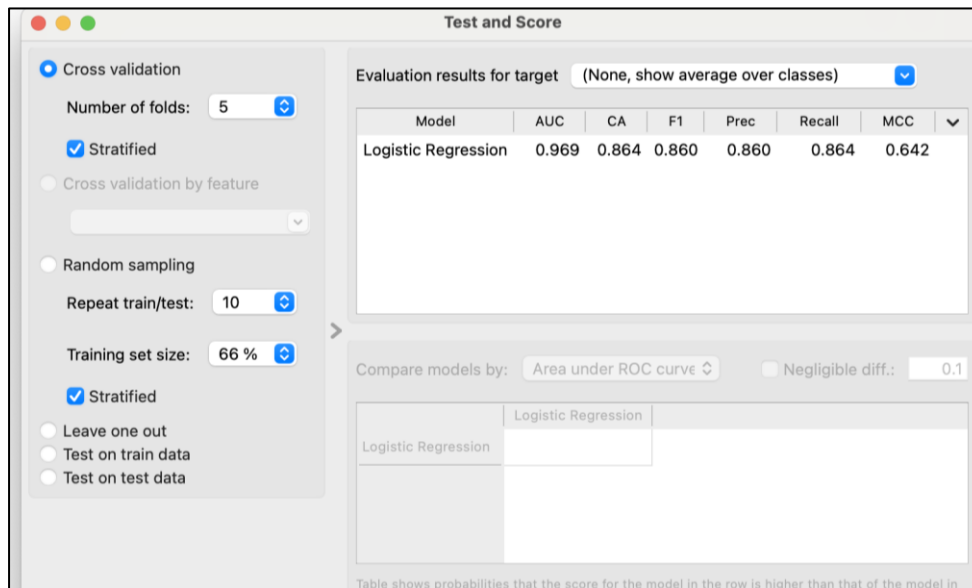
#### Step 4.2: Test the Model

1. Add the **Test and Score** widget.
2. Connect the **Test and Score** widget to:
  - a. The **Logistic Regression** widget (learner data)
  - b. The processed data.



### Step 4.3: Choose a Validation Method

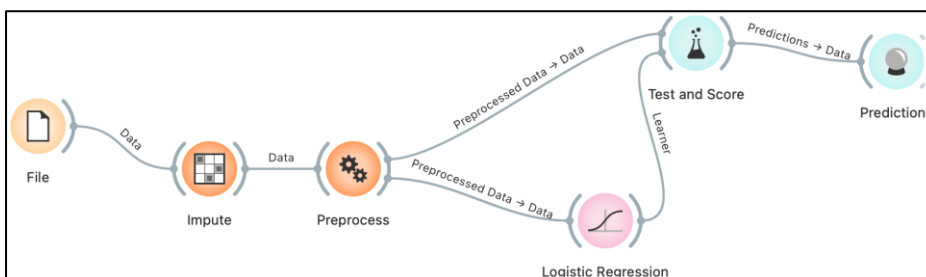
1. Double-click the **Test and Score** widget. Select a validation method (e.g., **Cross-Validation**).



### Step 4.4: Generate Predictions

Connect the **Predict** widget to the **Test and Score** widget.

Check the predictions generated using the Logistic Regression model.



## 5.7. Mining Data Streams

To understand mining data streams, we first understand what data stream is. A data stream is a continuous, real-time flow of data generated by various sources. These sources can include sensors, satellite image data, Internet and web traffic, etc.

Mining data streams refers to the process of extracting meaningful patterns, trends, and knowledge from a continuous flow of real-time data. Unlike traditional data mining, it processes data as it arrives, without storing it completely. An example of an area where data

stream mining can be applied is website data. Websites typically receive continuous streams of data daily. For instance, a sudden spike in searches for "election results" on a particular day might indicate that elections were recently held in a region or highlight the level of public interest in the results.

## 5.8. Future of Big Data Analytics

The future of Big Data Analytics is highly influenced by several key technological advancements that will shape the way data is processed and analyzed. A few of them are:

**Real-Time Analytics:** It will allow businesses to process data instantaneously, providing immediate insights for decision-making and enabling actions based on live data, such as monitoring customer behavior or tracking supply chain activities.

**Development of Advanced Models in Predictive Analytics:** Predictive analytics will evolve with the integration of more sophisticated machine learning and AI algorithms, enabling organizations to forecast trends and behaviors with greater precision.

**Quantum Computing:** Quantum computing promises to revolutionize Big Data analytics by offering unprecedented processing power. Quantum computers will be able to solve complex problems much faster than classical computers.

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### Activity 1: Note – This is a research-based group activity

- i) Watch this video using the link <https://www.youtube.com/watch?v=37x5dKW-X5U>
- ii) Form a group, explore the applications of Big Data & Data Analytics in the following fields, and fill in the table given below:

| Field                   | Video resource | Insights are drawn about this field and its futuristic development |
|-------------------------|----------------|--------------------------------------------------------------------|
| Education               |                |                                                                    |
| Environmental Science   |                |                                                                    |
| Media and Entertainment |                |                                                                    |

## 6.1. WHAT IS A NEURAL NETWORK?

A network of neurons is called “Neural Network”. “Neural” comes from the word neuron of the human nervous system. The neuron is a cell within the nervous system which is the basic unit of the brain used to process and transmit the information to all other nerve cells and muscles.

A similar behaviour of neurons is embedded by Artificial Intelligence into a Network called Artificial Neural Network. It can adapt to changing inputs and hence the network generates the best possible outcome without making any changes anywhere as the human brain does.

A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.

The main advantage of neural networks is that they can extract data features automatically without needing any input from the programmer. Artificial Neural Networks (ANN) are very famous these days and also considered to be a very interesting topic due to their applications in chat-bots. It also makes our work easy by auto replying the emails, suggesting email replies, spam filtering, Facebook image tagging, showing items of our interest in the e-shopping web portals, and many more. One of the best-known examples of a neural network is Google’s search algorithm.

### 6.1.1 PARTS OF A NEURAL NETWORK: -

Every neural network comprises layers of interconnected nodes — an input layer, hidden layer(s), and an output layer, as shown in Fig 6.1.

1. **Input Layer:** This layer consists of units representing the input fields. Each unit corresponds to a specific feature or attribute of the problem being solved.
2. **Hidden Layers:** These layers, which may include one or more, are located between the input and output layers. Each hidden layer contains nodes or artificial neurons, which process the input data. These nodes are interconnected, and each connection has an associated weight.
3. **Output Layer:** This layer consists of one or more units representing the target field(s). The output units generate the final predictions or outputs of the neural network.

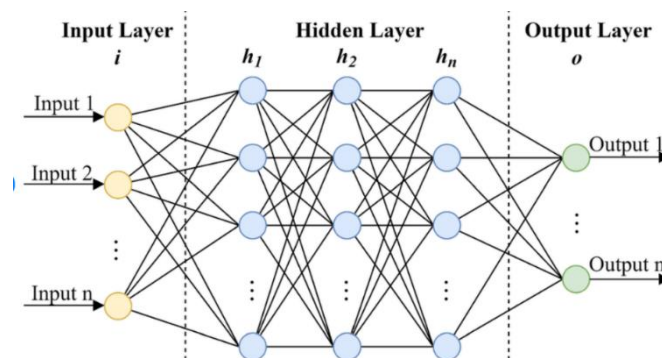


Fig 6.1 Layers of Neural Network

Figure Source : [https://www.researchgate.net/figure/General-Neural-Network-Architecture-45\\_fig3\\_355944019](https://www.researchgate.net/figure/General-Neural-Network-Architecture-45_fig3_355944019)

Each node is connected to others, and each connection is assigned a weight. If the output of a node exceeds a specified threshold value, the node is activated, and its output is passed to the next layer of the network. Otherwise, no data is transmitted to the subsequent layer.

An Artificial Neural Network (ANN) with two or more hidden layers is known as a **Deep Neural Network**. The process of training deep neural networks is called **Deep Learning**. The term “deep” in deep learning refers to the number of hidden layers (also called *depth*) of a neural network. A neural network that consists of more than three layers—which would be inclusive of the inputs and the output layers—can be considered a deep learning algorithm. A neural network that only has three layers is just a basic neural network.

## 6.2. COMPONENTS OF A NEURAL NETWORK

The key components of a neural network are as follows:

### 1. Neurons:

- Neurons (also known as nodes) are the fundamental building blocks of a neural network.
- They receive inputs from other neurons or external sources.
- Each neuron computes a weighted sum of its inputs, applies an activation function, and produces an output.

### 2. Weights:

- Weights represent the strength of connections between neurons.
- Each connection (synapse) has an associated weight.
- Weights convey the importance of that feature in predicting the final output.
- During training, neural networks learn optimal weights to minimize error.

### 3. Activation Functions

- Activation functions in neural networks are like decision-makers for each neuron. They decide whether a neuron should be activated (send a signal) or not based on the input it receives.
- Different types of Activation Functions are Sigmoid Function, Tanh Function, ReLU (Rectified Linear Unit), etc. (A detailed explanation of activation functions is beyond the scope of the syllabus)
- These functions help neural networks learn and make decisions by adding non-linearities to the model, allowing them to understand complex patterns in data.

### 4. Bias:

- Bias terms are constants added to the weighted sum before applying the activation function.
- They allow the network to shift the activation function horizontally.
- Bias helps account for any inherent bias in the data.

## 5. Connections:

- Connections represent the synapses between neurons.
- Each connection has an associated weight, which determines its influence on the output of the connected neurons.
- Biases (constants) are also associated with each neuron, affecting its activation threshold.

## 6. Learning Rule:

- Neural networks learn by adjusting their weights and biases.
- The learning rule specifies how these adjustments occur during training.
- Backpropagation, a common learning algorithm, computes gradients and updates weights to minimize the network's error.

## 7. Propagation Functions:

- These functions define how signals propagate through the network during both forward and backward passes. Forward pass is known as Forward Propagation and backward pass is known as Back Propagation.

### • Forward Propagation

In the forward propagation, input data flows through the layers, and activations are computed. Here, input data flows through the network layers, and activations are computed. The predicted output is compared to the actual target (ground truth), resulting in an error (loss).

### • Back Propagation

Backpropagation is the essence of neural network training. It is the practice of fine-tuning the weights of a neural network based on the error rate (i.e. loss) obtained in the previous epoch (i.e. iteration.) Proper tuning of the weights ensures lower error rates, making the model reliable by increasing its generalization and helping the network to improve its prediction over time.

Backpropagation, (short for “backward propagation of errors”) is an optimization algorithm used during neural network training. It adjusts the weights of the network based on the error (loss) obtained in the previous iteration (epoch).

In Back propagation, gradients are propagated to update weights using optimization algorithms (e.g., gradient descent).

## 6.3. WORKING OF A NEURAL NETWORK

Imagine each node as a simple calculator. It takes input numbers, multiplies them by certain values (weights), adds them together, adds an extra number (bias), and then gives an output. This output is used as input for the next node in the network. The formula would look something like this:



$$\sum w_i x_i + \text{bias} = w_1 x_1 + w_2 x_2 + w_3 x_3 + \text{bias}$$

output  $\rightarrow f(x) = 1$  , if  $\sum w_i x_i + b \geq 0$ ;  
 $f(x) = 0$  , if  $\sum w_i x_i + b < 0$

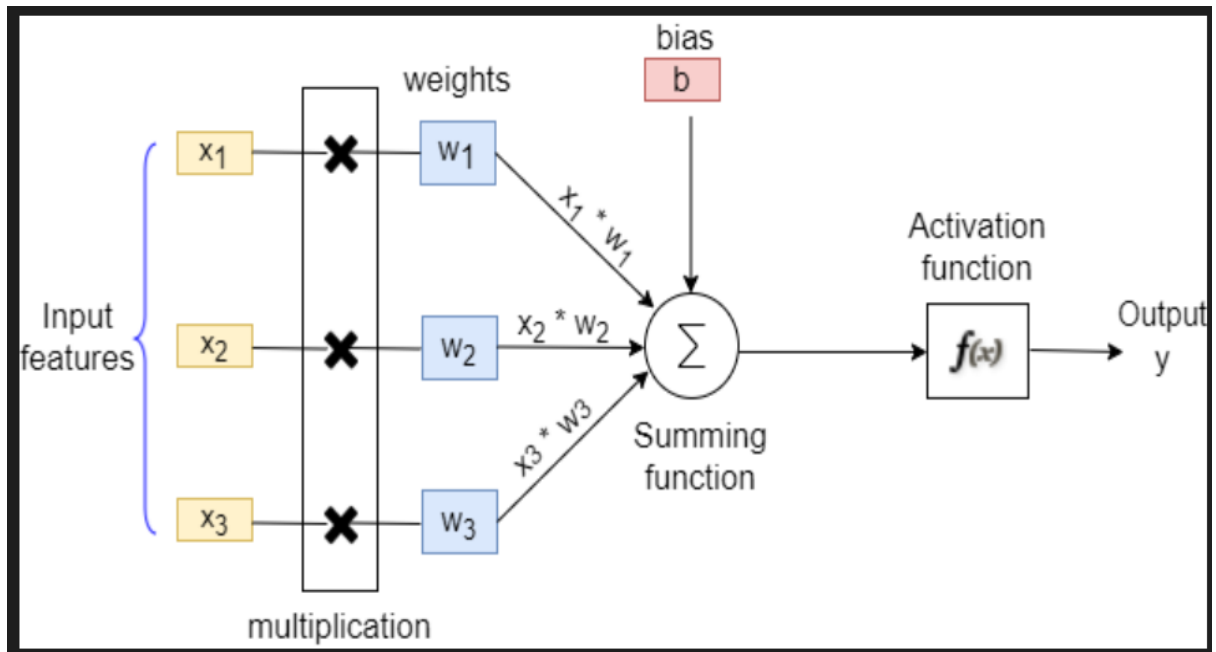


Fig 6.2 Image source : <https://www.bombaysoftwares.com/blog/feed-forward-propagation>

Fig 6.2 illustrates that the output of a neuron can be expressed as a linear combination of weight 'w' and bias 'b', expressed mathematically as  $w * x + b$ .

Each input is assigned a weight to show its importance. These weights are used to calculate the total input by multiplying each input with its weight and adding them together. This total input then goes through an activation function, which decides if the node should "fire" or activate based on the result. If it fires, the output is passed to the next layer. This process continues, with each node passing its output to the next layer, defining the network as a feedforward network. *One single node might look like using binary values.*

Let us see a simple problem.

CASE I: Let the features be represented as  $x_1, x_2$  and  $x_3$ .

Input Layer:

Feature 1,  $x_1 = 2$

Feature 2,  $x_2 = 3$

Feature 3,  $x_3 = 1$

Hidden Layer:

Weight 1,  $w_1 = 0.4$

Weight 2,  $w_2 = 0.2$

Weight 3,  $w_3 = 0.6$

bias = 0.1

threshold = 3.0

Output: Using the formula:

$$\sum w_i x_i + \text{bias} = w_1 x_1 + w_2 x_2 + w_3 x_3 + \text{bias}$$

$$= (0.4 \times 2) + (0.2 \times 3) + (0.6 \times 1) + 0.1$$

$$= 0.8 + 0.6 + 0.6 + 0.1$$

$$= 2.1$$

Now, we apply the threshold value:

If output > threshold, then output = 1 (active)

If output < threshold, then output = 0 (inactive)

In this case:

Output (2.1) < threshold (3.0)

So, the output of the hidden layer is:

Output = 0

This means that the neuron in the hidden layer is inactive.

## CASE II

Let's say we have another neuron in the output layer with the following weights and bias:

$w_1 = 0.7$

$w_2 = 0.3$

bias = 0.2

The output of the hidden layer (0) is passed as input to the output layer:

$$\text{Output} = w_1 \_x1 + w_2 \_x2 + \text{bias}$$

$$= 0.7\_0 + 0.3\_0 + 0.2$$

$$= 0.2$$

Let's assume the threshold value for the output layer is 0.1:

Output (0.2) > threshold (0.1)

So, the final output of the neural network is:

Output = 1

We can now apply this concept to a more tangible example: -

Imagine you are standing on the shore, contemplating whether to go for surfing or not. You glance out at the ocean, assessing the conditions. Are the waves big and inviting? Is the line-up crowded with fellow surfers? And perhaps most importantly, has there been any recent shark activity? Using Neural Network concept decide **“Whether you should go for surfing or not”**

Note: The decision to go or not to go is our predicted outcome,  $\hat{y}$  or y-hat. (The estimated or predicted values in a regression or other predictive model are termed the  $\hat{y}$ .)

If Output is 1, then Yes (go for surfing), else 0 implies No (do not go for surfing)

### Solution:

- Let us assume that there are three factors influencing your decision-making:
  - Are the waves good? (Yes: 1, No: 0)
  - Is the line-up empty? (Yes: 1, No: 0)
  - Has there been a recent shark attack? (Yes: 0, No: 1)
- Then, let us assume the following, giving us the following inputs:
  - $X_1 = 1$ , since the waves are pumping
  - $X_2 = 0$ , since the crowds are out
  - $X_3 = 1$ , since there has not been a recent shark attack
- Now, we need to assign some weights to determine importance. Larger weights signify that particular variables are of greater importance to the decision or outcome.
  - $W_1 = 5$ , since large swells do not come often
  - $W_2 = 2$ , since you are used to the crowds
  - $W_3 = 4$ , since you have a fear of sharks

| Factor             | Input | Weight |
|--------------------|-------|--------|
| Wave Quality       | 1     | 5      |
| Lineup Congestion  | 0     | 2      |
| Any Shark Activity | 1     | 4      |

- Finally, we will assume a threshold value of 3, which would translate to a bias value of  $-3$ . With all the various inputs, we can start to plug in values into the formula to get the desired output.

$$\hat{y} = (1*5) + (0*2) + (1*4) - 3 = 6$$

If  $\hat{y} > \text{threshold}$ , then output = 1

If  $\hat{y} < \text{threshold}$ , then output = 0

Since 6 is greater than 3, we can determine that the output of this node would be 1. **In this instance, you would go surfing.**

If we adjust the weights or the threshold, we can achieve different outcomes from the model.

## 6.4. TYPES OF NEURAL NETWORKS

Neural networks can be classified into different types, which are used for different purposes. While this is not a comprehensive list of types, the below would be representative of the most common types of neural networks that you will come across for its common use cases:

### 6.4.1 Standard Neural Network (Perceptron):

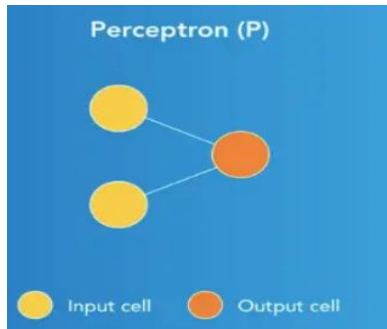


Fig 6.3

The perceptron, created by Frank Rosenblatt in 1958, is a simple neural network with a single layer of input nodes fully connected to a layer of output nodes (Fig 6.3). It uses Threshold Logic Units (TLUs) as artificial neurons.

**Application:** Useful for binary classification tasks like spam detection and basic decision-making.

### 6.4.2 Feed Forward Neural Network (FFNN):

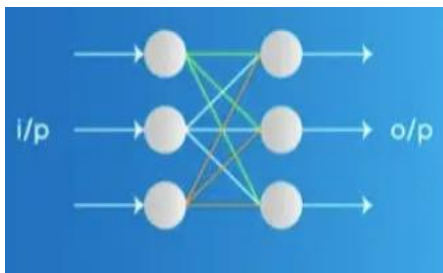


Fig 6.4

FFNN, also known as multi-layer perceptrons (MLPs) have an input layer, one or more hidden layers, and an output layer. Data flows only in one direction, from input to output. They use activation functions and weights to process information in a forward manner (Fig 6.4). FFNNs are efficient for handling noisy data and are relatively straightforward to implement, making them

versatile tools in

various AI applications.

**Application:** used in tasks like image recognition, natural language processing (NLP), and regression.

### 6.4.3 Convolutional Neural Network (CNN):

CNNs utilize filters to extract features from images, enabling robust recognition of patterns and objects (Fig 6.5). CNNs differ from standard neural networks by incorporating a three-dimensional arrangement of neurons, which is particularly effective for processing visual data.

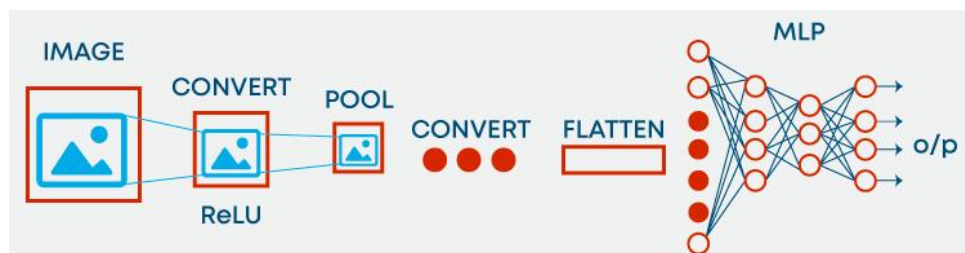
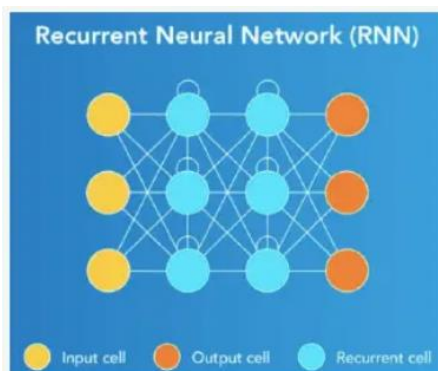


Fig 6.5 Image Source: <https://www.mygreatlearning.com/blog/types-of-neural-networks/>

**Application:** Dominant in computer vision for tasks such as object detection, image recognition, style transfer, and medical imaging.

[To see the working of a CNN, you may watch the video on  
[https://www.youtube.com/watch?v=K\\_BHmztRTpA&t=1s](https://www.youtube.com/watch?v=K_BHmztRTpA&t=1s) ]

#### 6.4.4 Recurrent Neural Network (RNN):

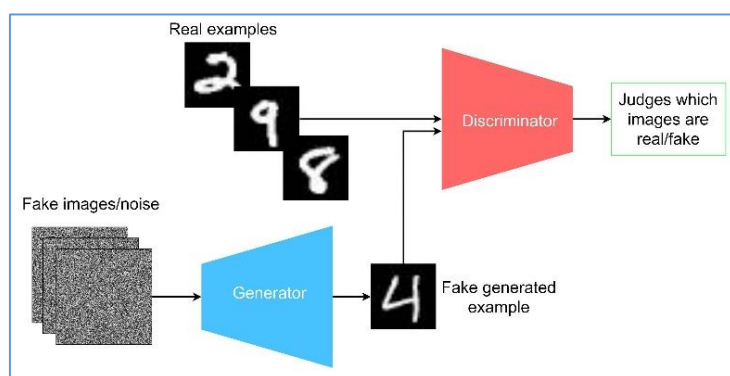


RNNs are designed for sequential data and feature feedback loops to allow information persistence across time steps. They have feedback connections that allow data to flow in a loop (Fig 6.6). If the prediction is wrong, the learning rate is employed to make small changes. Hence, making it gradually increase towards making the right prediction during the backpropagation.

**Application:** Used in NLP for language modeling, machine translation, chatbots, as well as in speech recognition, time series prediction, and sentiment analysis.

Fig 6.6

#### 6.4.5 Generative Adversarial Network (GAN):



GANs consist of two neural networks – a generator and a discriminator – trained simultaneously. The generator creates new data instances, while the discriminator evaluates them for authenticity. They are used for unsupervised learning and can generate new data samples. GANs

are used for generating realistic data, such as images and videos.

Fig 6.7

**Application:** Widely employed in generating synthetic data for various tasks like image generation, style transfer, and data augmentation.

## 6.5. FUTURE OF NN AND ITS IMPACT ON SOCIETY:

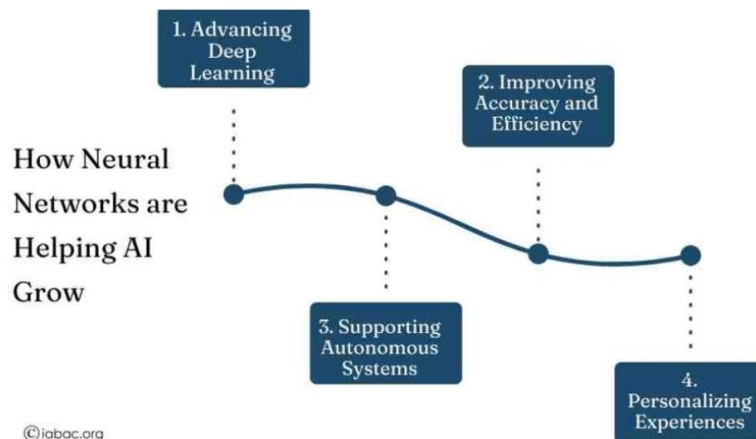


Fig 6.8 <https://iabac.org/blog/exploring-the-role-of-neural-networks-in-the-future-of-artificial-intelligence>

The widespread integration of neural networks has revolutionized society across various domains. These sophisticated algorithms have significantly enhanced efficiency and productivity by automating tasks, streamlining processes, and optimizing resource allocation in industries ranging from manufacturing to finance. Fig 6.8 illustrates how Neural network are helping A.I. grow tremendously.

Moreover, neural networks have personalized products and services by analysing vast datasets, leading to tailored recommendations and experiences that cater to individual preferences and needs. Economically, the adoption of neural networks has spurred innovation, driven economic growth, and created new job opportunities in burgeoning fields like data science and artificial intelligence.

However, alongside these benefits, concerns over ethical and societal implications such as data privacy, algorithmic bias, and job displacement highlight the need for careful consideration and regulation to ensure that neural networks serve the collective well-being of society.

## 6.6. PRACTICAL ACTIVITIES

### 6.6.1 Activity 1: (for Practical Concepts Only)

Machine Learning for Kids url: <https://machinelearningforkids.co.uk/>

Project: Identifying  
Animals & Birds

The screenshot shows the "Start a new machine learning project" form on the Machine Learning for Kids website. The form has the following fields: "Project Name" with the value "Animal & Bird", "Project Type" with the value "recognising text", "Language" with the value "English", and "Storage" with the value "In the cloud". There are "CREATE" and "CANCEL" buttons at the bottom right.

## 7.1 Introduction to Generative AI

In recent times, we all have come across these pieces of information widely being circulated in news or online media. Celebrities and famous people are being targeted with fake images, damaging their reputations and self-esteem. Additionally, some individuals are using AI tools to generate content and claiming it as their own, thereby misusing the power of AI technology. But do you know what is behind all of this? This chapter takes you on an interesting journey where you will explore a new dimension of AI, known as Generative AI.



Fig. 7.1

Generative AI, a facet of artificial intelligence, creates diverse content like audio, text, images, and more, aiming to generate new data resembling its training samples. It utilises machine learning algorithms to achieve this, learning from existing datasets. Examples include ChatGPT, Gemini, Claude, and DALL-E.

## 7.2 Working of Generative AI

Generative AI learns patterns from data and autonomously generates similar samples. It operates within the realm of deep learning, employing neural networks to understand intricate patterns. Models such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) facilitate tasks like image and text generation.

1. **Generative Adversarial Networks (GANs):** A Generative Adversarial Network is a type of neural network architecture. It consists of two networks, a generator, and a discriminator, that compete against each other. The generator generates new data samples, such as images or text (which are fake), while the discriminator evaluates these samples to distinguish between real and fake data. The generator aims to produce samples that are indistinguishable from real data, while the discriminator aims to differentiate between real and generated data. Through adversarial training, where



these networks challenge one another, GANs learn to generate increasingly realistic samples. GANs have been successfully applied in various domains, including image generation, style transfer, and data augmentation.

2. **Variational Autoencoders (VAEs):** VAEs, or Variational Autoencoders, are computer programs designed to learn from data uniquely. They consist of two parts: an encoder, which understands the data and converts it into a hidden space called a latent space (latent space is the compressed representation of the actual data), and a decoder, which translates the information back from this hidden space into its original form. Unlike some other similar programs, like GANs, VAEs focus on capturing the underlying patterns of the data to generate new samples. They find applications in various tasks such as Data generation, detecting anomalies in data, and filling in missing information.

Both GANs and VAEs are powerful generative models but have distinct strengths. GANs are excellent for creating visually realistic outputs, while VAEs are better suited for structured data generation and tasks requiring interpretable latent spaces.

## 7.3 Generative and Discriminative Models

With the vast amount of data generated globally every day, acquiring new data is straightforward. However, addressing this immense volume of data requires either the development of new algorithms or the scaling up of existing ones. These algorithms, rooted in mathematics—particularly calculus, probability, and statistics—can be broadly categorized into two types:

1. Discriminative models

Discriminative models focus on defining class boundaries within the data, making them suitable for tasks like classification. On the other hand, generative models seek to comprehend the underlying data distribution and generate new samples. In simple words, discriminative models are used to distinguish between different categories or classes. These models learn the boundary or difference between classes based on the features of the data. For example, we want to identify an email as either "spam" or "not spam". A discriminative model would learn the features like certain words or phrases that differentiate spam emails from non-spam emails.

2. Generative models

By efficiently tackling complex tasks, generative models play a crucial role in managing large datasets.

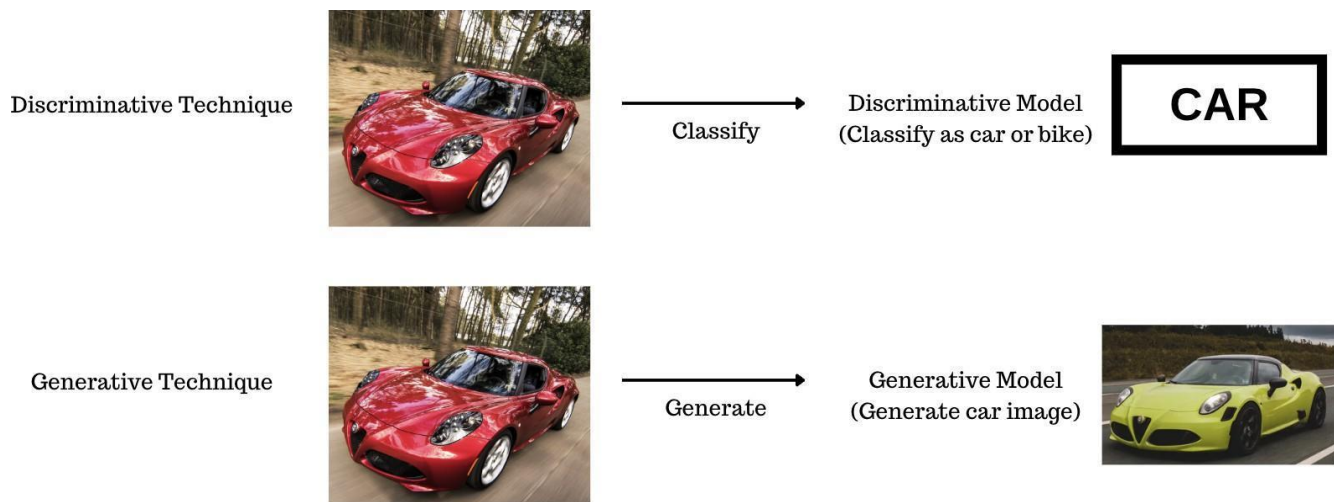


Fig. 7.2

### Differences between Generative AI and Discriminative AI:

|                                                                  | Generative AI                                                                                                                              | Discriminative AI                                                                                                                                            |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose<br/>(What is it for?)</b>                             | Helps create things like images and stories and finds unusual things. It learns from data without needing to be told precisely what to do. | Helps determine what something is or belongs to by looking at its features. It is good at telling different things apart and making decisions based on that. |
| <b>Models<br/>(What are they like?)</b>                          | Uses tricks like making things compete against each other or making guesses based on patterns to create new things.                        | Learn by finding rules to separate things and recognise patterns, like understanding whether something is a dog or a cat.                                    |
| <b>Training Focus<br/>(What did they learn during training?)</b> | Tries to understand what makes data unique and how to create new data that are similar but different.                                      | Focuses on learning how to draw lines or make rules to tell other things apart based on their features.                                                      |
| <b>Application<br/>(How are they used in the real world?)</b>    | Helps artists create new artworks, generate new ideas for stories, and find unusual patterns in data.                                      | Powers things like facial and speech recognition and helps make decisions like whether an email is spam or not.                                              |
| <b>Examples of Algorithms used</b>                               | Naïve Bayes, Gaussian discriminant analysis, GAN, VAEs, LLM, DBMs, Autoregressive models                                                   | Logistic Regression, Decision Trees, SVM, Random Forest                                                                                                      |

## 7.4 Applications of Generative AI

### Image Generation:

It involves creating new images based on patterns learned from existing datasets. These models analyse the characteristics of input images and generate new ones that exhibit similar features. It is like computers producing new pictures resembling ones they have seen before. For example, imagine a computer generating fresh cat images based on the ones it has encountered in its training data. Examples include Canva, DALL-E, Stability AI, and Stable Diffusion.

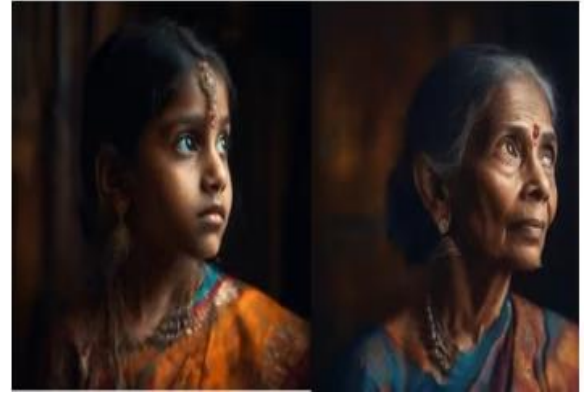
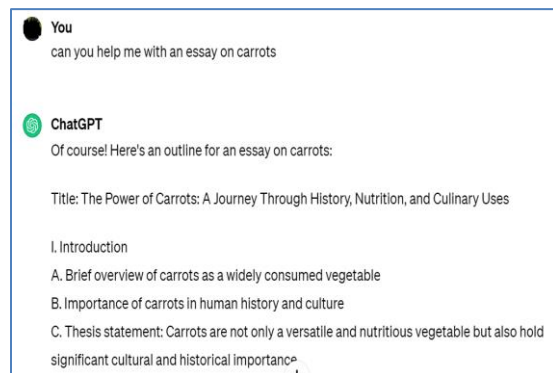


Fig 7.3



Source: Software Snapshot

Fig. 7.4

### Text Generation:

Text generation is when computers write sentences that sound like people wrote them. It involves creating written content that mimics human language patterns. These models analyse text data to produce coherent and contextually relevant text. They learn from many written words to create new sentences that make sense. For example, an AI tool/application that might compose a story that reads like a human authored it. Examples include OpenAI's ChatGPT,

Perplexity and Google's Bard (Gemini)

### Video Generation:

It involves creating new videos by learning from existing ones, including animations and visual effects. These models learn from videos to create realistic and unique visuals, producing new videos that look authentic. For instance, an AI tool/application might generate a movie scene that resembles professional filmmaking. Examples include Google's Lumiere and Deepfake algorithms for modifying video content.



A sample video generated by Meta's new AI text-to-video model, Make-A-Video. The text prompt used to create the video was "a teddy bear painting a portrait." Image: Meta

**Audio Generation:** Audio generation involves computers producing new sounds, such as music or voices, based on sounds they have heard. It involves generating fresh audio content, including music, sound effects, and speech, using AI models. These models derive inspiration from existing audio recordings to generate new audio samples. They learn from existing sounds to create new ones. For instance, an AI tool/application might compose a song that sounds like a real band performed it. Examples include Meta AI's Voicebox and Google's Music LM.

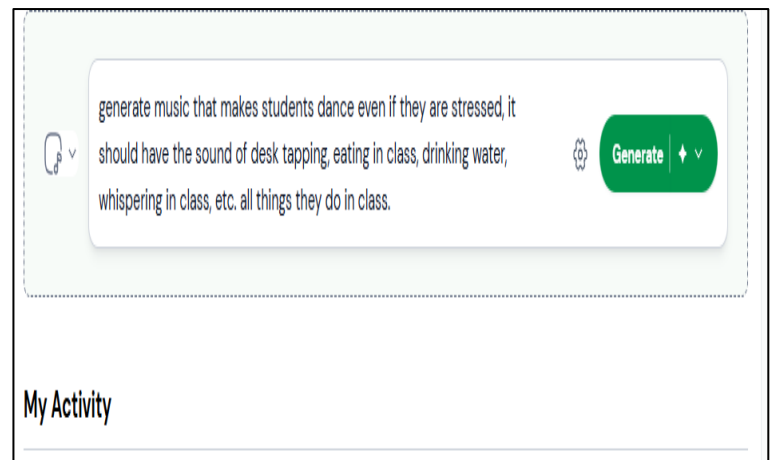


Figure 7.6 Google's Music LM to generate music from text.

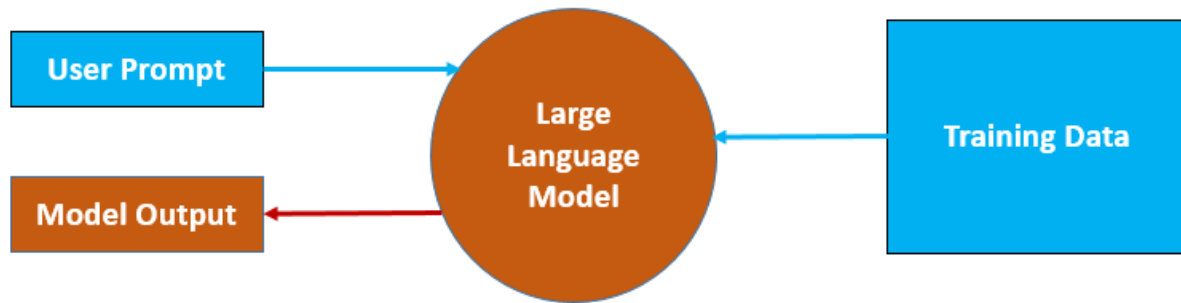
## 7.5 LLM- Large Language Model

A Large Language Model (LLM) is a deep learning algorithm that can perform a variety of Natural Language Processing (NLP) tasks, such as generating and classifying text, answering questions in a conversational manner, and translating text from one language to another.



Fig. 7.7

Large Language Models (LLMs) are called large because they are trained on massive datasets of text and code. These datasets can contain trillions of words, and the quality of the dataset will affect the language model's performance.



**Fig. 7.8**

Image source: <https://levelup.gitconnected.com/understanding-the-magic-of-large-language-model-architecture-transformer-models-exposed-f5164b5db174>

### Transformers in LLMs:

Transformers are a type of neural network architecture that has revolutionized the field of Natural Language Processing (NLP), particularly in the context of Large Language Models (LLMs). The primary use of transformers in LLMs is to enable efficient and effective learning of complex language patterns and relationships within vast amounts of text data.

Some leading Large Language Models (LLMs) are:

- OpenAI's GPT-4o: GPT-4 is multimodal and excels in processing and generating both text and images.
- Google's Gemini 1.5 Pro: Integrates advanced multimodal capabilities for seamless text, image, and speech understanding.
- Meta's LLaMA 3.1: Open-source and optimized for high efficiency in diverse AI tasks.
- Anthropic's Claude 3.5: Prioritizes safety and interpretability in language model interactions.
- Mistral AI's Mixtral 8x7B: Implements a sparse mixture of experts for superior performance with smaller model sizes.

### Applications of LLMs:

- **Text Generation:** LLMs are primarily employed for text generation tasks, including content creation, dialogue generation, story writing, and poetry generation. They can produce coherent and contextually relevant text based on given prompts or input. Other examples include
  - translating natural language descriptions into working code and streamlining development processes.
  - autocompleting text and generating continuations for sentences or paragraphs, enhancing writing tools, and email auto-completion.

- **Audio Generation:** While LLMs themselves do not directly generate audio signals, they can indirectly influence audio generation tasks through their text-to-speech (TTS) capabilities. By generating textual descriptions or scripts, LLMs enable TTS systems to synthesize natural-sounding speech from text inputs.
- **Image Generation:** LLMs have been adapted for image captioning tasks, where they generate textual descriptions or captions for images. While they do not directly generate images, their understanding of visual content can be leveraged to produce relevant textual descriptions, enhancing image accessibility and understanding.
- **Video Generation:** Similarly, LLMs can contribute to video generation tasks by generating textual descriptions or scripts for video content. These descriptions can be used to create subtitles, captions, or scene summaries for videos, improving accessibility and searchability.

#### **Limitations of LLM:**

- Processing text requires significant computational resources, leading to high response time and costs.
- LLMs prioritise natural language over the accuracy, which may result in generating factually incorrect or misleading information with high confidence.
- LLMs might memorize specific details rather than generalize, leading to poor adaptability.

#### **Risks associated with LLM:**

- Trained on Internet text, LLMs may exhibit biases, and concerns arise regarding data privacy when personal information is processed.
- Using sensitive data in training can inadvertently reveal confidential information.
- Inputs intentionally crafted to confuse the model may lead to harmful or illogical outputs.

#### ***Case Study: Demystifying LLaMA - A Publicly Trained Powerhouse in the LLM Landscape***

Large Language Models (LLMs) are a type of deep learning algorithm revolutionizing Natural Language Processing (NLP). These advanced AI models excel at various tasks, including generating human-quality text, translating languages, and answering questions in a conversational way. Their effectiveness hinges on the quality and size of the training data. Traditionally, LLMs are trained on massive, proprietary datasets, limiting transparency and accessibility. This case study explores LLaMA, a unique LLM developed by Meta AI that stands out for its approach to training data and architecture.

**LLaMA's Disruptive Training Approach:** LLaMA leverages publicly available text and code scraped from the internet for training. This *focus on open data, fosters transparency* within the research community and allows for wider accessibility. Additionally, LLaMA's developers implemented efficient training techniques, *requiring less computational power* compared to some LLMs. This translates to better scalability, making LLaMA a more feasible solution for deployment on a wider range of devices with varying processing capabilities.

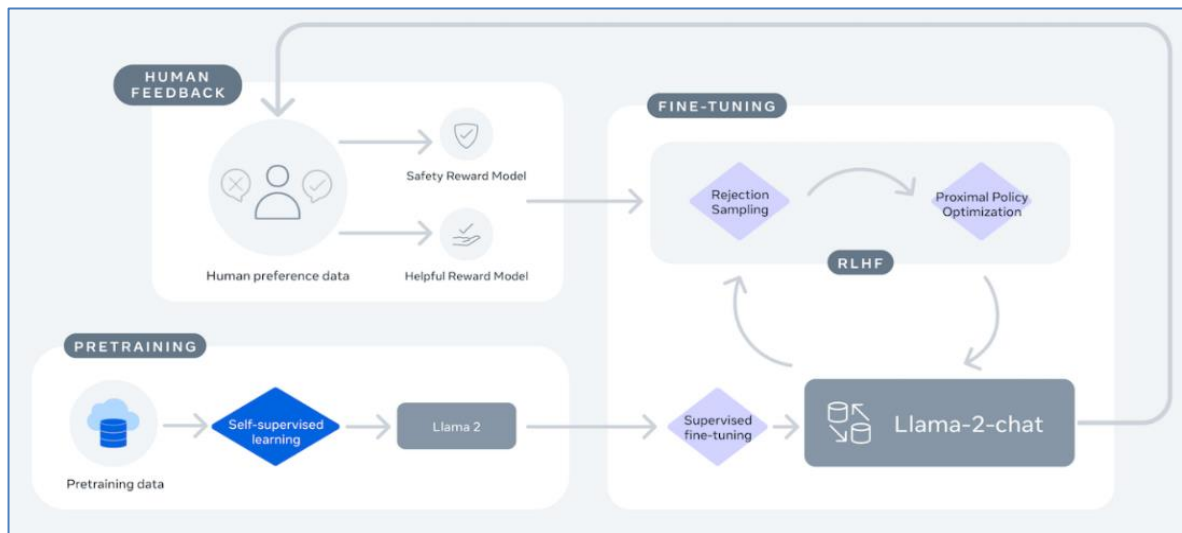


Fig. 7.8

**Flexibility Through Multi-Model Design:** LLaMA offers flexibility through its multi-model design. Meta releases LLaMA in various sizes, ranging from 7 billion to 65 billion parameters. This allows users to choose the model that best suits their specific needs and computational resources. Smaller models can be deployed on devices with lower processing power, ideal for everyday tasks. Conversely, larger models offer superior performance for complex tasks requiring more intensive processing power.

**Impressive Results Despite Publicly Trained Data:** Despite its focus on efficient training with publicly available data, LLaMA delivers impressive results. It benchmarks competitively or even surpasses some larger LLMs on various NLP tasks. Areas of particular strength include *text summarization and question answering*, showcasing LLaMA's ability to grasp complex information and generate concise, informative outputs. This positions LLaMA as a valuable tool with promising applications in education (*personalized learning materials*), content creation (*brainstorming ideas, generating drafts, translation*), and research assistance (*analysing vast amounts of data*).



## 7.6 Future of Generative AI

The future of AI focuses on evolving architectures to surpass current capabilities while prioritizing ethical development to minimize biases and ensure responsible use. Generative AI will address complex challenges in fields like healthcare and education, enhance NLP tasks like multilingual translation, and expand in multimedia content creation. Collaboration between humans and AI will deepen, emphasizing AI's role as a supportive partner across domains.

## 7.7 Ethical and Social Implications of Generative AI

Generative AI, with its ability to create realistic content such as images, videos, and text, brings about a multitude of ethical and social considerations. While the technology offers promising applications, its potential for misuse and unintended consequences raises significant concerns. In this context, understanding the ethical and social implications of generative AI becomes crucial for ensuring responsible development and deployment.

### 1. Deepfake Technology:

The emergence of deepfake AI technology, such as DeepFaceLab and FaceSwap, raises concerns about the authenticity of digital content. Deepfake algorithms can generate compelling fake images, audio, and videos, jeopardising trust in media integrity and exacerbating the spread of misinformation.

**Examples:** Deepfake AI tools, such as DeepArt's style transfer algorithms, can seamlessly manipulate visual content, creating deceptive and misleading media. For instance, deepfake videos have been used to superimpose individuals' faces onto adult content without their consent, leading to privacy violations and reputational damage.

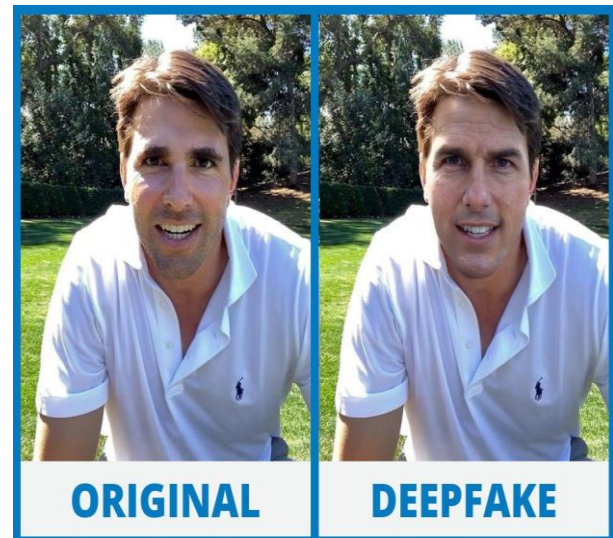


Fig. 7.8

### 2. Bias and Discrimination:

Generative AI models, exemplified by Clearview AI's facial recognition algorithms, have demonstrated biases that disproportionately affect certain demographic groups. These biases perpetuate social inequalities and reinforce stereotypes, posing ethical dilemmas regarding the fairness and impartiality of AI-generated outputs.

**Examples: The AI-powered hiring platform developed by HireVue has faced criticism for perpetuating recruitment bias.** The platform may inadvertently discriminate against underrepresented groups by relying on historical hiring data, hindering diversity and inclusivity.

in employment opportunities.

### 3. Plagiarism:

Presenting AI-generated content as one's work, whether intentionally or unintentionally, raises ethical questions regarding intellectual property rights and academic integrity. Moreover, if AI output significantly resembles copyrighted material, it could potentially infringe upon copyright laws, leading to legal ramifications.

### 4. Transparency:

Transparency in the use of generative AI is paramount to maintaining trust and accountability. Disclosing the use of AI-generated content, particularly in academic and professional settings, is essential to uphold ethical standards and prevent instances of academic dishonesty. Failure to disclose AI use can erode trust and undermine the credibility of research and scholarly work.

#### Points to Remember:

- Be cautious and transparent when using generative AI.
- Respect copyright and avoid presenting AI output as your own.
- Consult your teacher/institution for specific guidelines.

#### Citing Sources with Generative AI:

- **Intellectual Property:** Ensure proper attribution for AI-generated content to respect original creators and comply with copyright laws.
- **Accuracy:** Verify the reliability of AI-generated information and cite primary data sources whenever possible to maintain credibility.
- **Ethical Use:** Acknowledge AI tools and provide context for generated content to promote transparency and ethical use.

#### Citation Example:

1. **Treat the AI as author:** Cite the tool name (e.g., Bard) & "Generative AI tool" in the author spot.
2. **Date it right:** Use the date you received the AI-generated content, not any tool release date.
3. **Show your prompt:** Briefly mention the prompt you gave the AI for reference (optional).

*APA reference for text generated using Google Gemini on February 20, 2024:*

(Optional in parentheses): "Prompt: Explain how to cite generative AI in APA style."

Bard (Generative AI tool). (2024, February 20). How to cite generative AI in APA style.

[Retrieved from [invalid URL removed]]

**Note: Generative AI is a weak AI**

## 8.1 Introduction to Storytelling

“Once upon a time, there lived a King...”. “On a dark night, when it was raining...”. “Long long ago...”. These familiar sentences spark interest, enthusiasm, and curiosity in all of us. Aren’t they familiar?

Yes, they are usually used when telling stories. Stories have been a part of our lives since ancient times. They have existed from the time of cavemen to the present day. Stories are a way to share our imaginations, experiences, and thoughts with others.



Fig. 8.1 Evolution of Storytelling

source : <https://www.trueeditors.com/blog/the-evolution-of-storytelling/>

## So, what are Stories?

Stories are a valuable form of human expression. They connect us closely with one another and transport us to different places and times. Stories can be of various types, like folk tales, fairy tales, fables, and real-life stories. Each type of story creates a sense of connection, and folk tales, in particular, strengthen our sense of belonging to our community and help establish our identity.

Every story has a theme or topic. There is always a storyteller and a listener, and sometimes the listeners can be a group of people. According to the dictionary, a 'story' is a 'factual or fictional narrative,' meaning it tells about an event that can be true or made up, in a way that the listener experiences or learns something. Stories can be used to share information, experiences, or viewpoints.

## What are the benefits of stories?

Stories have a strong impact on human minds as they motivate, ignite, and change our perspectives. **The process of telling a story narratively is known as Storytelling.** It encourages people to make use of their imagination and creativity to express themselves, which improves their communication skills. Storytelling can be in different forms- oral, digital, and written.

## Why is storytelling a powerful tool?

Storytelling is a potent tool for several reasons, such as –

- It generates interest, captivates audiences, and draws them to the narrative.
- It captures our attention, keeping us engaged and focused.
- It communicates meaning, making complex ideas more accessible and understandable.
- It inspires us, evokes emotions, and motivates us to take action.

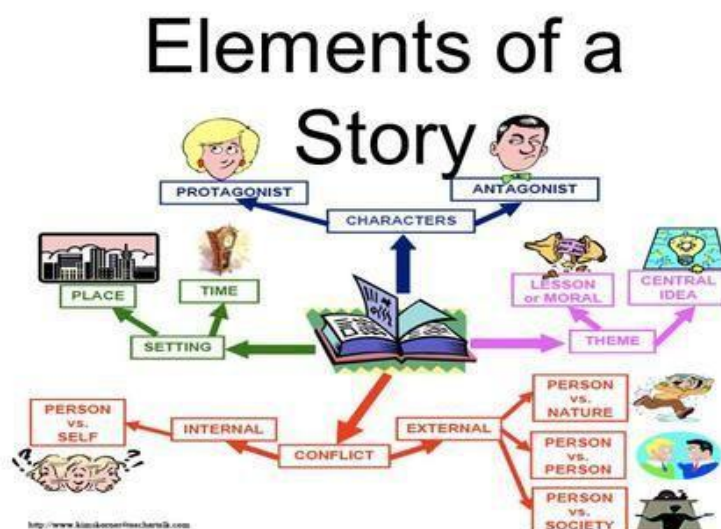


Fig. 8.2 Elements of a story

Source: <https://in.pinterest.com/pin/778419116828184060/>

1. **Characters:** The characters are the people or animals or some things or objects which are featured in a story. They perform the actions and drive the story.
2. **Plot/setting:** Setting refers to the time or location in which the story takes place. Plot refers to the sequence of the events of the story.
3. **Conflict:** It is the problem or the situation the characters are dealing with. It drives the story forward which makes the story engaging and a key element for the characters.
4. **Resolution:** It is the end of the story where the characters arrive at a particular situation to resolve the conflict. It is the stage after climax which is the peak or height of any story.
5. **Insights:** The ability to have a clear, deep, and sometimes sudden understanding of a complicated problem or situation.

### Activity 1:

Think of different types of stories, a real story, a mythological story, a fiction story, folk tale, and then complete the table according to the given headings.

| Name of the story | Type of the story | Characters | Insight gathered/moral |
|-------------------|-------------------|------------|------------------------|
|                   |                   |            |                        |
|                   |                   |            |                        |
|                   |                   |            |                        |
|                   |                   |            |                        |

### Introduction to Data Storytelling

According to Wikipedia, **data are individual facts, statistics, or items of information, often numeric.** In a more technical sense, data is a set of qualitative or quantitative variables about one or more persons or objects. So, keeping these definitions in mind, when we connect logically related data together, they tell us something. The collection of data when represented in a better way has a greater impact which can be engaging, entertaining, thought provoking, helps in better decision making and can make way for big changes.

So, when we interpret this data in a systematic way, then this concept is known as Data Storytelling. It is a practice that is used lately by analysts and data scientists

to communicate their findings and observations from data to technical and non-technical business stakeholders who are in general called audience.

**Data storytelling is the art and practice of translating complex data and analytics into a compelling narrative that is easily understandable and relatable to various audiences.**

Data can be in the simple form of numbers and digits. This data when it is pictorially represented is known as **Data Visualization**. It can be in the form of different types of charts or graphs. Depending on the requirements, data can be interpreted in the form of narratives known as Data Stories which can reduce ambiguity. It can be clear with respect to context and convey the right meaning which can be used for an effective decision-making process.

### **Need for Data Storytelling**

*"Sometimes reality is too complex. Stories give it form." — Jean-Luc Godard, film director, screenwriter*

### **Activity 2:**

Read this passage aloud in class:

Rahul eagerly wanted to share the exciting news with his sister, Priya, and his wife, Smita. Priya's friend, Anil, had promised to arrange an interview for Rahul at the software company where he worked. This job was particularly significant to Rahul as it aligned perfectly with his area of expertise and offered a competitive salary. However, just as Rahul was preparing for the interview, he received a call from another company, where Smita had applied. They offered her a position as well, but it required them to relocate to a different city. This sudden twist left Rahul torn between pursuing his dream job and supporting his wife's career aspirations. Unable to reach Priya, Rahul left her a voicemail, brimming with anticipation. A few hours later, when Priya returned his call, she sounded even more thrilled than Rahul. "Guess what," Priya exclaimed, "Anil just helped me secure a position at the software company."

Now, ask the following questions:

Who is Anil's friend?

Who is Rahul's sister?

Who arranged an interview for whom?



### Activity 3:

Show the following content to the students on screen for approximately 2 minutes.

A police officer pulls over a car for driving too slowly on the highway. “This is a highway, you must drive at least **80 km/hr.**”

“But the sign says **20!**” said the driver. “This is highway route **20**, not speed limit **20.**”

The officer then sees the passenger is unconscious. “Is everything OK?”

“She’s been like that since we turned off highway **180.**”

An established finding in traffic safety is that vehicles should maintain speeds consistent with those of neighboring vehicles. For this reason, federal agencies recommend a minimum speed of eighty kilometers per hour to be adopted by local state, and provincial governments. Nearly all local governments have conformed to this recommendation as it promotes safety and minimizes confusion when passing from one jurisdiction to another or when merging from one highway onto another.

Now, ask them “What is the minimum speed limit on a highway?”

With these two activities, we understand that data, when presented in a narrative format, is better absorbed, retained, and understood compared to a collection of disjointed facts or figures. Just like the effectiveness of storytelling in memory retention, data storytelling enhances the comprehension and impact of data insights by providing context and structure.

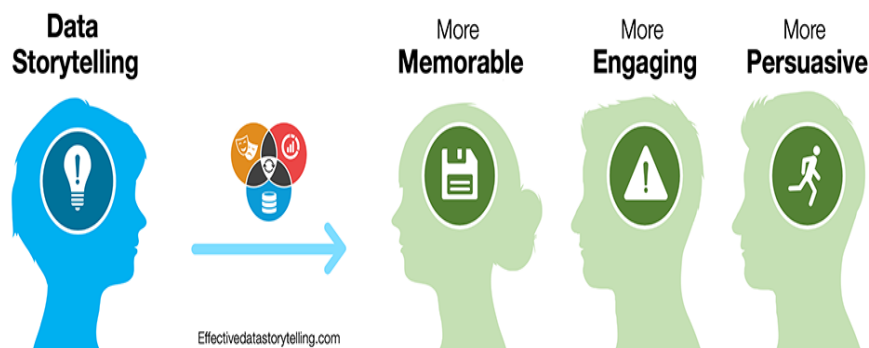
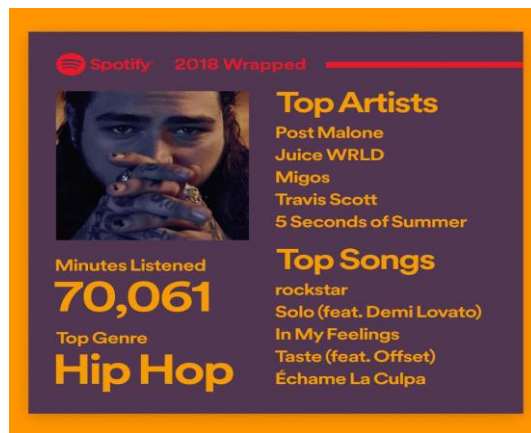


Fig. 8.3 Data Storytelling: Makes information more memorable, engaging, and persuasive

The need for data storytelling is gaining importance in all fields. Many companies and brands are using data storytelling as an effective method of conveying their message and gaining client loyalty. Data storytelling makes complex data more accessible and understandable, allowing audiences or stakeholders to grasp insights easily. Engaging narratives and compelling visuals keep audiences engaged, increasing retention and attention. Storytelling with data empowers better decision-making by presenting evidence-based insights.



## Examples of some famous brand Data Stories



Spotify



Uber

### Why has Data Storytelling become very powerful now?

In today's business context, data storytelling becomes so critical due to these characteristics:

1. It makes the insights and key findings memorable to the audience.
2. It is a persuasive way of communicating key insights and findings to both business stakeholders and technical stakeholders.
3. It is also important that the story is engaging to the audience.

Today, data storytelling is filling a vital gap in business as well as the technical analytics process between technology and people. By combining an increasingly essential resource (data) with a familiar and time-tested form of communication (storytelling), this skill can help more people translate their insights into action. From a data literacy perspective, once an organization's people are comfortable with reading and working with data, they should also learn how to communicate insights effectively.

### 8.2 Essential elements of Data Storytelling

The three key elements of Data Storytelling are:

1. Data
2. Narrative
3. Visuals

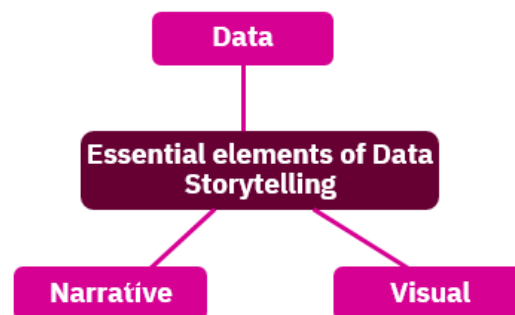
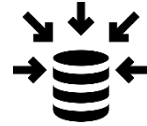


Fig. 8.4 Essential elements of Data Storytelling

**Data:** Basic facts or raw facts about any entity is known as Data. Data is the primary building block of every data story. It serves as the foundation for the narrative and visual elements of your story.



**Narrative:** It refers to the structure or storyline that is crafted to present insights derived from data in a clear, engaging, and informative manner to the audience. It involves identifying and organizing the key information in a linear and coherent fashion, ensuring that the storyline is crisp, and contextual. A well-defined narrative allows audience to understand the significance of insights and how they relate to the broader context of the data analysis.



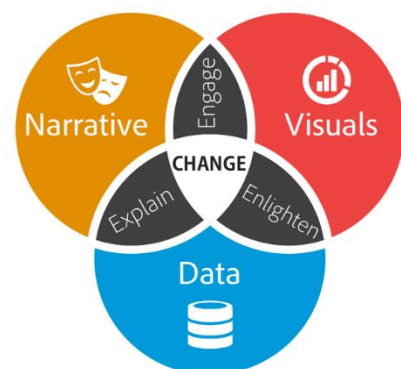
**Visuals:** Visuals refer to the pictorial representations of data using various graphs, charts, and diagrams. They serve as a scene of the data presented in a graphical format, helping to convey complex information more clearly and effectively. They enable the audience to visualize trends, patterns, and anomalies in the data, facilitating better understanding and interpretation of the insights being communicated.



### How are these three interlinked with each other?

- When we explain data observations, it helps the audience understand how the data behaves in different situations and why certain insights are generated.
- Visualizing data through charts and graphs allows the audience or stakeholders to see the data from a different perspective, helping them analyse and make the right decisions.
- Combining narrative and visuals can engage or even entertain an audience.

Each element—data, narrative, or visuals—can be powerful on its own. You can achieve some success with a compelling story, an interesting statistic, or impressive data visualization. However, the real impact comes from skillfully blending data, narrative, and visuals in a data story. When you combine the right visuals and narrative with the right data, you create a data story that can influence and drive change.



## Narrative structure of a data story

Most stories follow a common arc – a protagonist who faces a complication goes on a journey of resolving a difficulty before returning to their normal lives. Building on Aristotle’s simple model, Freytag developed a more robust narrative framework to better understand the arc or progression of a story. The “pyramid-based” dramatic structure with five key stages:

1. **Introduction:** The beginning of the story when the setting is established, and main characters are introduced. It provides the audience with ample background information to understand what is going to happen.
2. **Rising action:** The series of events that build up to the climax of the story.
3. **Climax:** The most intense or important point within the story. It is often an event in which the fortune of the protagonist turns for the better or worse in the story.
4. **Falling action:** The rest of the events that unravel after the main conflict has occurred, but before the final outcome is decided.
5. **Conclusion:** The conclusion of the story where all of the conflicts are resolved and outstanding details are explained.

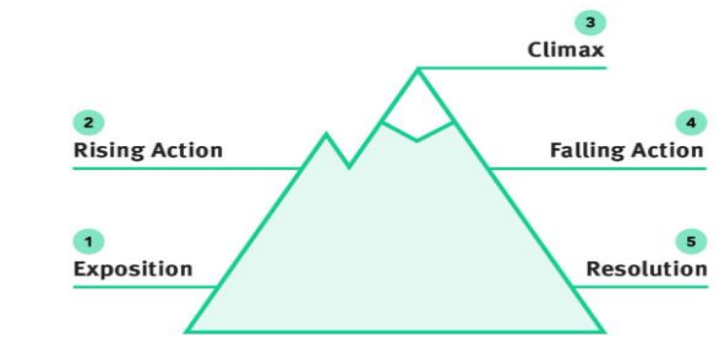


Fig.8.5 Freytag's Pyramid

Image Source: <https://www.linkedin.com/pulse/freytags-pyramid-data-storytelling-unlocking-power-ahsan-khurram/>

In the context of data storytelling, Freytag's Pyramid can be used as a framework to structure the presentation of data and insights in a way that captivates the audience's attention and guides them through the narrative journey.

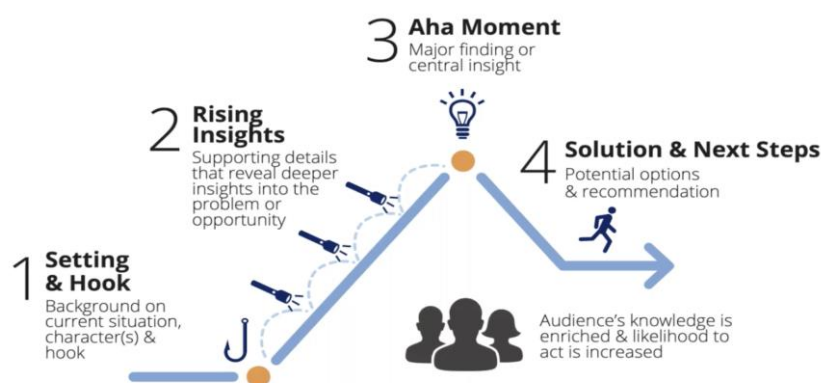
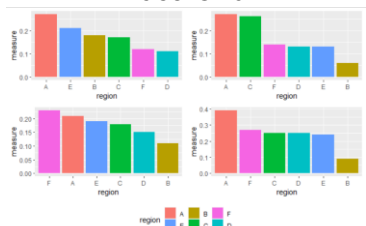
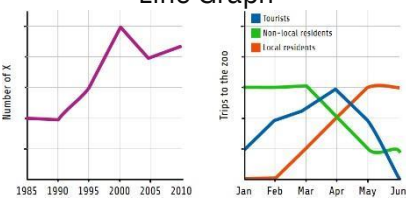
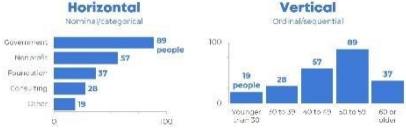
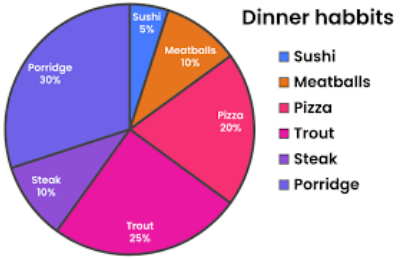


Fig. 8.6 The Data Storytelling Arc: Crafting Insightful Narratives

## Visualizations for different data

Data visualization is a powerful way to show context. Data charts can reveal crucial deviations or affinities in the data that can lead to insights.

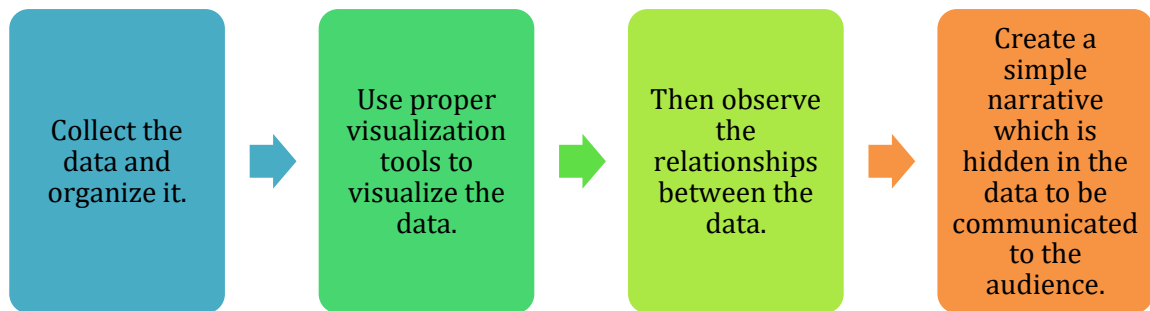
| Type         | Visualization Type                                                                                                        | Description                                                                                  |
|--------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Text Data    | <p>Word Cloud</p>                        | Visual representation of word data where word size indicates frequency and importance.       |
| Mixed Data   | <p>FacetGrid</p>                         | Multi-axes grid with subplots to visualize distribution and relationships between variables. |
| Numeric Data | <p>Line Graph</p>                      | Illustrates data changing over time, useful for trends.                                      |
|              | <p>Bar Chart</p>                       | Compares data between categories using bars.                                                 |
|              | <p>Pie Chart</p> <p>Dinner habits</p>  | Circular chart illustrating numerical proportions.                                           |

|            |                                                                               |                                                                                                                  |
|------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|            | <p><b>Scatter Plot</b></p> <p>Performance of analysts after pilot program</p> | Visualizes relationships and trends between two variables.                                                       |
|            | <p><b>Histogram</b></p> <p>Kobe Bryant Career Playoff Scoring</p>             | Represents distribution of continuous data through bars.                                                         |
|            | <p><b>Heat Map</b></p>                                                        | Compares data across categories using color to identify strong and weak categories.                              |
| Stocks     | <p><b>Candlestick Chart</b></p>                                               | Visual aid for decision-making in stock, forex, commodity, and option trading.                                   |
| Geographic | <p><b>Map Chart</b></p>                                                       | Utilizes geographical maps to display data points or statistical information associated with specific locations. |

## Steps to create a story through data

If the data collected is represented in just a series of graphs and charts, it will not serve the purpose to any organization. It should be communicated well with proper narrative, with proper context and meaning, relevance and clarity. The narrative should be able to take the focus of the audience to the correct spot and not

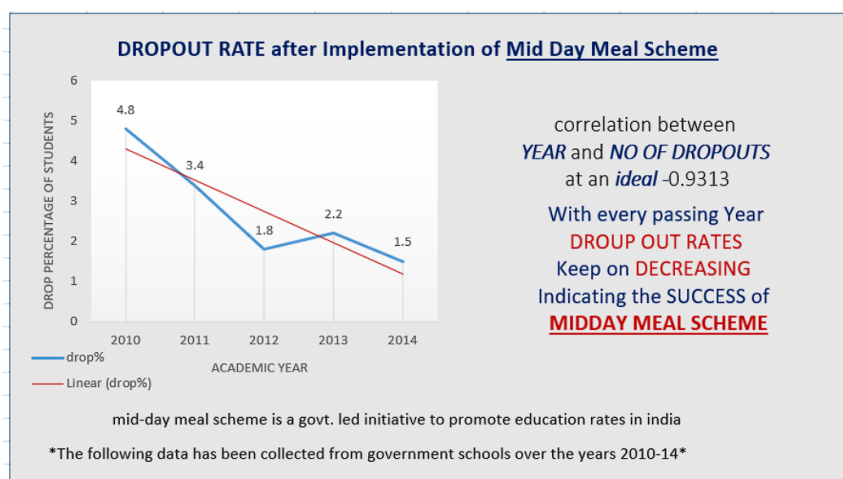
miss out on important facts. To find compelling stories in data sets, the following steps are to be followed:



### Example 1:

Using available data on student enrollment, attendance, and dropout rates, create a compelling data story that explores the impact of the Mid-Day Meal Scheme (MDMS) since its launch in 1995. Uncover trends, patterns, and correlations in the data to tell a story about how the implementation of the MDMS may have influenced dropout rates in the state over the years. Consider incorporating visualizations, charts, and graphs to effectively communicate your findings. Additionally, analyze any external factors or events that might have played a role in shaping these trends. Your goal is to provide a comprehensive narrative that highlights the relationship between the MDMS and student dropout rates in the state.

| Year | Enrolled | No_of_Dropout | Dropout_percentage |
|------|----------|---------------|--------------------|
| 2010 | 3720068  | 178490        | 4.8                |
| 2011 | 3753087  | 128185        | 3.4                |
| 2012 | 3728540  | 68860         | 1.8                |
| 2013 | 3531426  | 76204         | 2.2                |
| 2014 | 3403395  | 50373         | 1.5                |



## Example 2:

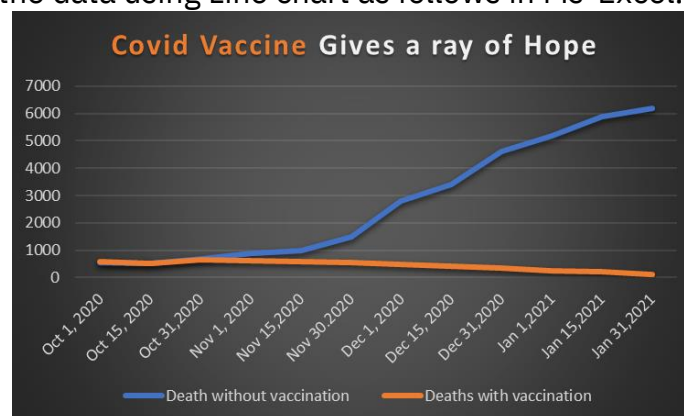
Let us do an activity now to create a data story with the information given below. We have collected the data. Use the above steps to create an effective Data Story.

| Months and Year | Death without vaccination | Deaths with vaccination |
|-----------------|---------------------------|-------------------------|
| Oct 1, 2020     | 560                       | 565                     |
| Oct 15, 2020    | 500                       | 502                     |
| Oct 31,2020     | 670                       | 650                     |
| Nov 1, 2020     | 890                       | 600                     |
| Nov 15,2020     | 1000                      | 580                     |
| Nov 30.2020     | 1500                      | 550                     |
| Dec 1, 2020     | 2800                      | 485                     |
| Dec 15, 2020    | 3400                      | 400                     |
| Dec 31,2020     | 4600                      | 350                     |
| Jan 1,2021      | 5200                      | 250                     |
| Jan 15,2021     | 5900                      | 200                     |
| Jan 31,2021     | 6200                      | 120                     |

### Solution:

Step 1 : Prepare this data sheet in MS-Excel.

Step 2: Visualize the data using Line chart as follows in Ms-Excel.



Step 3: Narrative:

### Covid Vaccine- Gives a ray of Hope

#### Ethics in Data Storytelling

Data storytelling is a powerful tool, but it also requires careful consideration of ethics. Each of the three key elements of data storytelling - Data, Narrative, and Visuals - presents unique ethical challenges that must be addressed to ensure responsible and trustworthy storytelling.

1. Accuracy: Ensure that the data is accurate, reliable, and truthful. Avoid manipulating data to support a predetermined narrative.
2. Transparency: Clearly cite the sources of the data, methods used for analysis, and any limitations or biases. Be transparent about the story's purpose and potential conflicts of interest.
3. Respect for Privacy: Protect the privacy of individuals and groups represented in the data. Avoid sharing personal or sensitive information without consent.



## Conclusion

A data story does not just happen on its own—it must be curated and prepared by someone for the benefit of other people. When we effectively combine the right insights with the right narratives and visuals, we can communicate the data in a manner that can inspire change. The data stories can help other people to understand a problem, risk, or opportunity in a meaningful way that compels them to act on it. So, we can define Data storytelling as a persuasive, structured approach for communicating insights using narrative elements and explanatory visuals to inform decisions and drive changes.

## Exercises

### A. Multiple choice questions

1. Which of the following best describes data storytelling?
  - a) Presenting raw data without any analysis
  - b) Communicating insights through data in a compelling narrative format
  - c) Creating colorful charts and graphs
  - d) Analyzing data without any visual aids
2. What is the primary goal of data storytelling?
  - a) To confuse the audience with complex charts
  - b) To entertain the audience with anecdotes
  - c) To communicate insights and findings effectively using data
  - d) To hide information from the audience
3. Which of the following is NOT a component of effective data storytelling?
  - a) Compelling visuals
  - b) Clear narrative structure
  - c) Overcomplicating the message
  - d) Insightful analysis
4. What role do visuals play in data storytelling?
  - a) They make the presentation look fancy
  - b) They distract the audience from the data
  - c) They help convey complex information quickly and effectively
  - d) They are not necessary in data storytelling
5. Why is it important to know your audience when creating a data story?
  - a) To impress them with your knowledge
  - b) To tailor the message and visuals to their level of understanding and interests
  - c) To exclude certain data points that might confuse them
  - d) To ignore their feedback and preferences